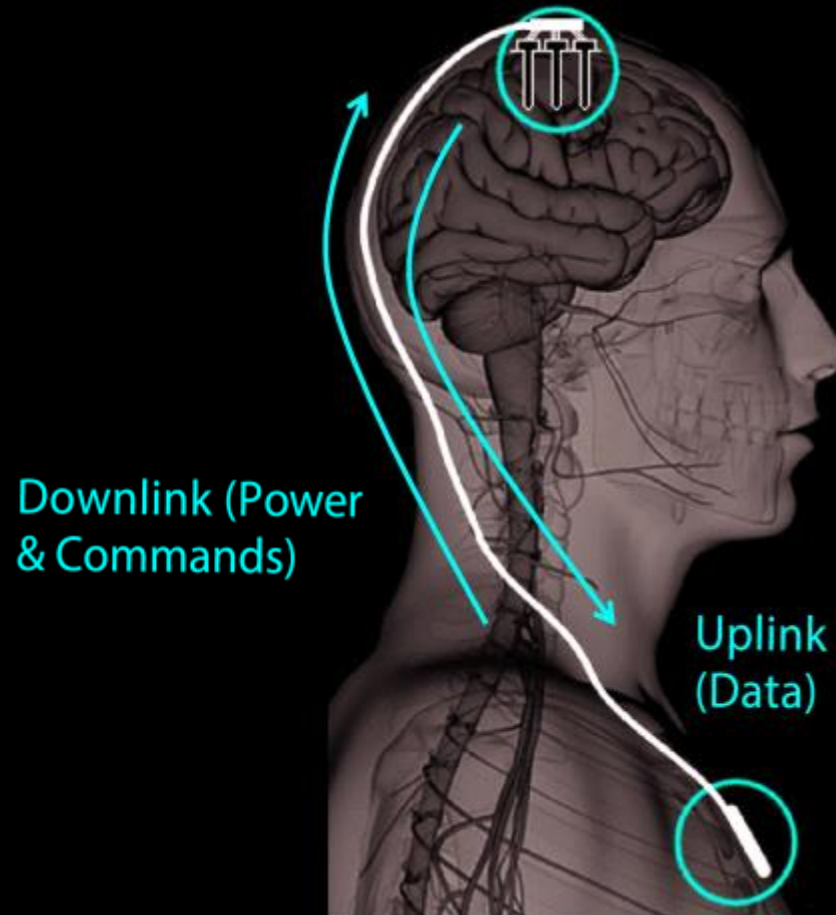




AIC – Analog Integrated Circuits

Low Power LDO  
for a biomedical implantable  
4-wire interface



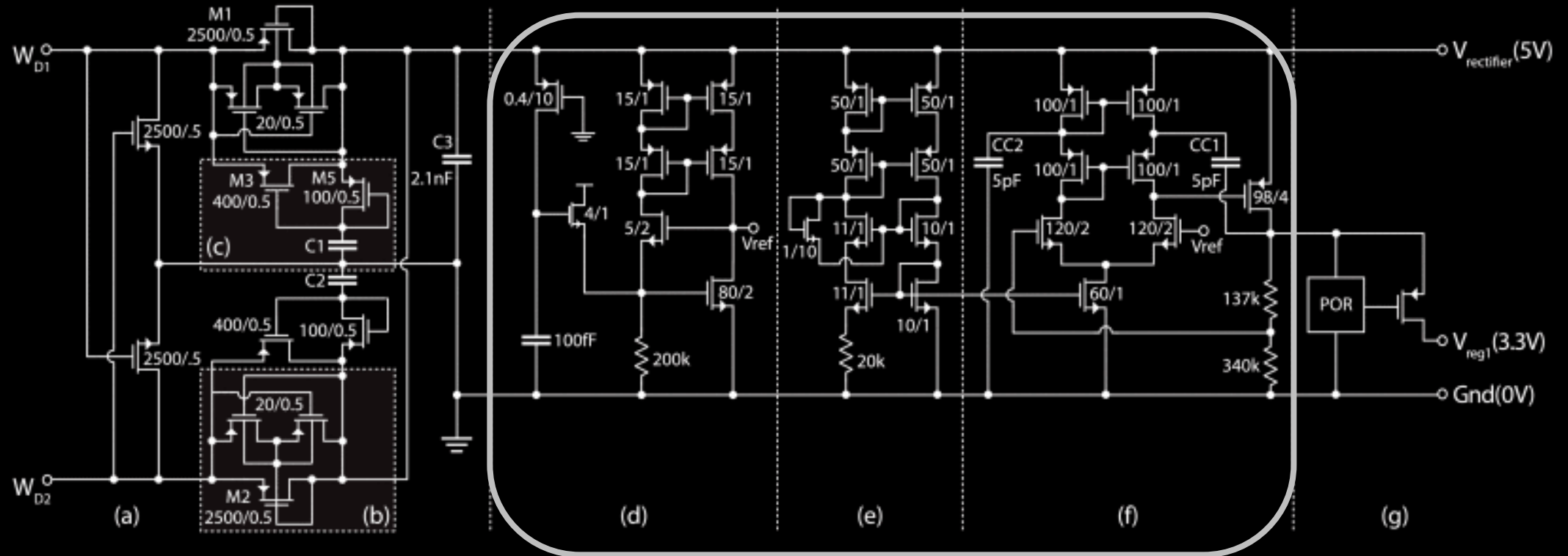
# The starting point...

The following project starts from an IEEE article\* about the needing for an energy efficient wired interface for biomedical multi-implant devices. The basic concept is to design an integrated chip capable of handling both communication and power sharing between an Uplink (coming from the slave sensors implant) and a Downlink (coming from the master chest implant with integrated battery).

The chip has been fabricated by using the ON C35B4 5V compliant technology and presents very low power consumption, requiring a 91.2  $\mu\text{A}$  quiescent current.

The following project is about a re-design of the integrated LDO exploited in the above mentioned article by using the ON C5N technology.

- Introduction
- Project structure



Passive full-wave rectifier with charge-storage technique. It allows the extraction of a DC 5V supply from the Dowlink square wave signal.

The circuitry in the box is the part of interest and it consists of an LDO stage (f) implemented with a PMOS pass-transistor topology, a BIAS generator (e) exploiting a  $\beta$  multiplier approach and a voltage REFERENCE (d). These three circuits have been designed and implemented at schematic level while the voltage reference has been completed with two layout versions.

Power-On-Reset circuit has a logic which monitors the regulator output enabling the final transistor when needed.

# Current budget

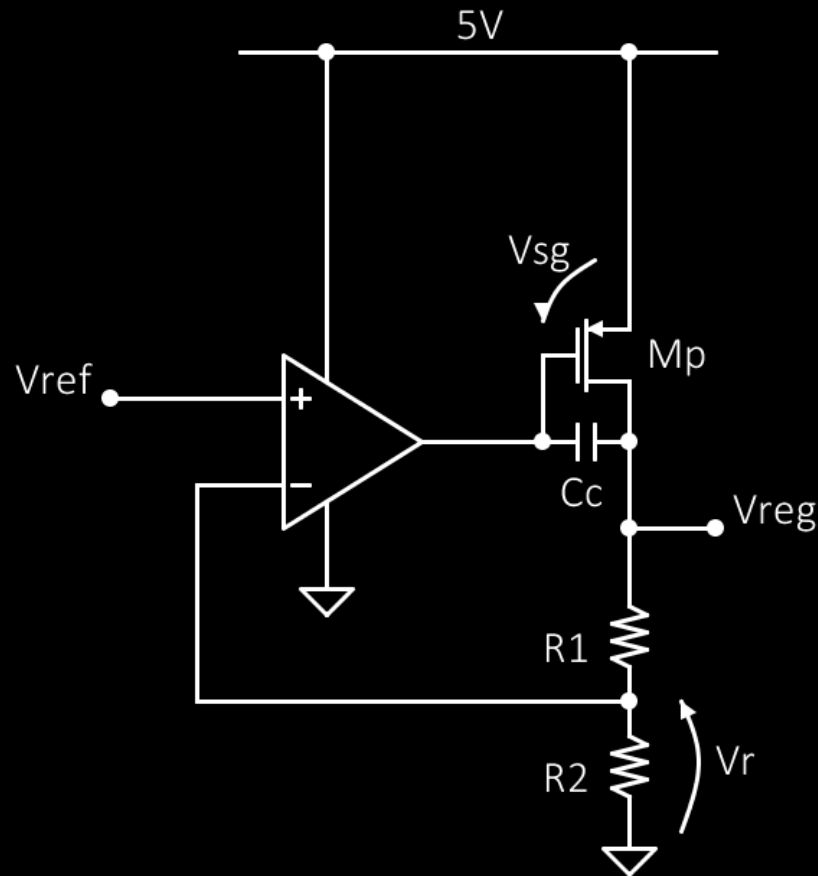
Since the total current of the IC is about 92  $\mu\text{A}$ , the budget for the considered power section has been set to  $\frac{1}{3}$  of the total which leads to about 30  $\mu\text{A}$  of available current. The latter can be rationed as follows:

- 12  $\mu\text{A}$  for the entire LDO stage ( 8  $\mu\text{A}$  for the resistors branch and 4  $\mu\text{A}$  for the differential amplifier);
- 8  $\mu\text{A}$  for the voltage reference (4  $\mu\text{A}$  per branch) which is an important part of the design for DC PSRR performances;
- the remaining budget is to supply the bias circuit (8  $\mu\text{A}$ ) and to keep a safety margin about possible variations.

Having in mind the above limits, the design can start from the LDO power stage and its differential amplifier.

- Introduction
- Project structure
- Current budget
- LDO design I

# LDO design



The specification is to design a 3.3 V regulator capable of delivering up to **2 mA** current to the load.

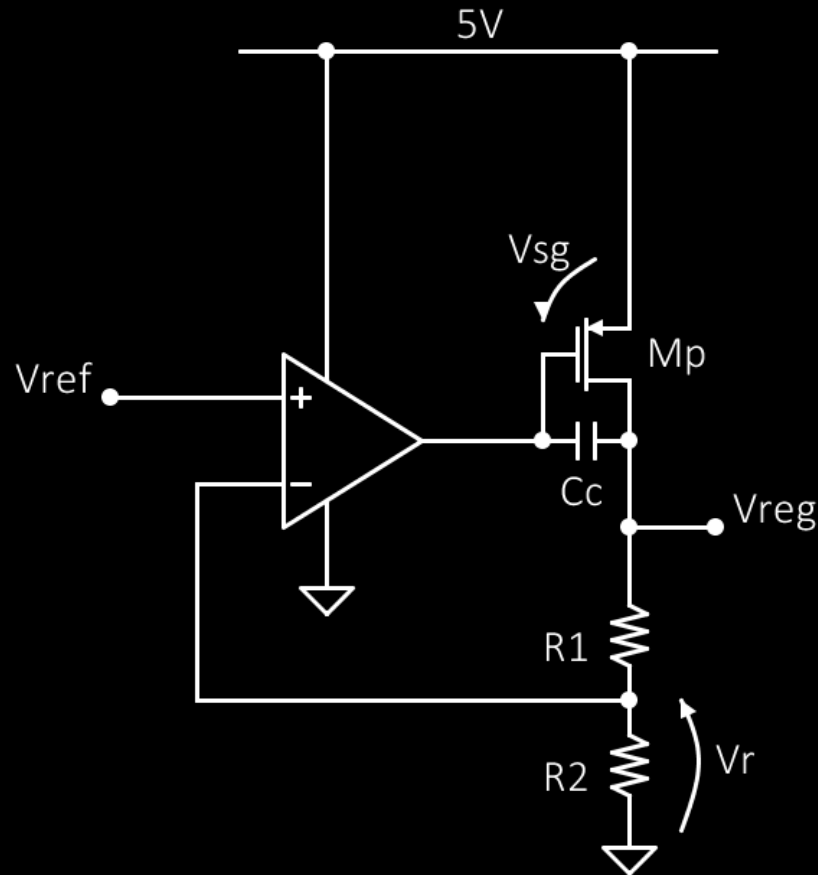
To design R1 and R2 it is requested the specific about quiescent current of the branch. Since the current assigned to the resistors branch is  $8 \mu\text{A}$ , the total resistance value is given by:

$$R = \frac{3.3 \text{ V}}{8 \mu\text{A}} = 412.5 \text{ k}\Omega$$

The feedback voltage depends on the resistors partition ratio and it sets the  $V_{ref}$  value too. In this work it has been chosen a  $\frac{1}{2}$  factor then the two resistors must have the same value:

$$R1 = R2 = \frac{412.5 \text{ k}\Omega}{2} \cong 206 \text{ k}\Omega$$

- Introduction
- Project structure
- Original technology
- LDO design I
- **LDO design II**



The remaining parameter to be designed is the  $M_p$  transistor dimension. Considering:

$$L = 3 * L_{min} = 1.8 \mu m, I_{max} = 2 \text{ mA}, K'_p = -18.9 \text{ mA}$$

it follows that:

$$W/L|_p = \frac{I_{max}}{|K'_p| * (V_{sg_{max}} - |V_{TH_p}|)^2} \quad (1)$$

assuming  $M_p$  in saturation region. From (1) can be observed that the output swing of the differential amplifier must be computed in order to estimate the  $V_{sg_{max}}$  value which must be chosen such that it is within the amplifier output swing while maintaining the saturation condition. The latter means that the op-amp design must be performed before moving forward.

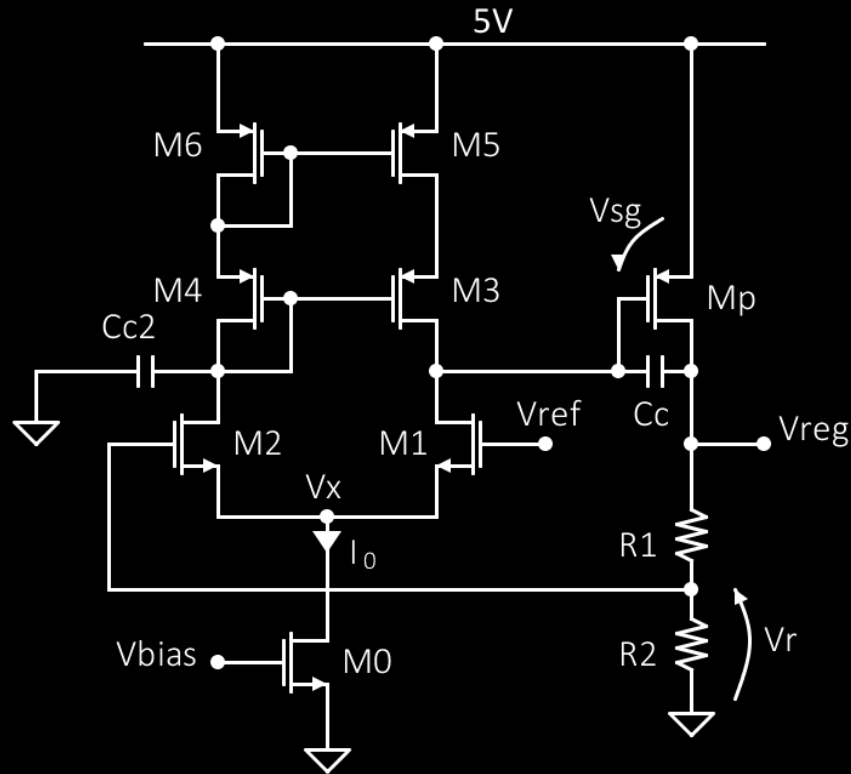
From the previous calculations the  $V_r$  or feedback voltage value has been set to  $\frac{1}{2}$  of the requested output which means:

$$V_r = 1.65 \text{ V}.$$

It will be exploited to set the CMR- specific of the differential amplifier.

- Introduction
- Project structure
- Original technology
- LDO design I
- LDO design II
- LDO design III

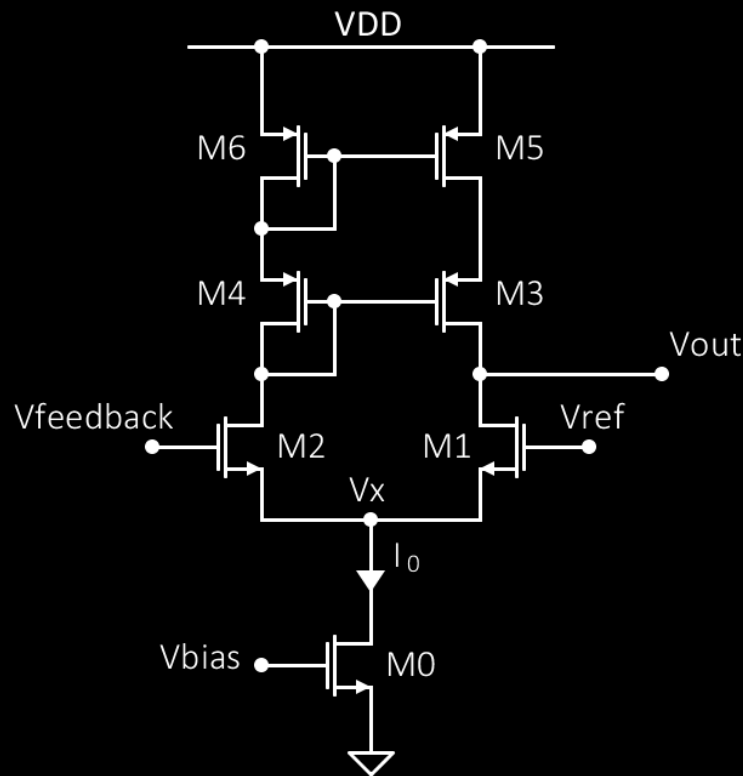
# Op-Amp specs



- $A_v > 10000$  because for a loop connected amplifier it is important to show a very high gain;
- $V_{DD} = 5\text{ V}$  and  $V_{SS} = 0\text{ V}$ ;
- $C_c = C_{c2} = 5\text{ pF}$  from the original design: this value has been kept as it should take into account the expected capacitive load which is not reported within the reference article;
- $V_{in\text{ range}} = [1,3]\text{ V}$  since the  $V_r$  value has been set to  $1.65\text{ V}$ ;
- $GB = 10\text{ MHz}$ , which is an arbitrary value chosen to achieve a reasonable frequency behaviour.
- $I_0 = 4\text{ }\mu\text{A}$  from the current budget requirement: this means that SR will not be a driving parameter in this design having the compensation capacitance and the current already fixed.

$$SR = 4\text{ }\mu\text{A} / 5\text{ pF} = 0.8\text{ V}/\mu\text{s}$$

- Introduction
- Project structure
- Original technology
- LDO design I
- LDO design II
- LDO design III
- **LDO design IV**



Let's start from the cascode current mirror dimensioning:

$$\begin{cases} W/L|_3 = W/L|_4 = W/L|_5 = W/L|_6 \\ W/L|_4 = \frac{I_0}{|K'_p| * (VDD - CMR^+ - |V_{TH4}| - |V_{TH6}| + V_{TH2})^2 / 2} \end{cases}$$

substituting the variables it comes out that:

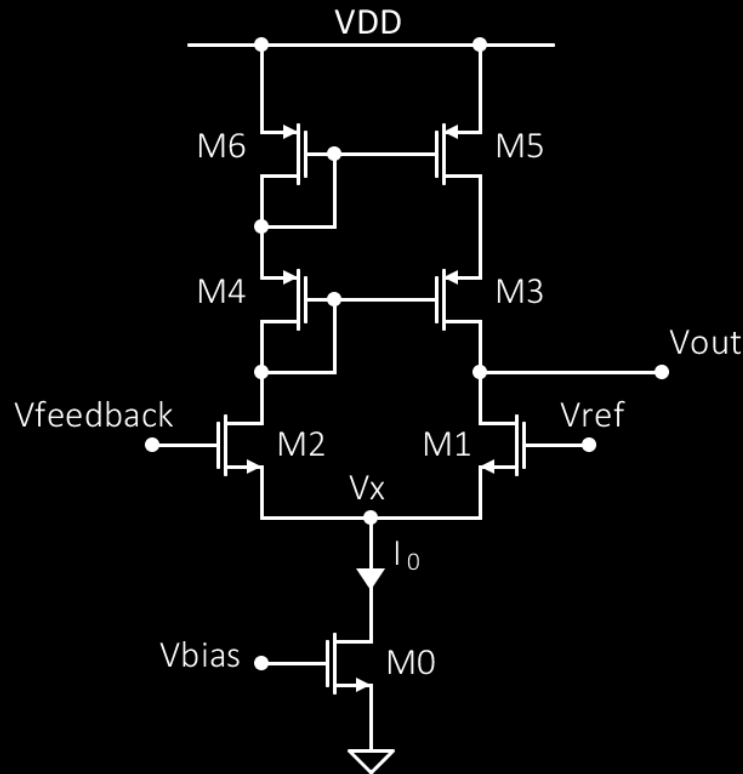
$$W/L|_4 = \frac{2 * 4 \mu}{18.9 \mu * (5 - 3 - 0.9 - 0.9 + 0.76)^2} \cong 1.35$$

Since this is the minimum value to keep the mirror working correctly and it is useful to have a larger margin due to the huge importance of current pairing in a differential stage, the chosen ratio is 2 times the computed one. So that  $W/L|_4 = 2.7$  which leads to :

$$\begin{cases} L_4 = L_3 = L_5 = L_6 = 1.8 \mu m \\ W_4 = W_3 = W_5 = W_6 = 4.8 \mu m \end{cases}$$

The next step is to check the pole associated to  $M_4$  to be larger than GB product.

- Introduction
- Project structure
- Original technology
- LDO design I
- LDO design II
- LDO design III
- LDO design IV
- **LDO design V**



The next step is to check the pole associated to  $M_4$  to be larger than GB product.

$$p_4 = \frac{g_{m_4}}{2 * C_{gs_4}}$$

Considering a worst case for the drain-source voltage which is to present a drop equal to half of VDD, it results  $g_m = \beta * V_{ds} = 2.17 * 10^4 \Omega^{-1}$ . The gate capacitance can be computed knowing that the poly/active layers shows about  $2.4 \text{ fF}/\mu\text{m}^2$  leading to  $C_{gs_4} = 2.4 \frac{\text{fF}}{\mu\text{m}^2} * 7.2 \mu\text{m}^2 = 17.28 \text{ fF}$ .

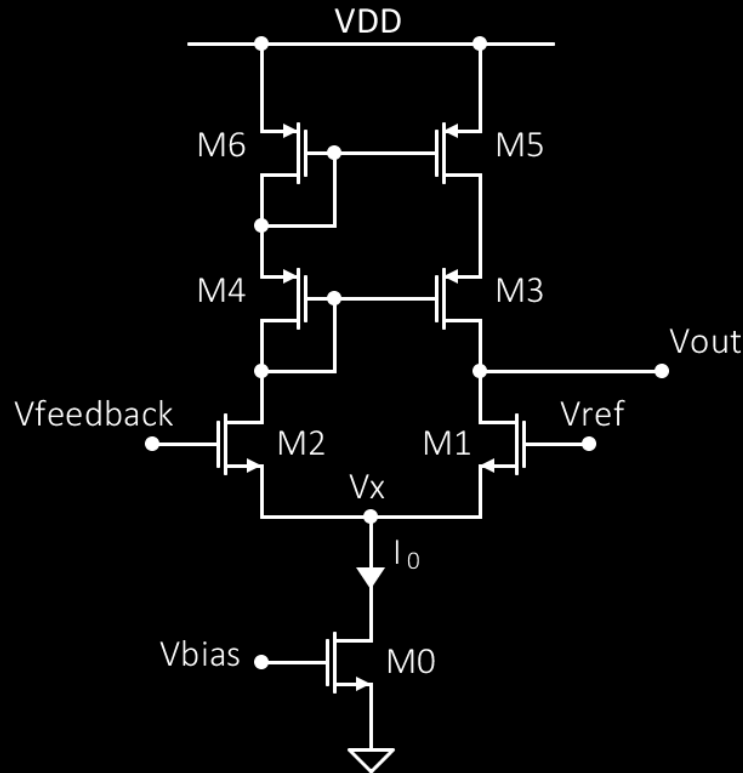
The pole value is thus  $p_4 = 6.29 * 10^9 \text{ Hz} \gg 10 \text{ MHz}$ .

To make things work the differential transistors must have the same  $g_m$  whose value depends on  $GB * C_c$ :

$$g_m = 10 \text{ MHz} * 5 \text{ pF} = 5 * 10^{-5} \Omega^{-1}$$

$$W/L|_1 = W/L|_2 = \frac{g_m^2}{2 * K'_n * I_0} = 5.4 \rightarrow 6 \quad \left\{ \begin{array}{l} L_1 = L_2 = 1.8 \mu\text{m} \\ W_1 = W_2 = 10.8 \mu\text{m} \end{array} \right.$$

- Introduction
- Project structure
- Original technology
- LDO design I
- LDO design II
- LDO design III
- LDO design IV
- LDO design V
- **LDO design VI**



The last transistor to be designed is  $M_0$ . This step makes use of the CMR-specification to compute the minimum drain-source voltage at which the transistor must work in saturation region.

$$W/L|_0 = \frac{I_0}{K'_n * V_{ds0sat}^2}$$

Being  $V_{ds0sat} = CMR^+ - VSS - \sqrt{\frac{I_0/2}{K'_n * W/L|_2}} - V_{TH2} \cong 170 \text{ mV}$ , the dimensions ratio of  $M_0$  must be greater than:

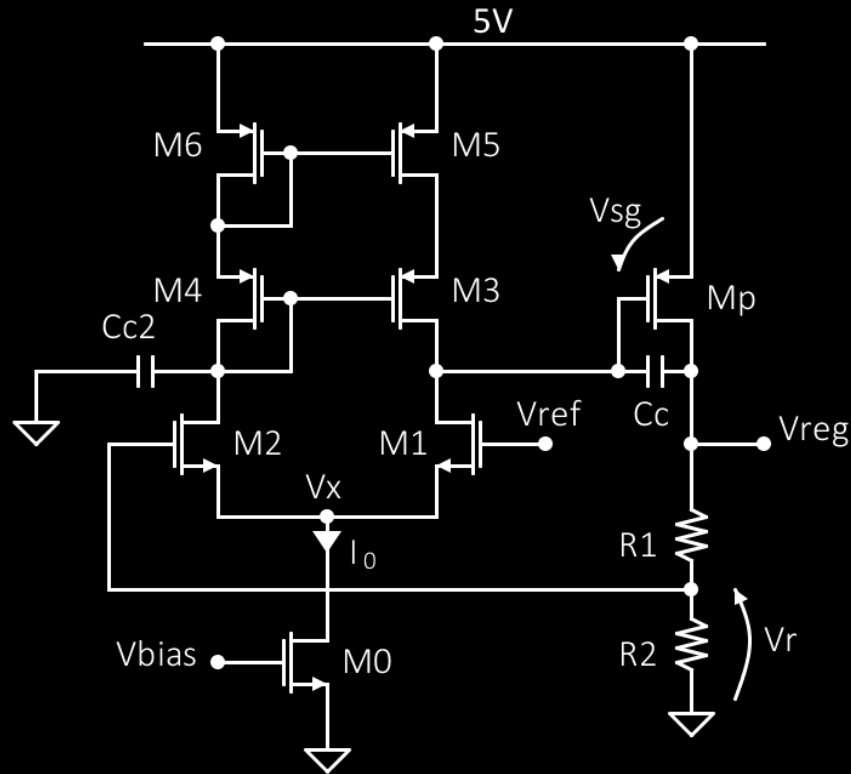
$$W/L|_0 = \frac{4 \mu A}{57.8 \mu A / V^2 * (170 \text{ mV})^2} = 2.39 \rightarrow 3$$

So:

$$\begin{cases} L_0 = 3.6 \mu m \\ W_0 = 10.8 \mu m \end{cases}$$

Having all the transistors dimensions set it is possible to compute the output swing needed for the PMOS pass transistor design.

- Introduction
- Project structure
- Original technology
- LDO design I
- LDO design II
- LDO design III
- LDO design IV
- LDO design V
- LDO design VI
- **LDO design VII**



The lower limit of the output swing sets the maximum gate voltage of  $M_p$  and it depends on  $M_0$  and  $M_1$  drain-source voltage to keep them working in saturation.

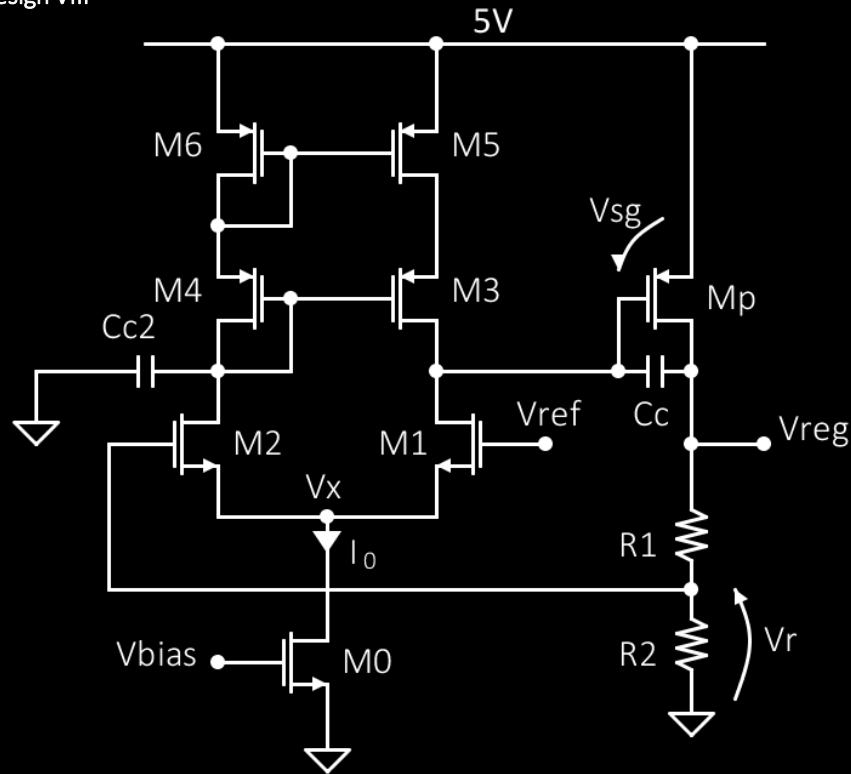
In general:

$$\begin{cases} V_{ds} > (V_{gs} - V_{THn}) \\ V_{gs} = \sqrt{\frac{I_0}{K'_n} * \frac{L}{W}} + V_{THn} \rightarrow V_{ds_{min}} = \sqrt{\frac{I_0}{K'_n} * \frac{L}{W}} = \Delta V \\ K'_n = 57.8 \mu A/V^2 \end{cases}$$

Since the current  $I_0$  is  $4 \mu A$ , it follows that:

$$\begin{cases} V_{out_{min}} = \Delta V_0 + \Delta V_1 \cong 227 mV \\ V_{out_{MAX}} = VDD - 2 * \Delta V_3 = 4.60 V \end{cases}$$

- Introduction
- Project structure
- Original technology
- LDO design I
- LDO design II
- LDO design III
- LDO design IV
- LDO design V
- LDO design VI
- LDO design VII
- **LDO design VIII**



this leads to  $V_{sg_{MAX}} = V_{DD} - V_{out_{min}}$ , which means:

$$V_{sg_{MAX}} = 5\text{ V} - 227\text{ mV} = 4.77\text{ V}$$

But, the saturation condition imposes  $V_{sg} > V_{ds_{MAX}} - |V_{TH_p}|$ :

$$V_{sg_{sat}} = V_{DD} - V_{reg} + |V_{TH_p}| = 2.6\text{ V} < 4.77\text{ V}$$

The above computed value allows the design of  $M_p$  resulting in the following expression:

$$\left. \frac{W}{L} \right|_p = \frac{I_{ds_{MAX}}}{|K'_p| * (V_{sg_{sat}} + |V_{TH_p}|)^2}$$

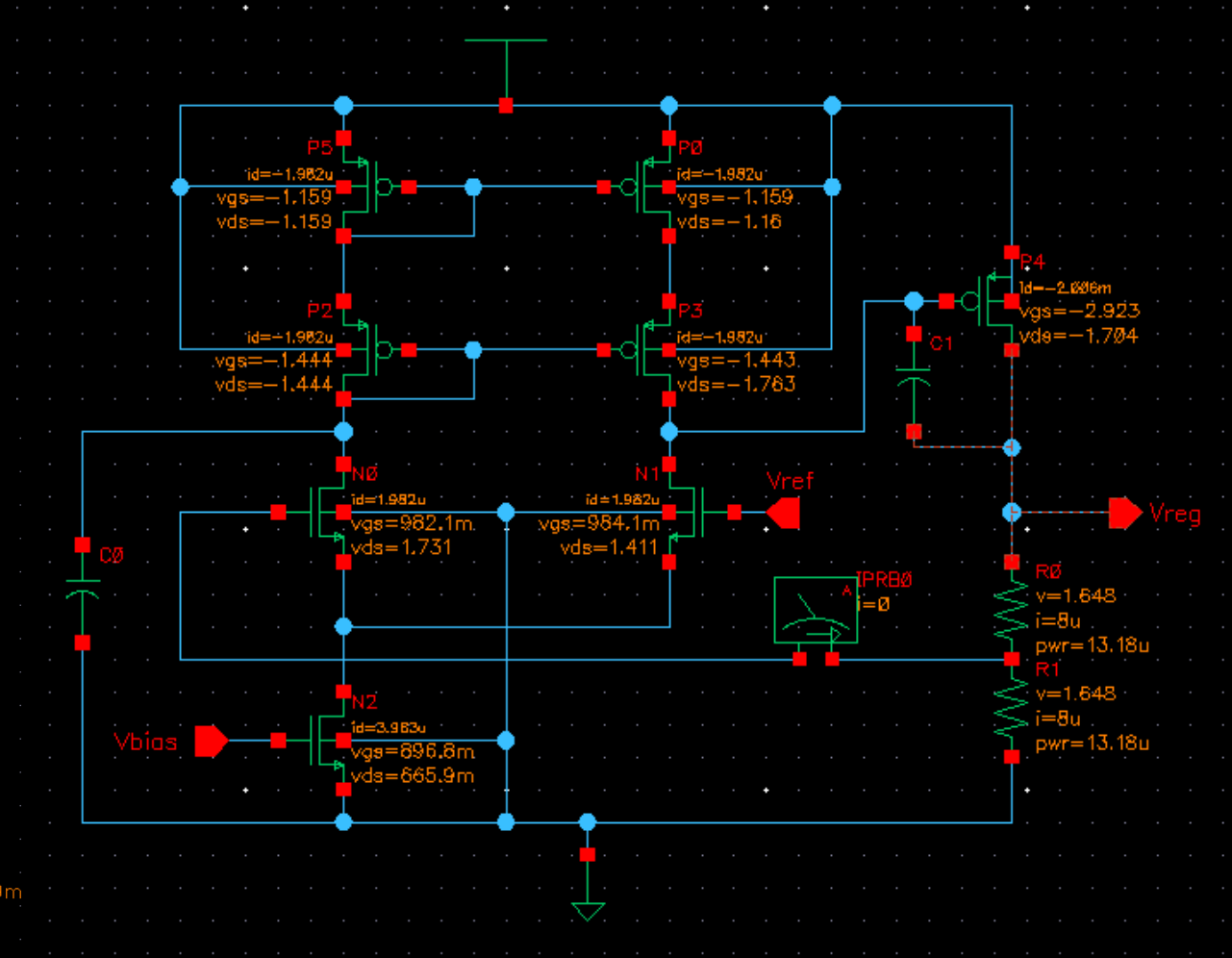
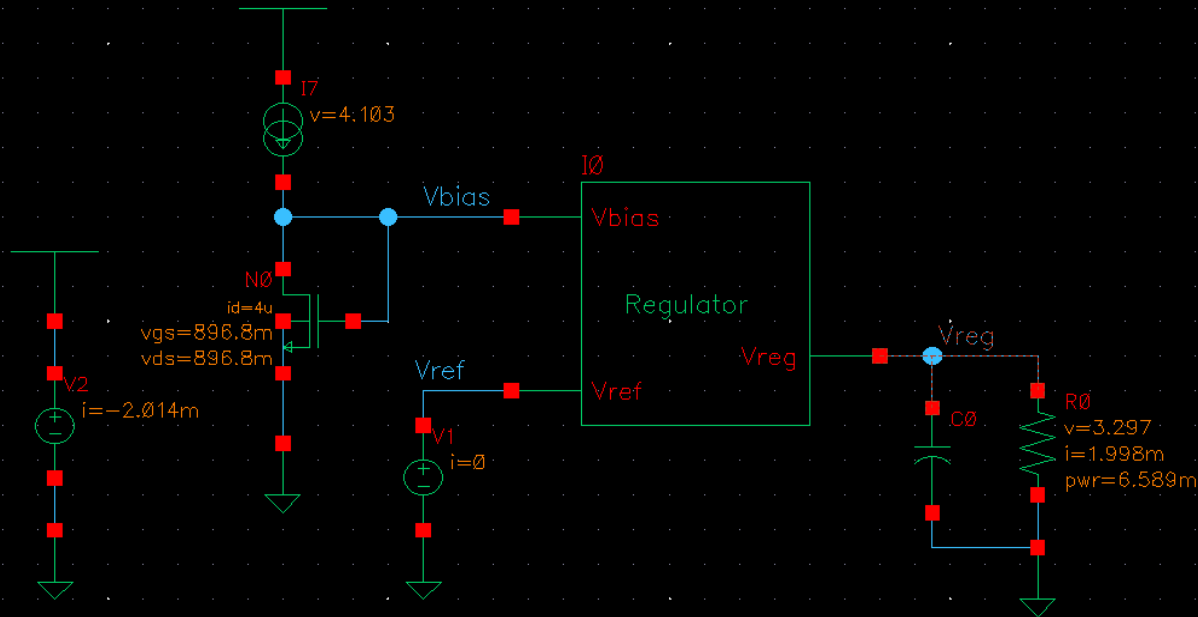
Considering  $I_{ds_{MAX}} = 2\text{ mA}$  the ratio is:

$$\left. \frac{W}{L} \right|_p = 36.6 \rightarrow 40 \rightarrow \begin{cases} L_p = 1.8\text{ }\mu\text{m} \\ W_p = 72\text{ }\mu\text{m} \end{cases}$$

- Introduction
- Project structure
- Original technology
- LDO design I
- LDO design II
- LDO design III
- LDO design IV
- LDO design V
- LDO design VI
- LDO design VII
- LDO design VIII
- LDO DC Sim

# LDO DC Operating Point @ 2mA

Here at side is shown the DC operating point of each component within the designed LDO while in the bottom it is reported the simulation environment exploited for the device testing. All transistors are working in saturation region as expected.



- Introduction
- Project structure
- Original technology
- LDO design I
- LDO design II
- LDO design III
- LDO design IV
- LDO design V
- LDO design VI
- LDO design VII
- LDO design VIII
- LDO DC Sim
- LDO Load reg

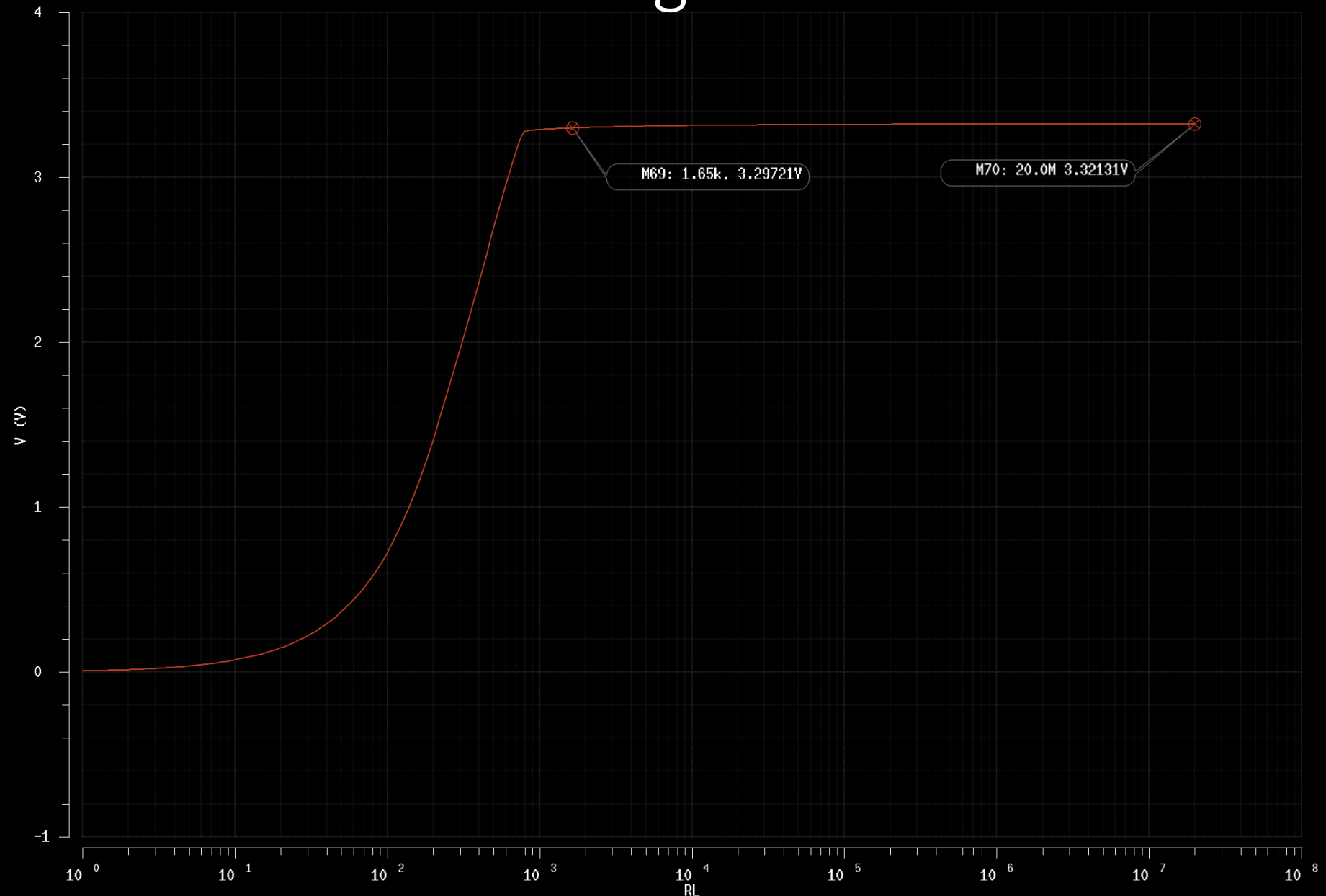
| Name                                     | Vis                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------|
| <span style="color: red;">■</span> /Vreg |  |

# LDO Load regulation

In figure two markers are shown to highlight the regulated voltage when the load is sinking 2 mA and when its current request is negligible with respect to the LDO resistors branch (@160 nA).

The variation can be computed as percentage ratio:

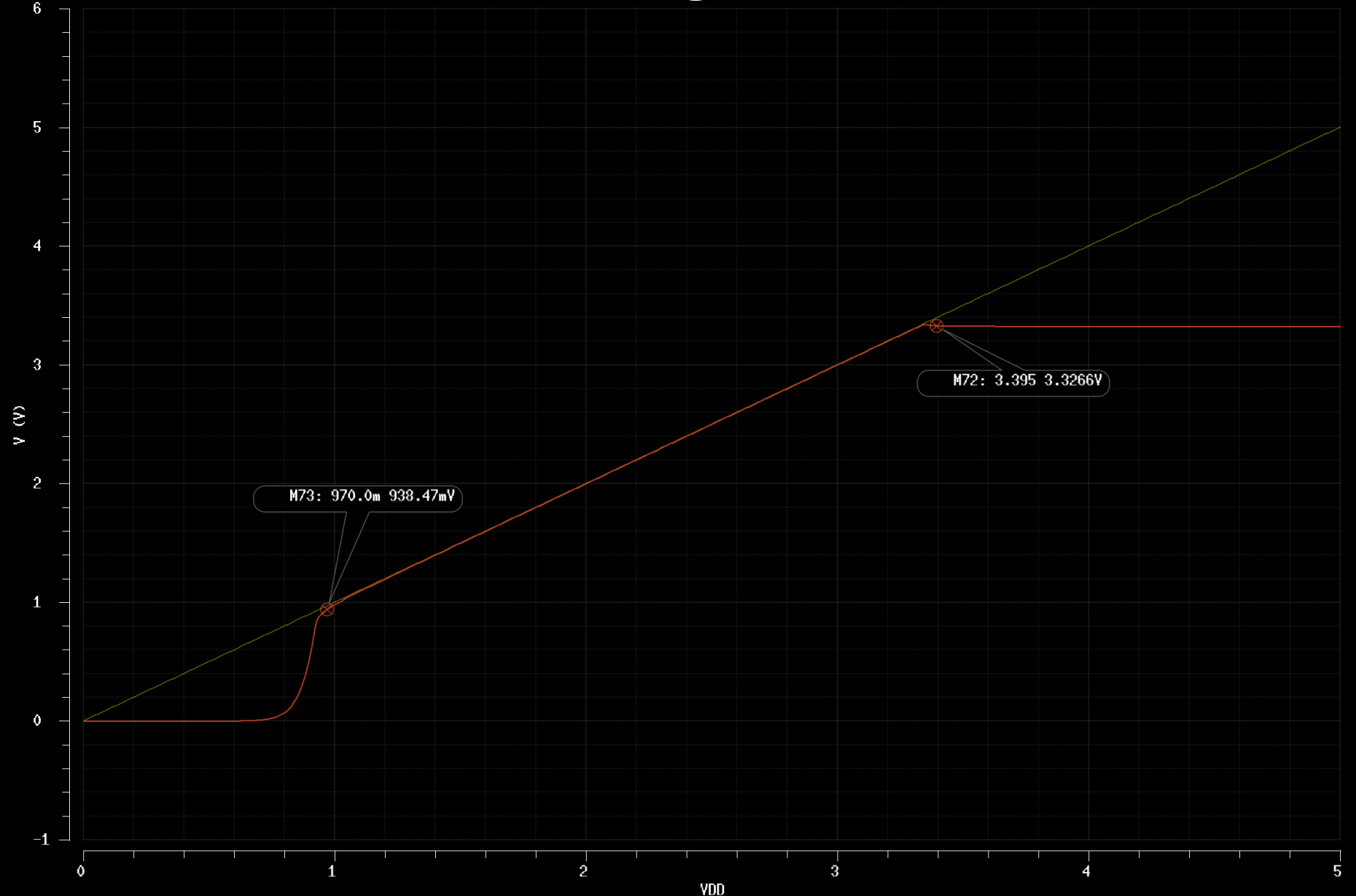
$$\frac{3.321 - 3.297}{3.321} * 100 \cong 0.7 \%$$



- Introduction
- Project structure
- Original technology
- LDO design I
- LDO design II
- LDO design III
- LDO design IV
- LDO design V
- LDO design VI
- LDO design VII
- LDO design VIII
- LDO DC Sim
- LDO Load reg
- **LDO Line reg**

| Name                                      | Vis                                 |
|-------------------------------------------|-------------------------------------|
| <span style="color: blue;">■</span> /vddI | <input checked="" type="checkbox"/> |
| <span style="color: red;">■</span> /Vreg  | <input checked="" type="checkbox"/> |

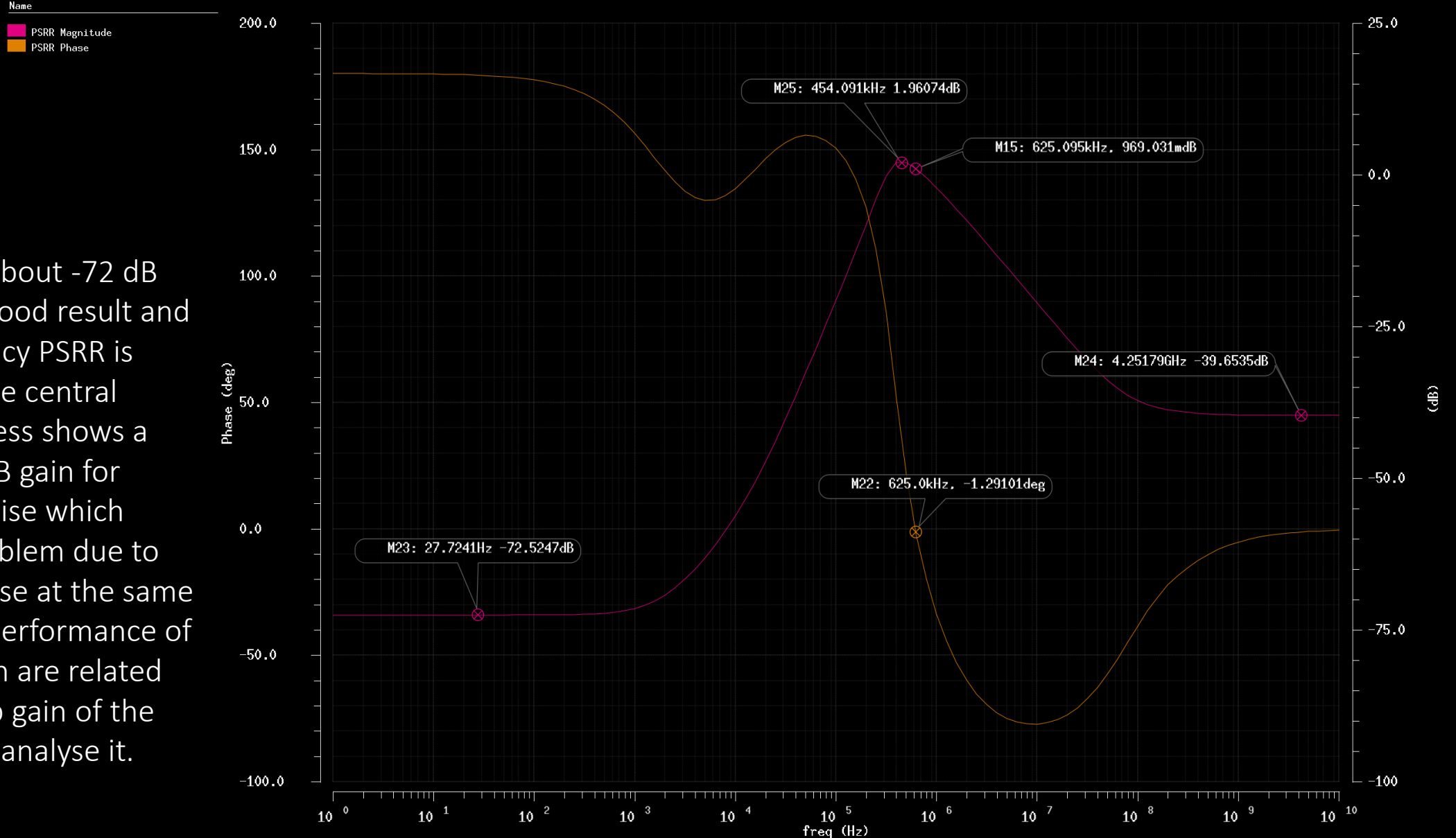
# LDO Line regulation



The LDO turns on when VDD approaches 1 V (PMOS pass transistor and works like a resistor until the power supply reaches 3.4 V. From this point on the regulation function starts to be effective.

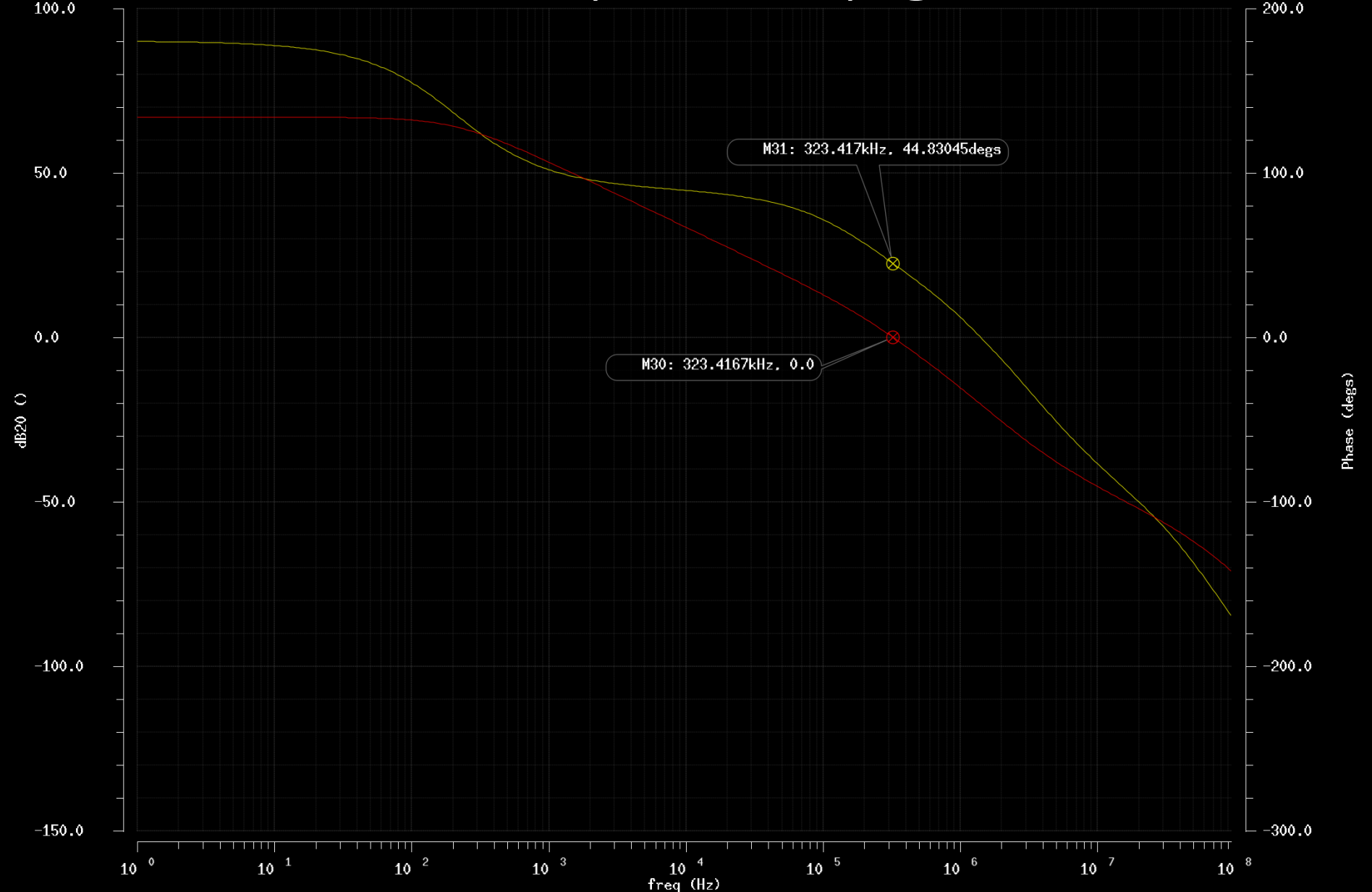
- Project structure
- Original technology
- LDO design I
- LDO design II
- LDO design III
- LDO design IV
- LDO design V
- LDO design VI
- LDO design VII
- LDO design VIII
- LDO DC Sim
- LDO Load reg
- LDO Line reg
- **LDO PSRR**

The DC PSRR is about -72 dB which is a very good result and the high frequency PSRR is about -40 dB. The central region nonetheless shows a greater than 1 dB gain for power supply noise which represents a problem due to the inverted phase at the same frequency. The performance of the centre region are related to the open-loop gain of the op-amp. So let's analyse it.

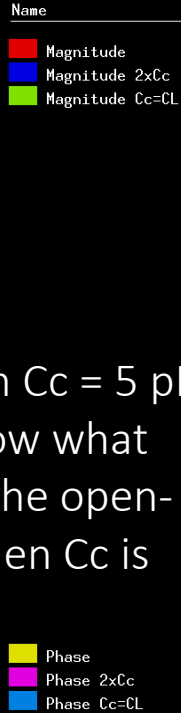


- Original technology
- LDO design I
- LDO design II
- LDO design III
- LDO design IV
- LDO design V
- LDO design VI
- LDO design VII
- LDO design VIII
- LDO DC Sim
- LDO Load reg
- LDO Line reg
- LDO PSRR
- **LDO open-loop**

| Name      | Vis                                 |
|-----------|-------------------------------------|
| Magnitude | <input checked="" type="checkbox"/> |
| Phase     | <input checked="" type="checkbox"/> |



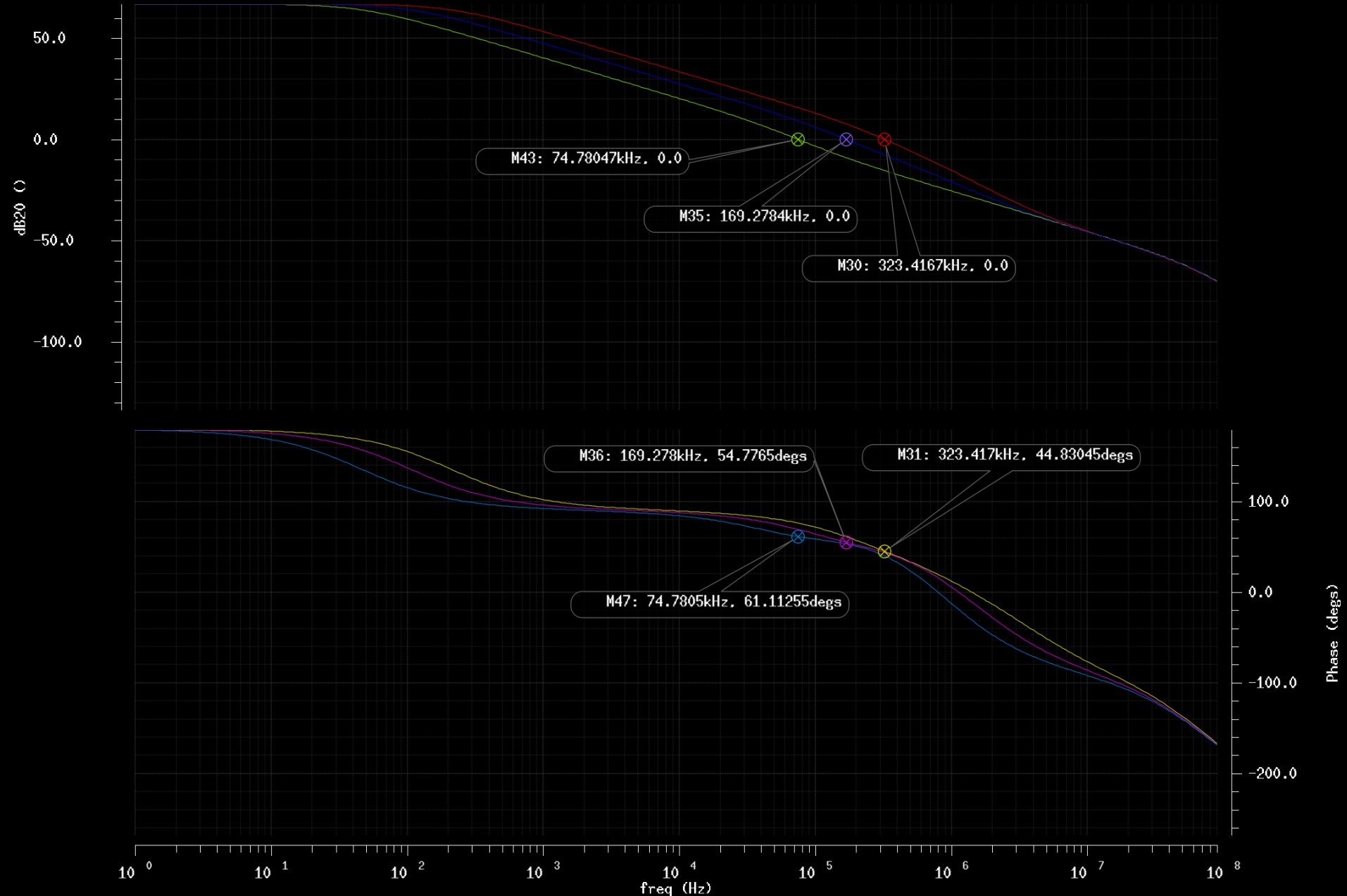
- LDO design I
- LDO design II
- LDO design III
- LDO design IV
- LDO design V
- LDO design VI
- LDO design VII
- LDO design VIII
- LDO DC Sim
- LDO Load reg
- LDO Line reg
- LDO PSRR
- LDO open-loop
- LDO open-loop adj



Starting from  $C_c = 5 \text{ pF}$  the plots show what happens to the open-loop gain when  $C_c$  is increased.

Keeping this value it is possible to look at the impact on PSRR. When  $C_c = CL = 23 \text{ pF}$  the phase margin is greater than  $60^\circ$ .

## LDO Open-Loop gain analysis vs $C_c$



- LDO design II
- LDO design III
- LDO design IV
- LDO design V
- LDO design VI
- LDO design VII
- LDO design VIII
- LDO DC Sim
- LDO Load reg
- LDO Line reg
- LDO PSRR
- LDO open-loop
- LDO open-loop adj
- **LDO PSRR adj**

Name

PSRR Magnitude Cc=CL

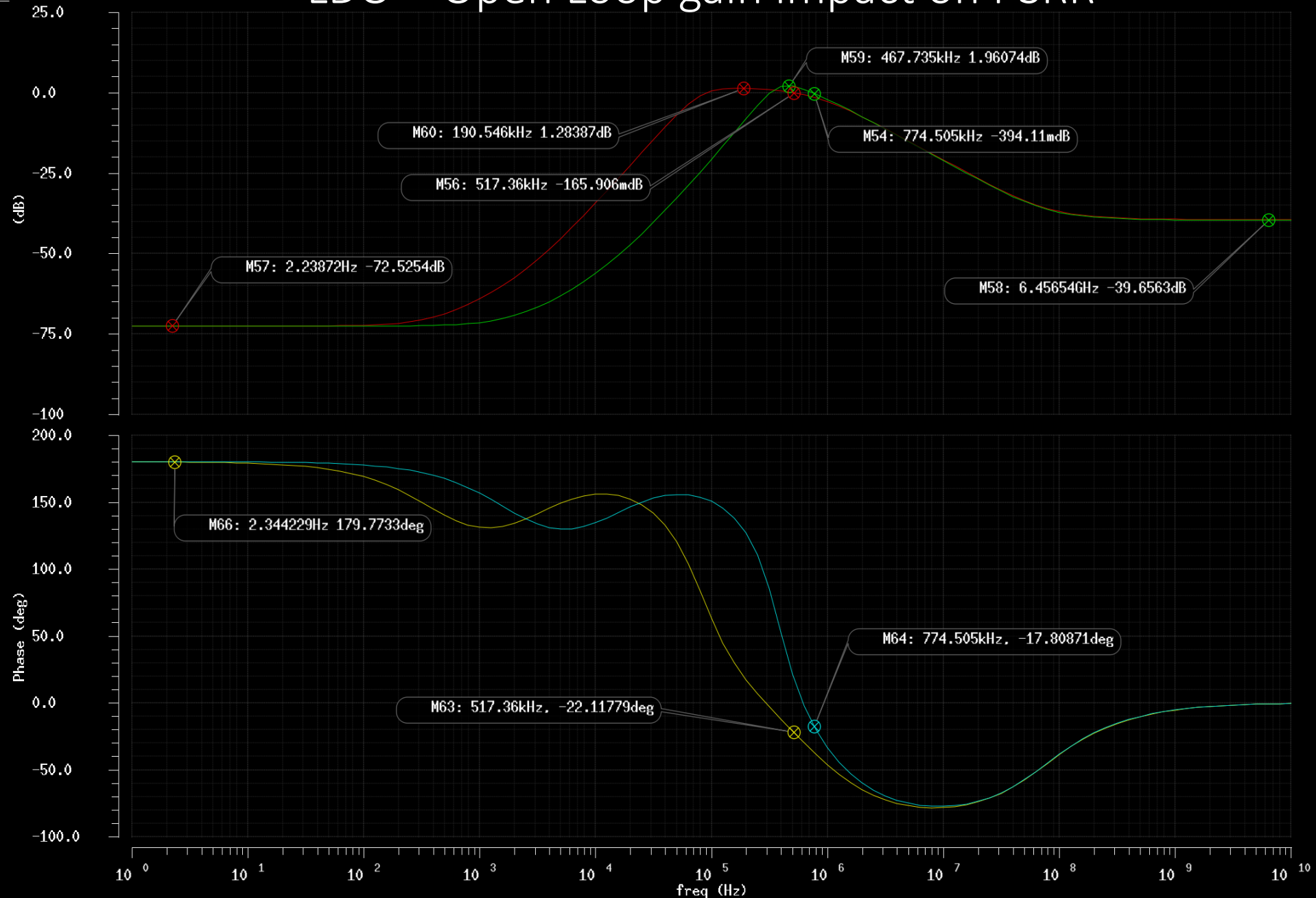
PSRR Magnitude

PSRR Phase Cc=CL

PSRR Phase

In this figure can be observed a small reduction of the noise gain peak but the frequency at which it occurs is reduced too. Considering the results, the compensation capacitor has been kept at the original 5 pF value

## LDO – Open Loop gain impact on PSRR



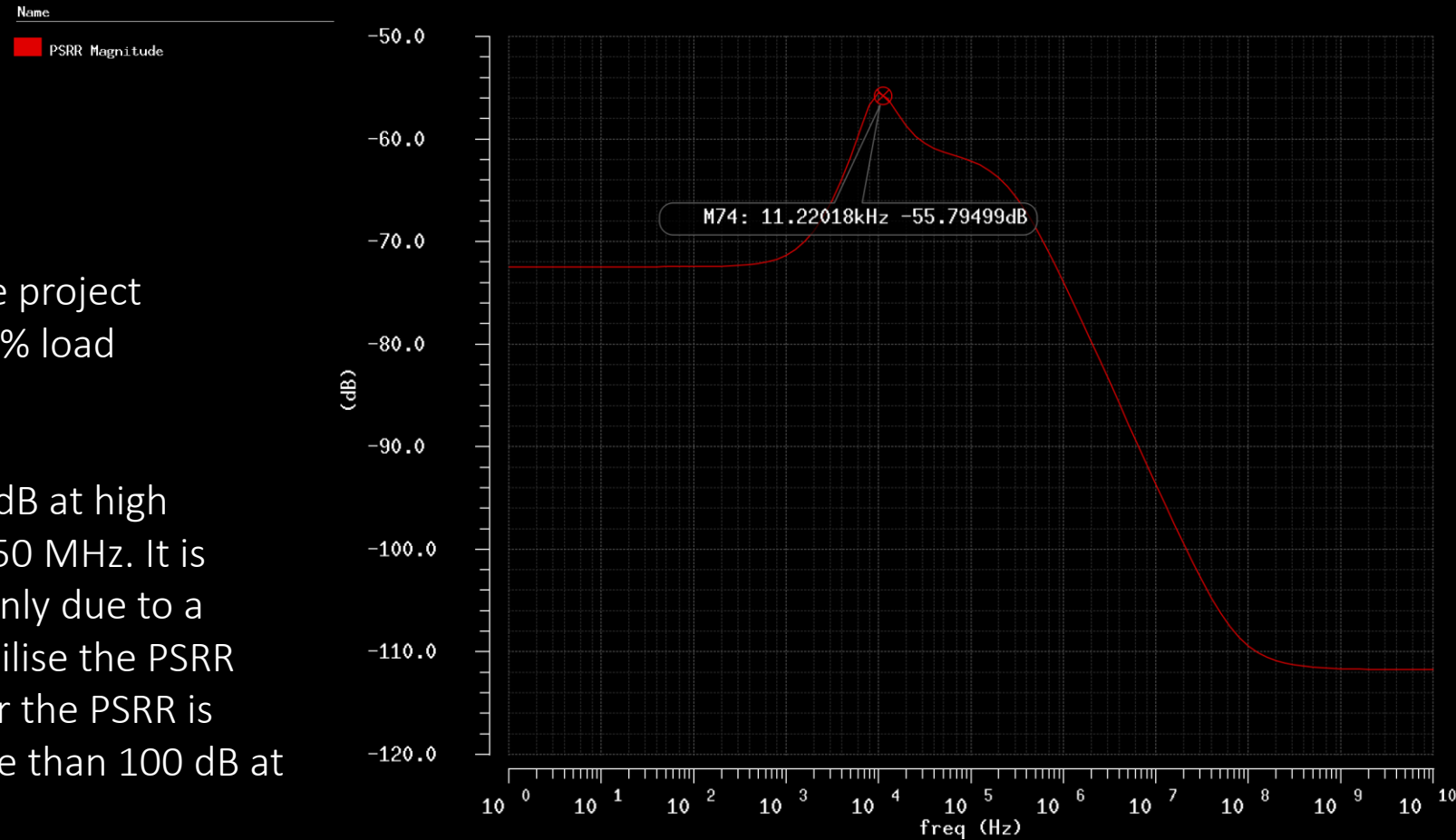
- LDO design III
- LDO design IV
- LDO design V
- LDO design VI
- LDO design VII
- LDO design VIII
- LDO DC Sim
- LDO Load reg
- LDO Line reg
- LDO PSRR
- LDO open-loop
- LDO open-loop adj
- LDO PSRR adj
- **LDO outcomes**

# LDO outcomes

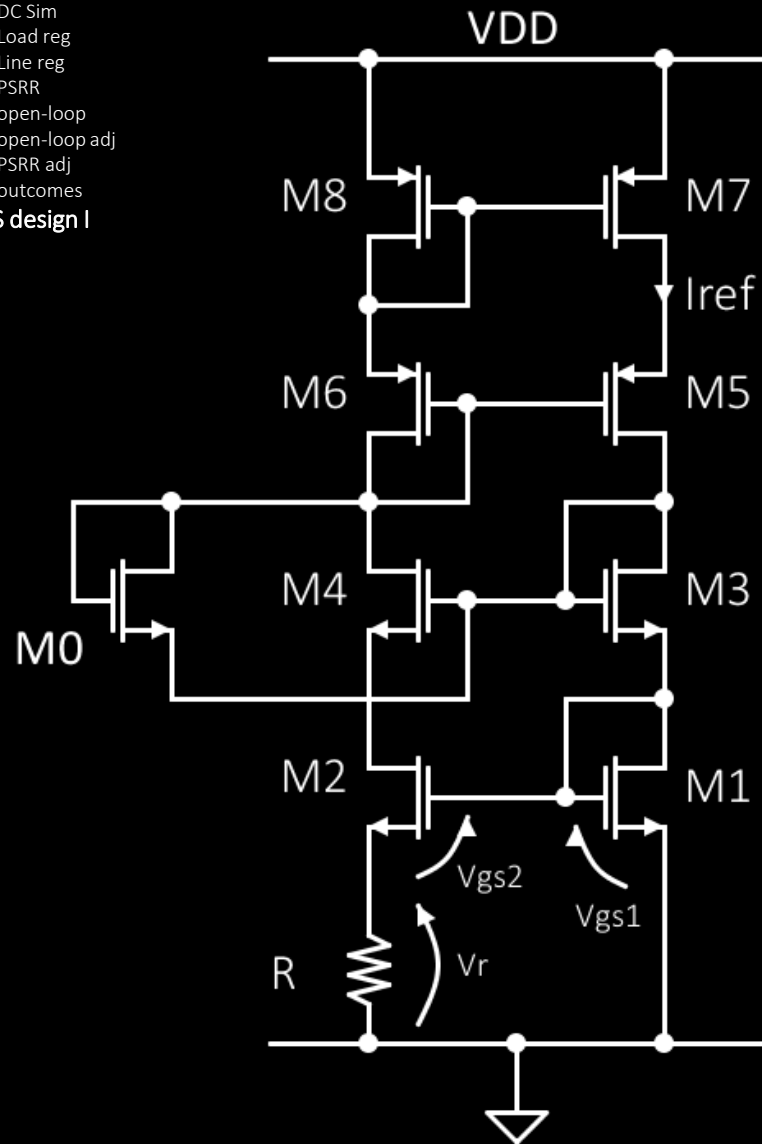
The final result is an LDO which respects all the project specifications delivering up to 2 mA with a 0.7 % load regulation.

The PSRR is about -70 dB at DC and about -40 dB at high frequency but shows a positive peak around 450 MHz. It is important to remind that these results are mainly due to a missing external capacitance which would stabilise the PSRR behaviour. For example with a 100 nF capacitor the PSRR is almost stable around -70 dB and grows to more than 100 dB at high frequency.

The final performances with a schematic level BIAS circuit and a layout level voltage reference will be evaluated later on.



- LDO design IV
- LDO design V
- LDO design VI
- LDO design VII
- LDO design VIII
- LDO DC Sim
- LDO Load reg
- LDO Line reg
- LDO PSRR
- LDO open-loop
- LDO open-loop adj
- LDO PSRR adj
- LDO outcomes
- **BIAS design I**



# BIAS circuit design

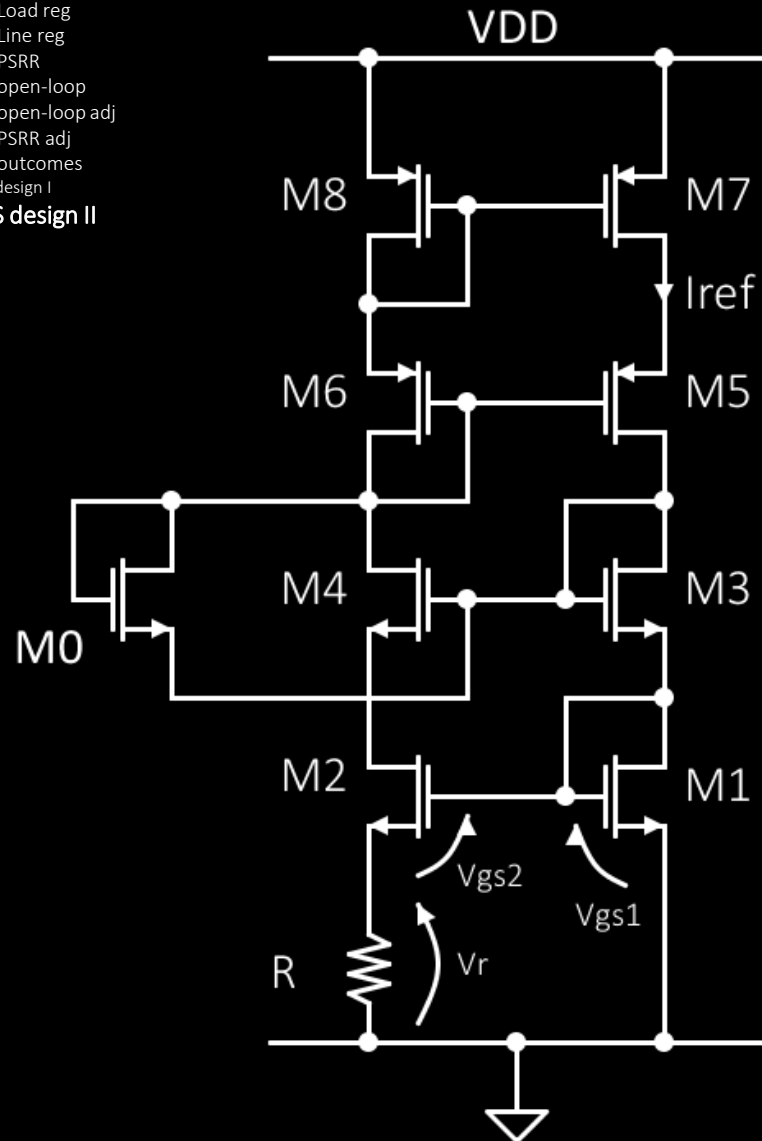
The BIAS circuit design starts from the following equations set:

$$\begin{cases} V_{gs1} = V_{gs2} + R * I_{ref} & (1) \\ V_{gs1} > V_{gs2} \rightarrow \beta_2 = k * \beta_1 \\ V_{gs} = \sqrt{\frac{2 * I_{ref}}{\beta}} + V_{THn} \end{cases}$$

Since  $I_{ref} = 4 \mu A$ ,  $V_{THn} = 0.76 V$ ,  $\beta = 115.6 \mu A/V^2 * W/L$ , some other parameter must be fixed to complete the following design equation derived from (1):

$$R = \sqrt{\frac{2}{\beta_1 * I_{ref}}} * \left(1 - \frac{1}{\sqrt{k}}\right)$$

- LDO design V
- LDO design VI
- LDO design VII
- LDO design VIII
- LDO DC Sim
- LDO Load reg
- LDO Line reg
- LDO PSRR
- LDO open-loop
- LDO open-loop adj
- LDO PSRR adj
- LDO outcomes
- BIAS design I
- **BIAS design II**



To have a better match of the BIAS current between this circuit and the LDO differential amplifier, the dimension ratio of  $M_1$  can be the same of  $M_0$  within the LDO which is:

$$\left. \frac{W}{L} \right|_1 = \left. \frac{W}{L} \right|_3 = 3 \rightarrow \begin{cases} L_1 = L_3 = 3.6 \mu m \\ W_1 = W_3 = 10.8 \mu m \end{cases}$$

Setting the parameter  $k = 1.1$  the resistor value becomes:

$$R = \sqrt{\frac{2}{346.8 \mu * 4 \mu}} * \left(1 - \frac{1}{\sqrt{1.1}}\right) = 1.767 k\Omega \text{ and } \begin{cases} L_2 = L_4 = 3.6 \mu m \\ W_2 = W_4 = 11.85 \mu m \end{cases}$$

The cascode current mirror transistors can be designed to keep working in saturation when the voltage drop above them is 2 V (arbitrary choice which can be replaced from a  $2 * \Delta V_n + 7mV$  minimum drop on  $M_2$  and  $M_4$  plus an arbitrary safety voltage or the min VDD allowed) , considering half of the voltage on each transistor:

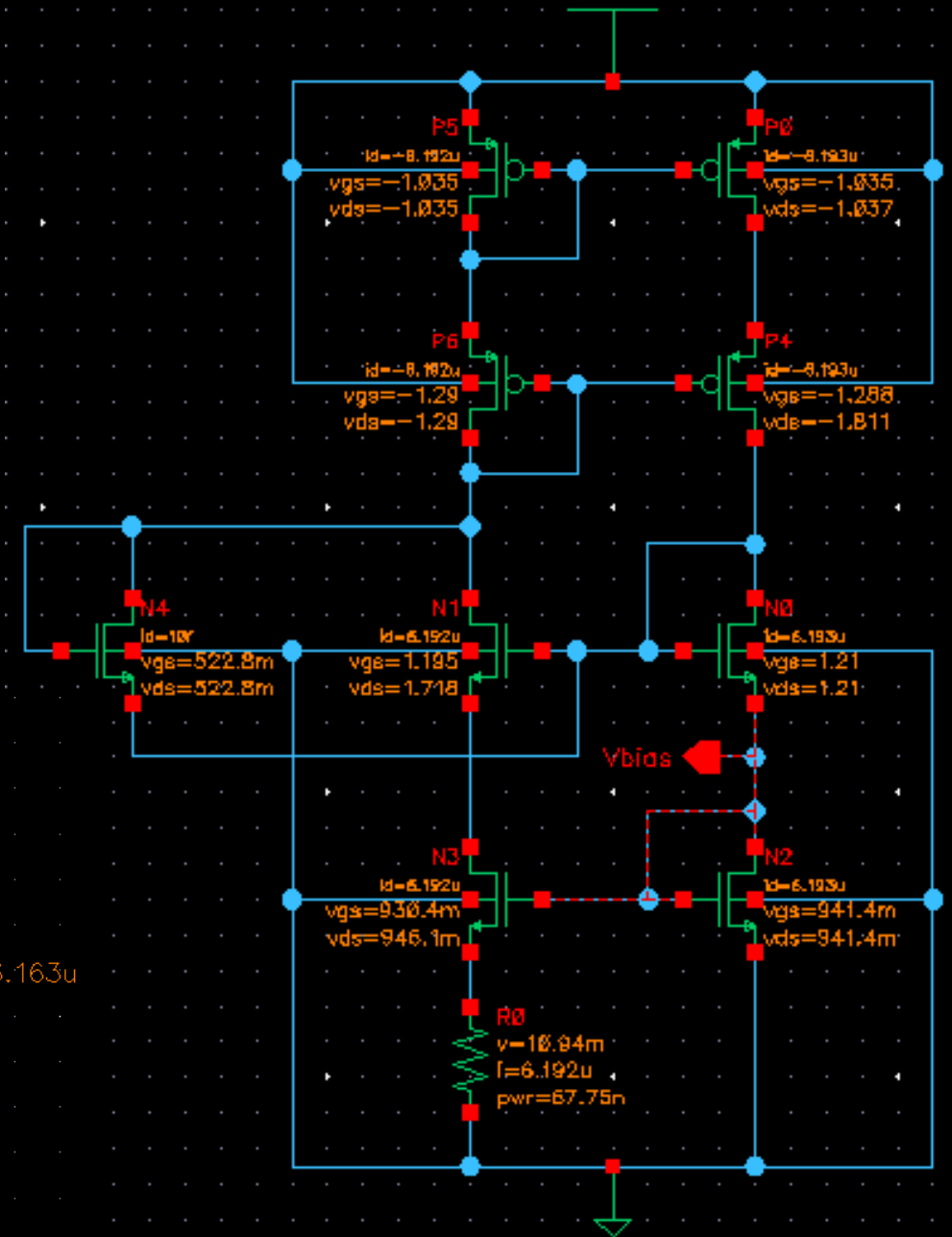
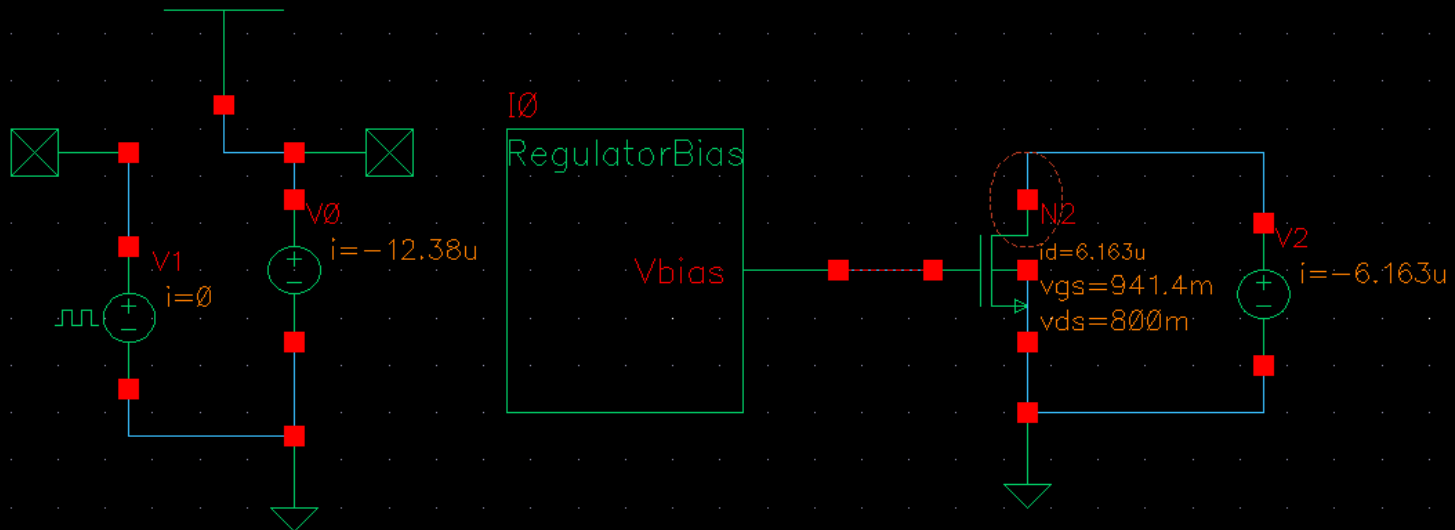
$$\left. \frac{W}{L} \right|_8 = \left. \frac{W}{L} \right|_7 = \left. \frac{W}{L} \right|_6 = \left. \frac{W}{L} \right|_5 = \frac{I_{ref}}{|K_p'| * (1 - 0.9)^2} \cong 21$$



- LDO design VII
- LDO design VIII
- LDO DC Sim
- LDO Load reg
- LDO Line reg
- LDO PSRR
- LDO open-loop
- LDO open-loop adj
- LDO PSRR adj
- LDO outcomes
- BIAS design I
- BIAS design II
- BIAS design III
- **BIAS DC sim**

# BIAS DC Simulation

The branch current has a value of about  $6 \mu\text{A}$  which is higher than the target value. To adjust this result it is possible to act on the resistor performing a parameter sweep analysis varying its resistance.



- LDO design VIII
- LDO DC Sim
- LDO Load reg
- LDO Line reg
- LDO PSRR
- LDO open-loop
- LDO open-loop adj
- LDO PSRR adj
- LDO outcomes
- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- **BIAS DC adj I**

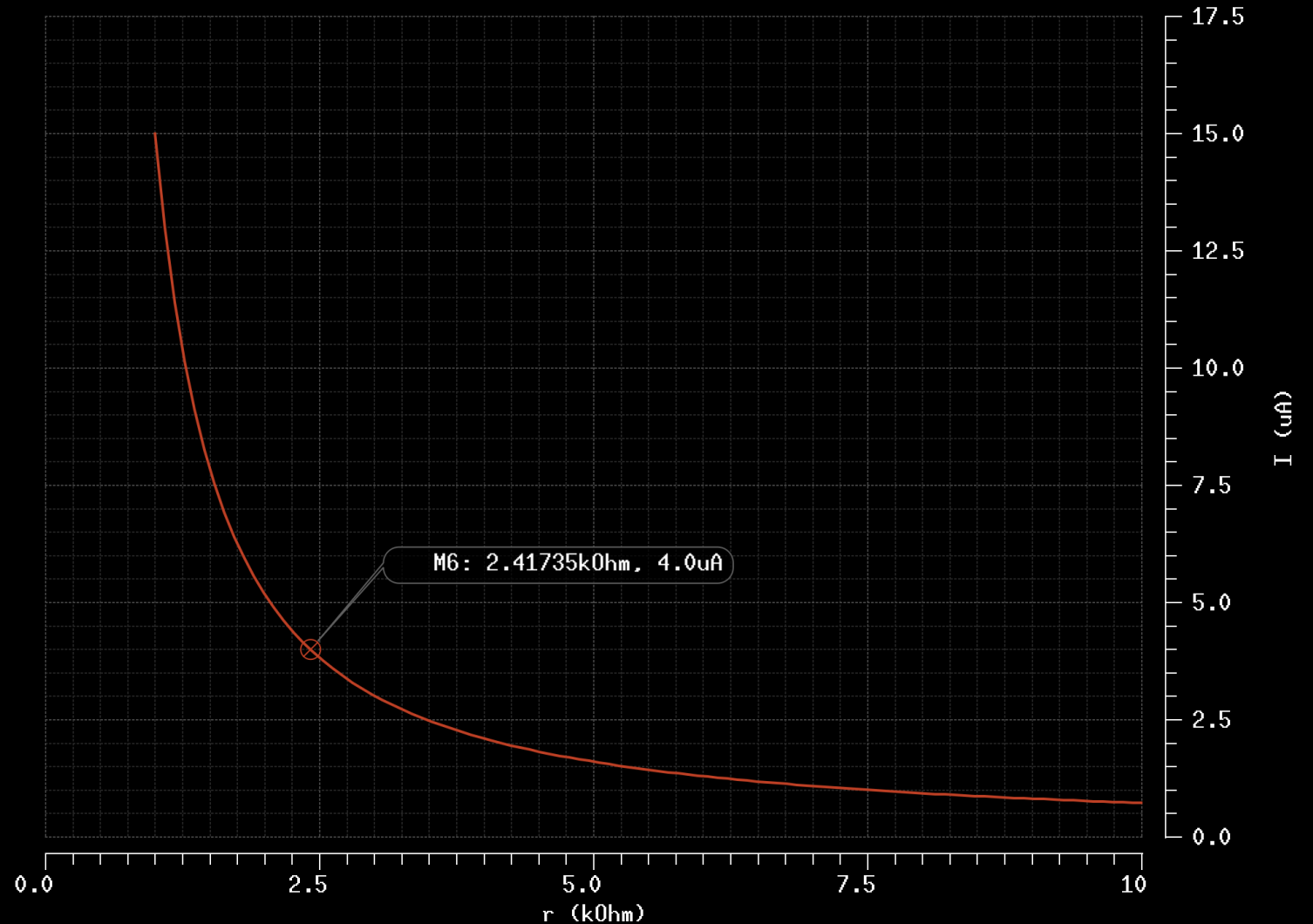
# BIAS DC adjust

Name

Branch Current

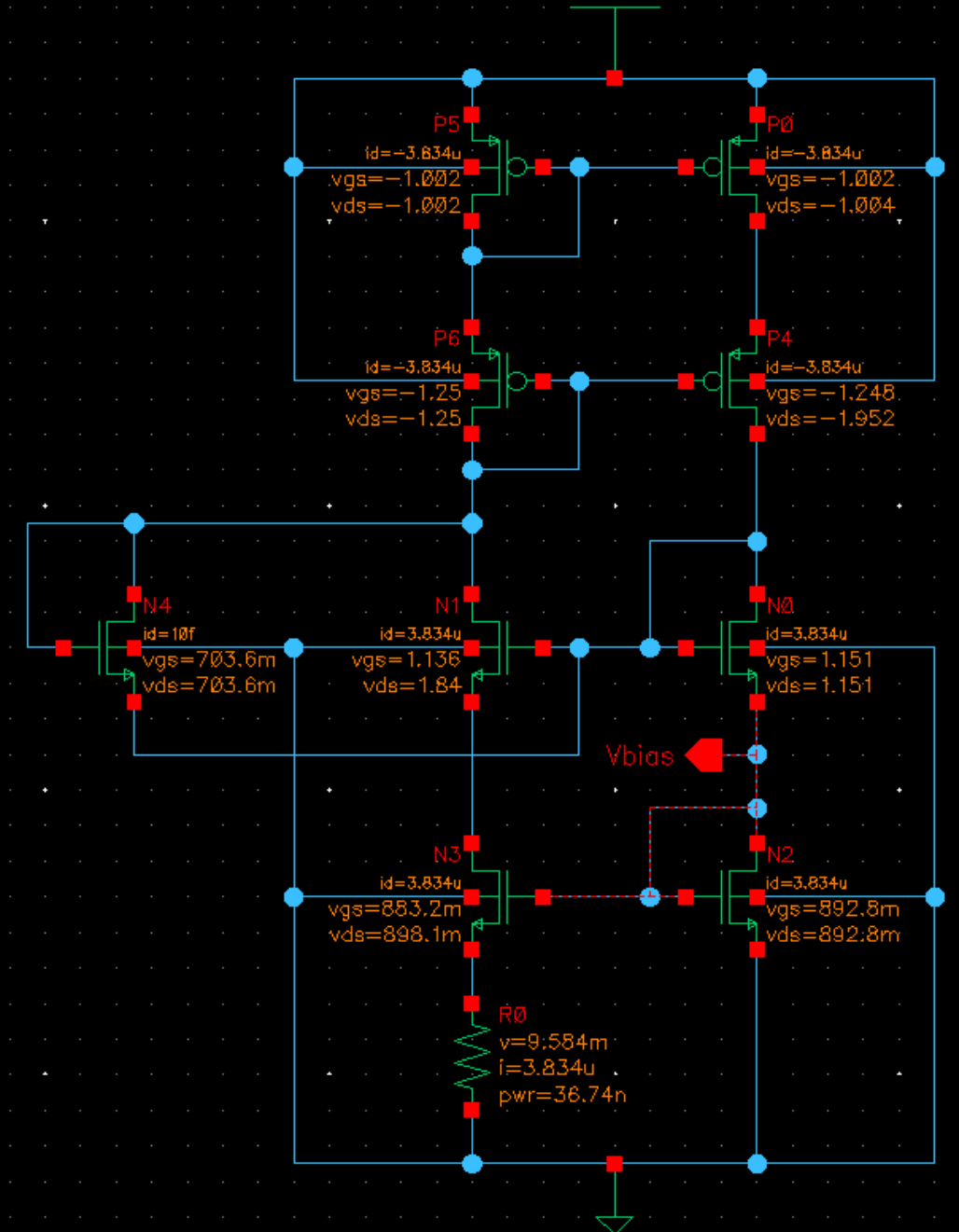
The marker highlights the resistance value to be implemented in order to have a branch current of about  $4 \mu A$ . Then a new DC simulation is performed to verify the correct behaviour of the circuit.

$$\rightarrow R = 2.5 \text{ k}\Omega$$



- LDO DC Sim
- LDO Load reg
- LDO Line reg
- LDO PSRR
- LDO open-loop
- LDO open-loop adj
- LDO PSRR adj
- LDO outcomes
- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- **BIAS DC adj II**

The final circuit drives  $3.83 \mu A$  in each branch. This result is acceptable and the simulation verifies that all the transistors are working in saturation region and above threshold. The following slides will report a set of simulations to characterize the designed circuit.



- LDO Load reg
- LDO Line reg
- LDO PSRR
- LDO open-loop
- LDO open-loop adj
- LDO PSRR adj
- LDO outcomes
- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- **BIAS temp sweep**

Name

Branch Current

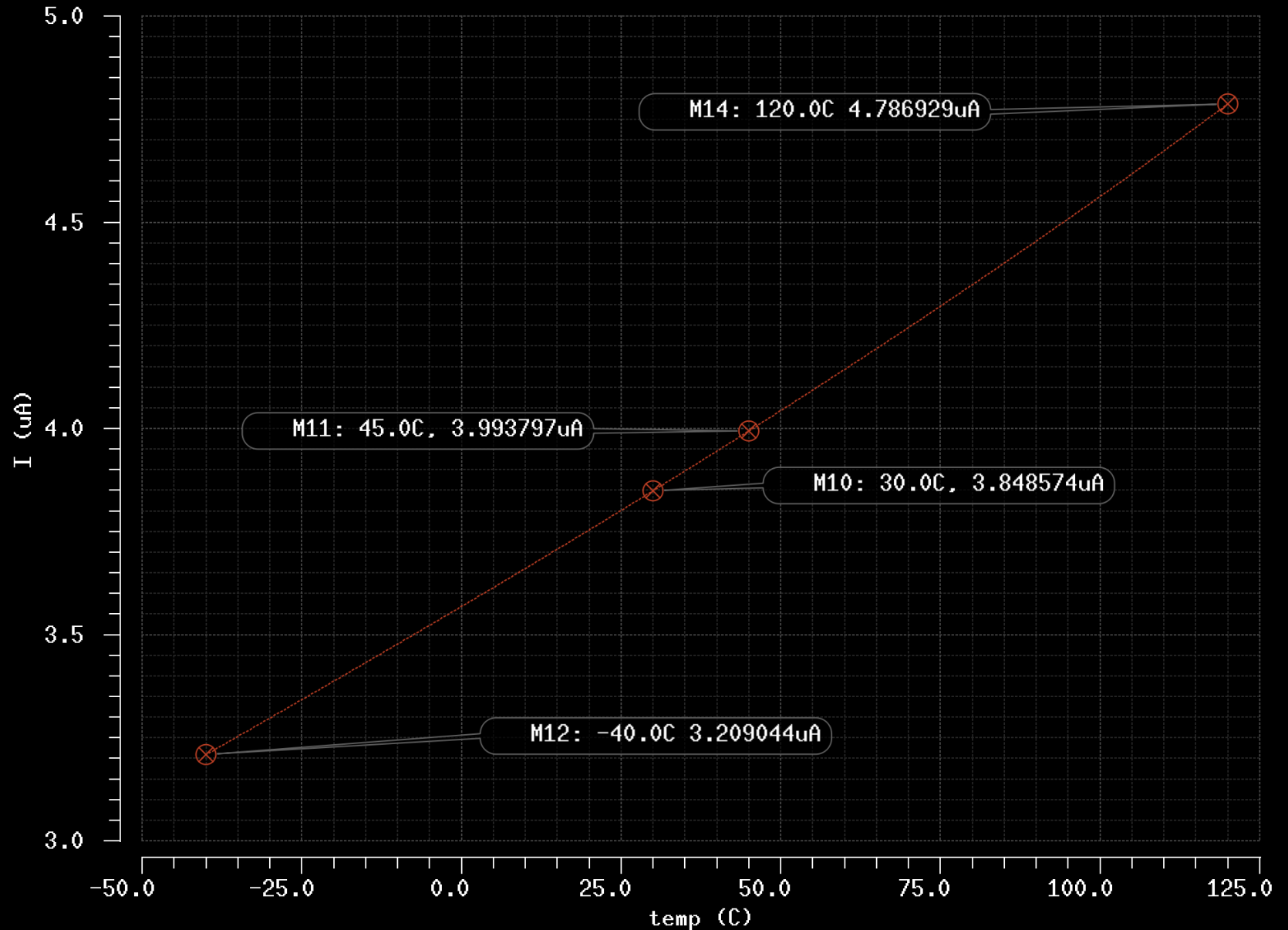
The considered range is the one delimited by M10 and M11 which goes from 30°C to 45°C.

The reason for this short range is the target application of the device which is intended to work with human implanted circuitry, meaning that the body temperature is the reference for this project.

The variation of the BIAS current can be computed as:

$$\frac{3.99 - 3.84}{3.99} * 100 = 3.8 \%$$

## BIAS temperature sweep



- LDO Line reg
- LDO PSRR
- LDO open-loop
- LDO open-loop adj
- LDO PSRR adj
- LDO outcomes
- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- **BIAS Line reg**

Name

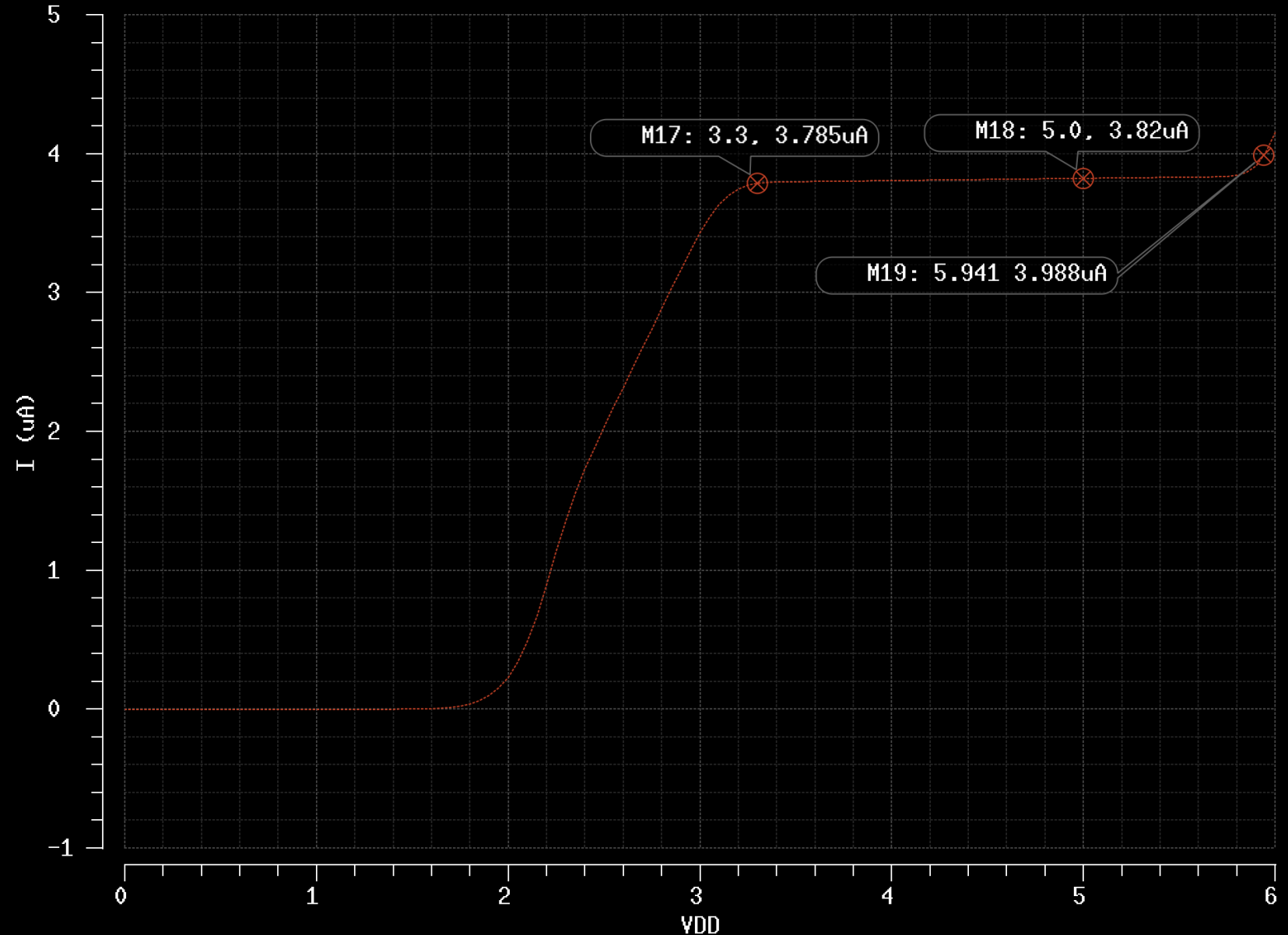
Branch Current

There is no particular specification about the supply voltage range but the plot shows a stable behaviour between 3.3V and about 5.9V.

To get some quantitative value, the variation of bias current between 3.3V and 5V can be computed:

$$\frac{3.82 - 3.785}{3.82} * 100 \cong 0.9 \%$$

# BIAS Line Regulation



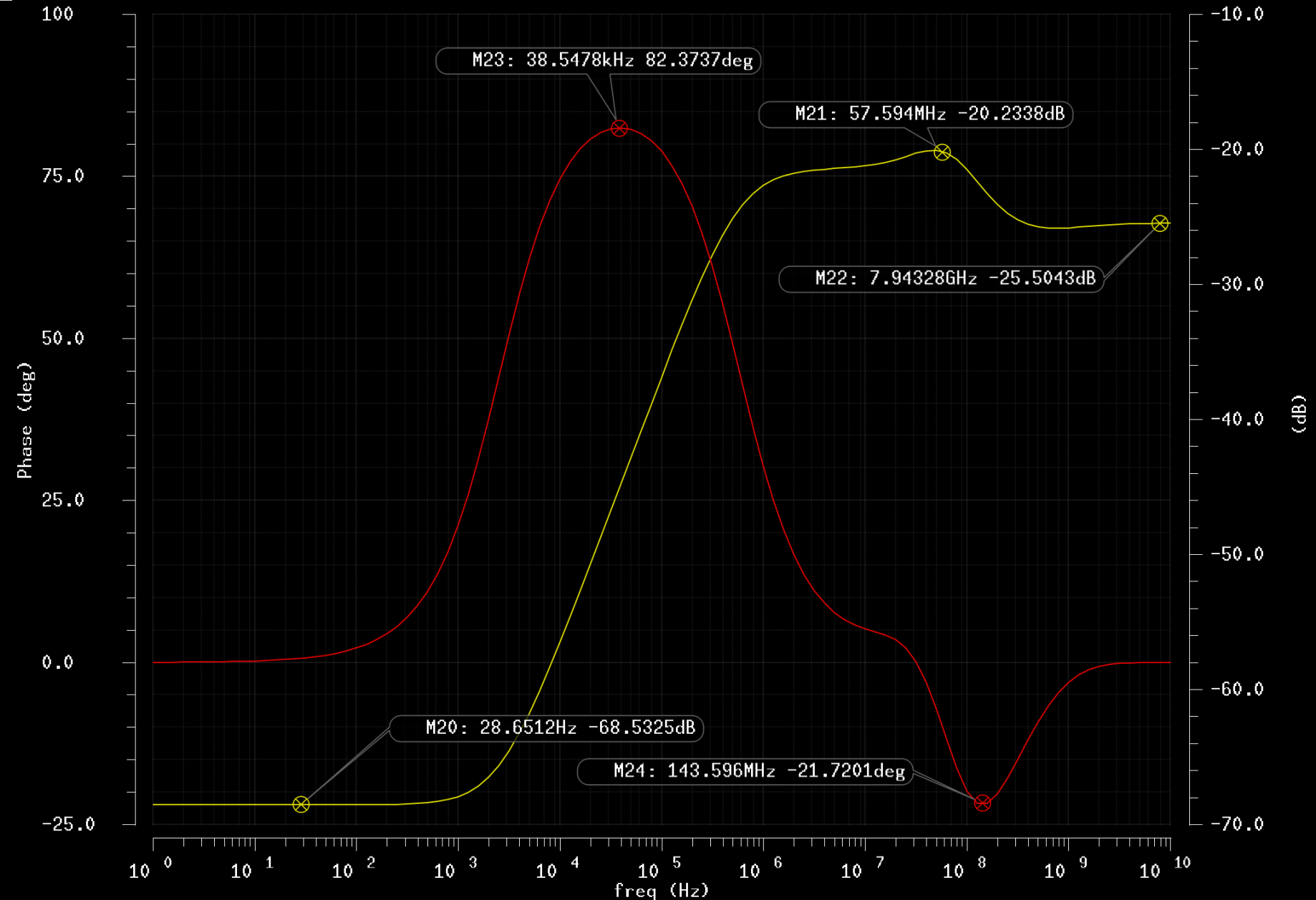
- LDO PSRR
- LDO open-loop
- LDO open-loop adj
- LDO PSRR adj
- LDO outcomes
- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- **BIAS PSRR**

| Name           |  |
|----------------|--|
| PSRR Phase     |  |
| PSRR Magnitude |  |

The BIAS circuit presents a good PSRR behaviour:

- -68 dB @ DC
- -20 dB peak @ 58MHz
- -25 dB @ HF

## BIAS PSRR



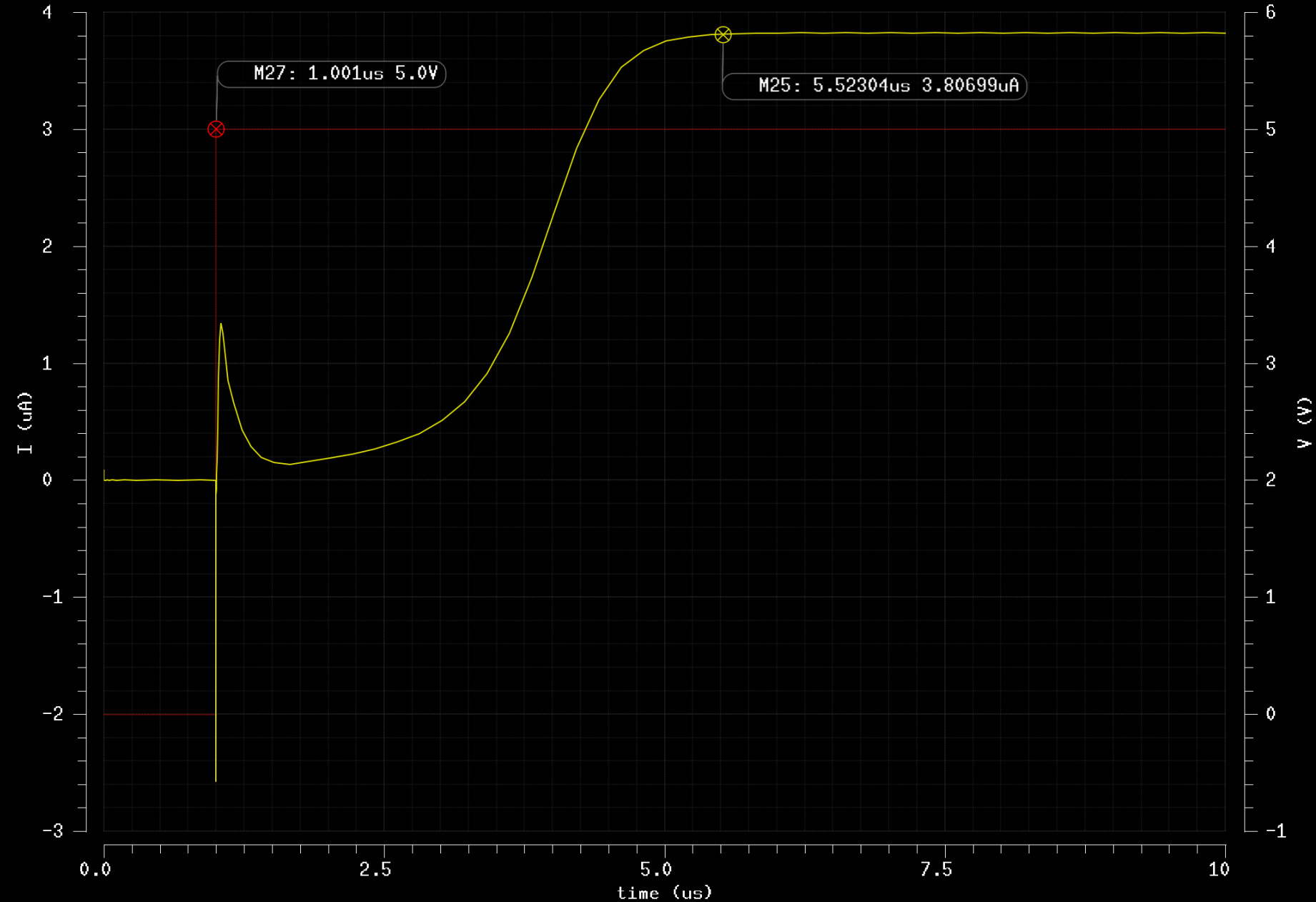
- LDO open-loop
- LDO open-loop adj
- LDO PSRR adj
- LDO outcomes
- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- **BIAS transient I**

Name

■ Iref

■ /vddI

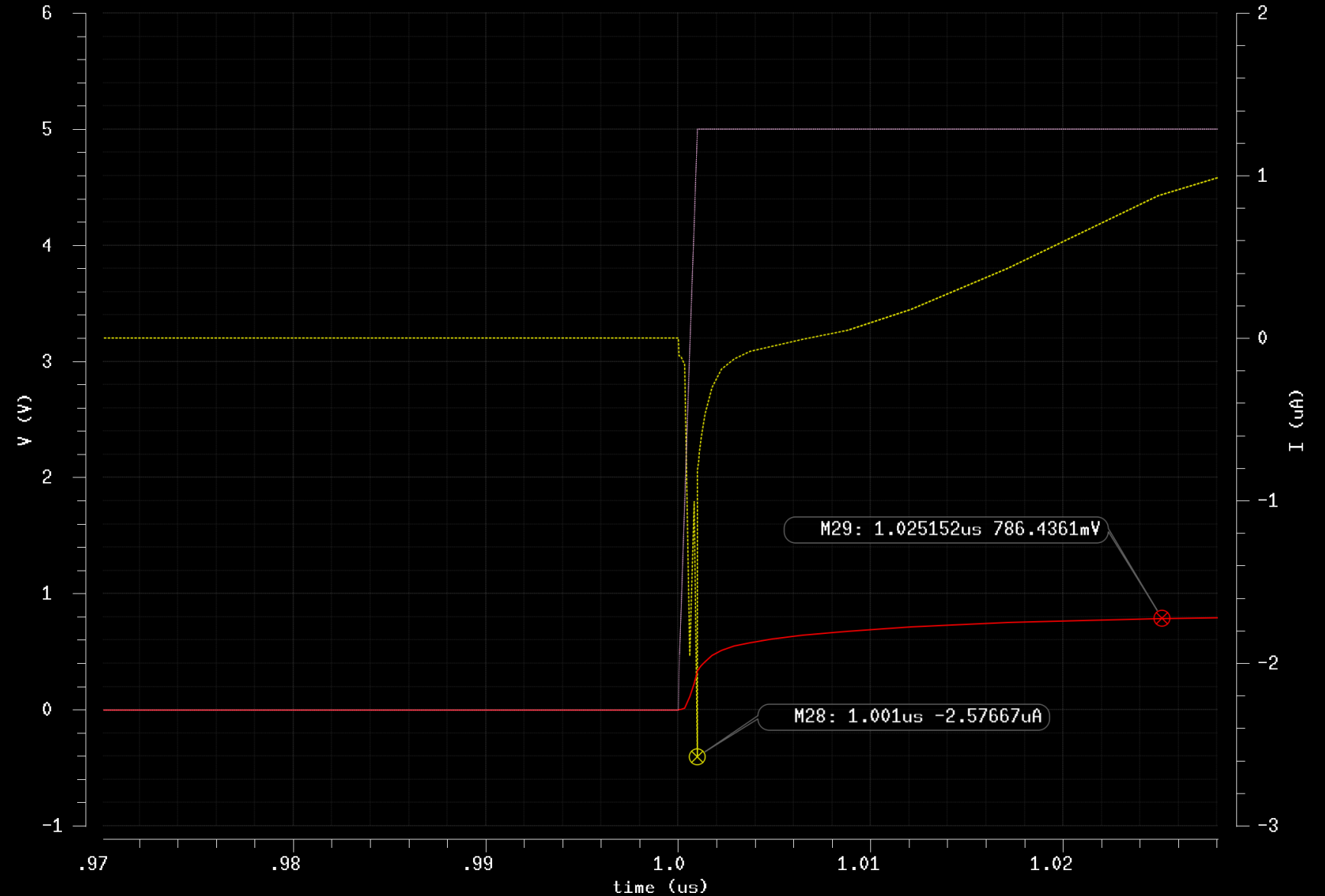
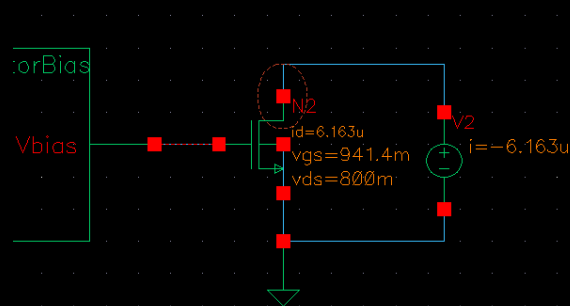
The start-up time is about  $4.5 \mu\text{s}$  and the transient presents a first negative peak followed by an opposite one when VDD steps up. It can be observed in detail in the next slide...



- LDO open-loop adj
- LDO PSRR adj
- LDO outcomes
- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- **BIAS transient II**

| Name  | Vis                                                                               |
|-------|-----------------------------------------------------------------------------------|
| Iref  |  |
| Vbias |  |
| /vddI |  |

In this plot the overlapping of the VDD step and the inverse current peak is highlighted. It could be caused by the fixed voltage imposed at the drain-source terminals of the copied transistor.



- LDO PSRR adj
- LDO outcomes
- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- **Reference design I**

# Voltage reference design

The last block to be designed is the voltage reference to be connected to the non-inverting pin of the LDO differential amplifier. Here will be listed the set of design equation and the assumption needed to get the final circuit.

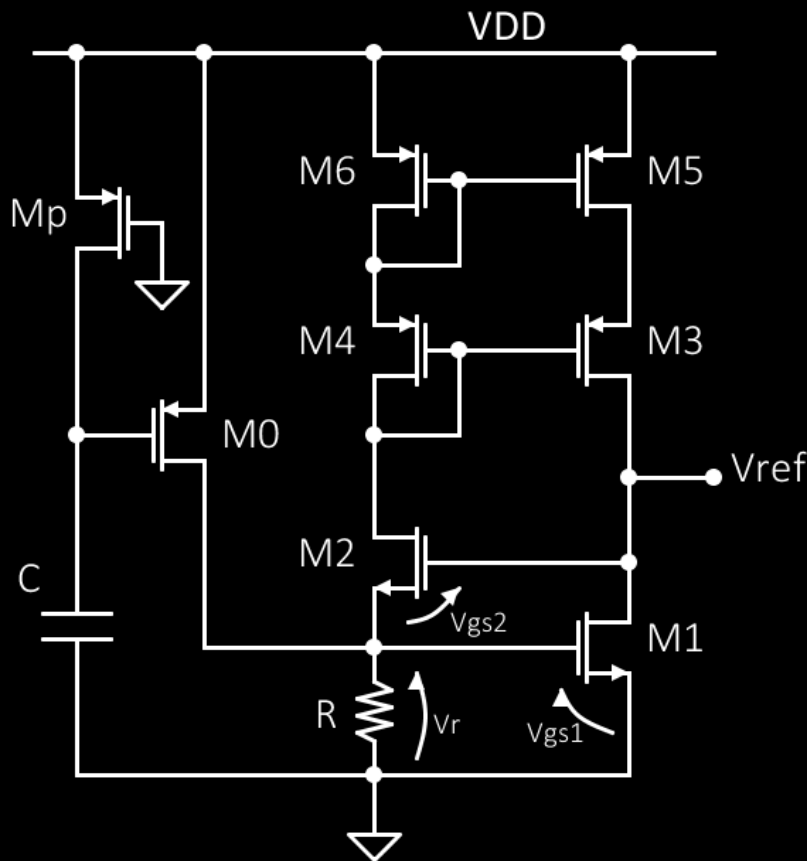
The driving parameter are the target  $V_{ref} = 1.65\text{ V}$  and the current budget assigned at the beginning of the project which is  $8\text{ }\mu\text{A}$ . The latter sets the branch current to be  $4\text{ }\mu\text{A}$ .

The last degree of freedom is about the voltage drop above R. It must be, of course, above threshold and in this case it will be  $V_r = 0.81\text{ V}$ . Since  $V_r = V_{gs1}$ :

$$\left. \frac{W}{L} \right|_1 = \frac{I_{branch}}{K'_n * (V_r - V_{TH_n})^2} = \frac{4\text{ }\mu}{57.8\text{ }\mu * (0.81 - 0.76)^2} = 27.68$$

Since this circuit will be implemented at layout level too, its design will consider the multiplier of each transistor to.

$$\begin{cases} W_1 = 49.8\text{ }\mu\text{m} \\ L_1 = 1.8\text{ }\mu\text{m} \end{cases} \rightarrow \begin{cases} W_1 = 8 * 6.15\text{ }\mu\text{m} \\ L_1 = 1.8\text{ }\mu\text{m} \end{cases}$$



- LDO outcomes
- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- **Reference design II**

The resistor value can be computed by means of the simple Ohm's law:

$$R = V_r / I_{branch} = 0.81 / 4 \mu = 202.5 \text{ k}\Omega \cong 200 \text{ k}\Omega$$

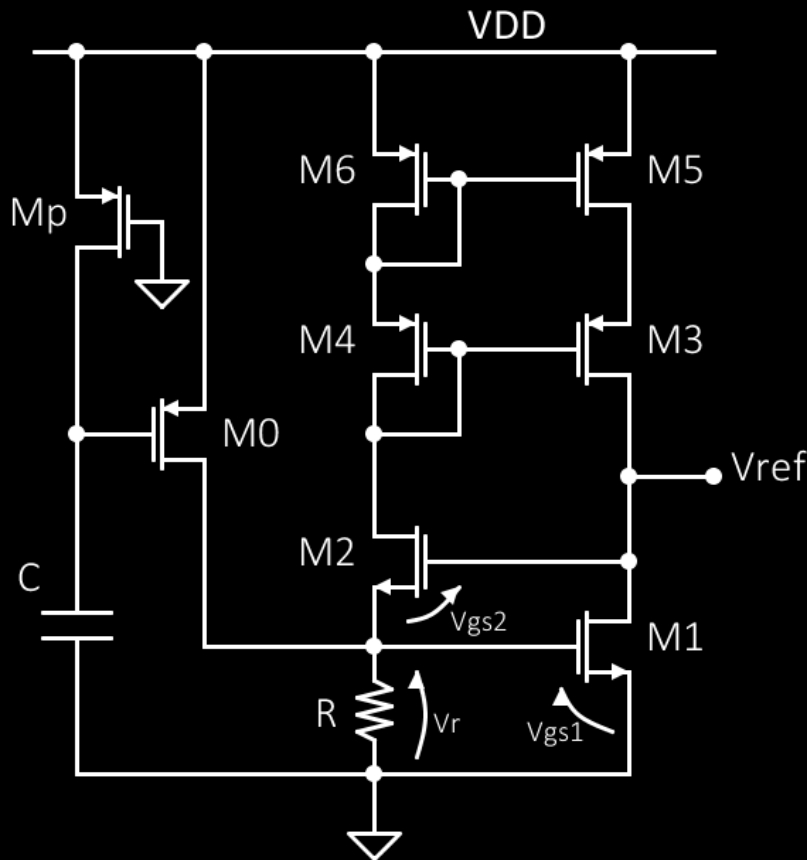
Due to better layout implementation, this value will be split into 3 parts having a  $66.6 \text{ k}\Omega$  resistance.

The next step is about the transistor  $M_2$  dimensions. The gate-source voltage of this device depends on the target  $V_{ref}$  being  $V_{gs2} = V_{ref} - V_r = 0.84 \text{ V}$ , then:

$$W/L|_2 = \frac{I_{branch}}{K'_n * (V_{gs2} - V_{TH_n})^2} = \frac{4 \mu}{57.8 \mu * (0.84 - 0.76)^2} \cong 11$$

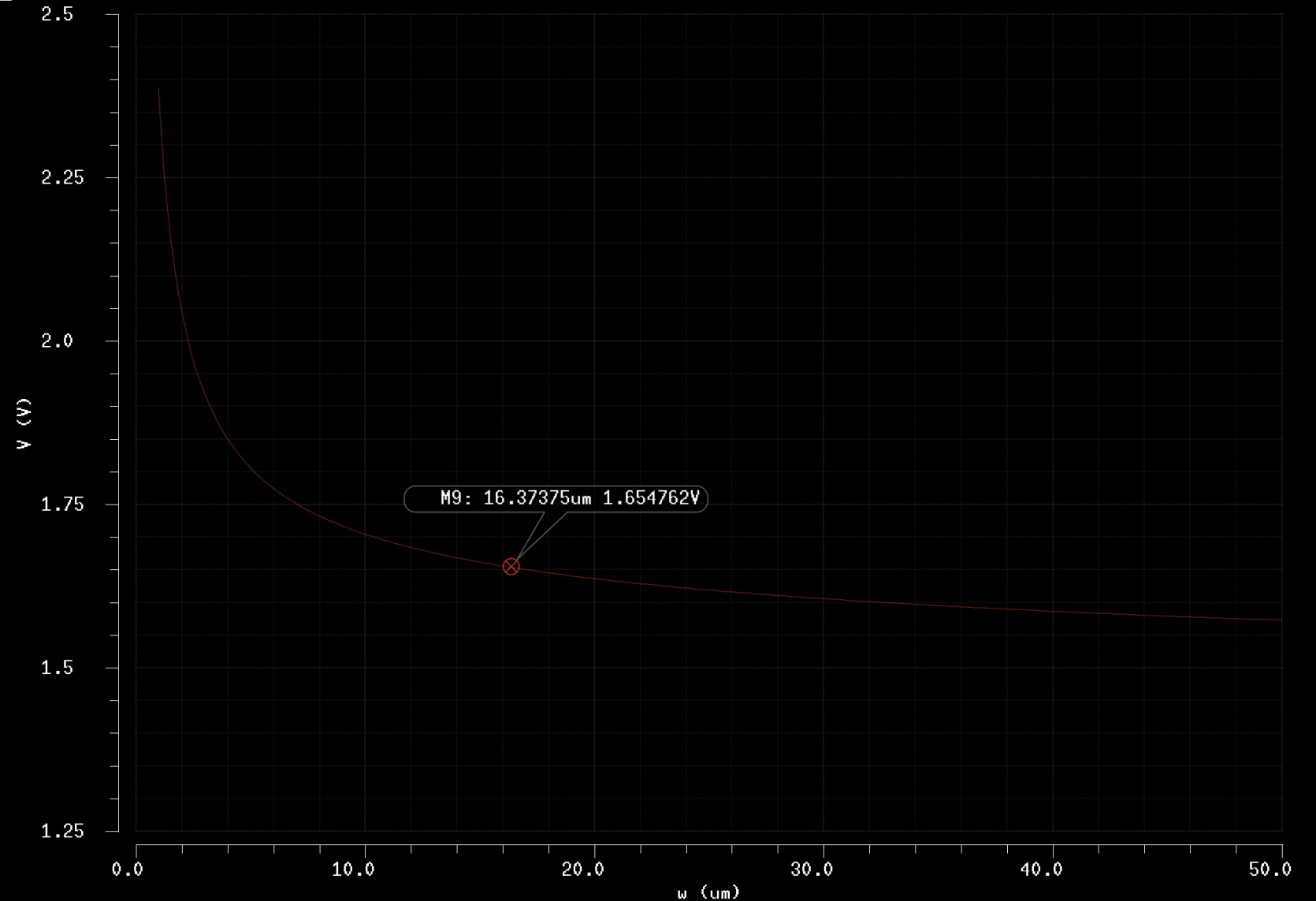
$$\begin{cases} W_2 = 19.8 \mu m \\ L_2 = 1.8 \mu m \end{cases} \rightarrow \begin{cases} W_2 = 8 * 2.5 \mu m \\ L_2 = 1.8 \mu m \end{cases}$$

Unfortunately, this transistor is strongly affected by body effect which will undermine the validity of the above computed value. For this reason parametric simulation of the width has been performed to adjust the final output voltage value.



- LDO outcomes
- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- **Reference design II**

| Name                                     | Vis                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------|
| <span style="color: red;">■</span> /Vref |  |



The output approaches the desired value when  $M_2$  fingers dimension is around  $16 \mu m$ , the final dimensions are:

$$\begin{cases} W_2 = 8 * 16.05 \mu m \\ L_2 = 1.8 \mu m \end{cases}$$

The resulting width is very much larger than the computed one but this was expected due to the strong BULK effect on the transistor.

- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- Reference design II
- **Reference design III**

The current mirrors and the start-up circuit must be defined.  
On the left side the minimum voltage for saturation working is:

$$VDD - V_r - \sqrt{\frac{I_{branch}}{K'_n * W/L|_2}} = 4.11 V @ VDD = 5 V$$

On the right side the voltage drop above the cascode current mirror is:

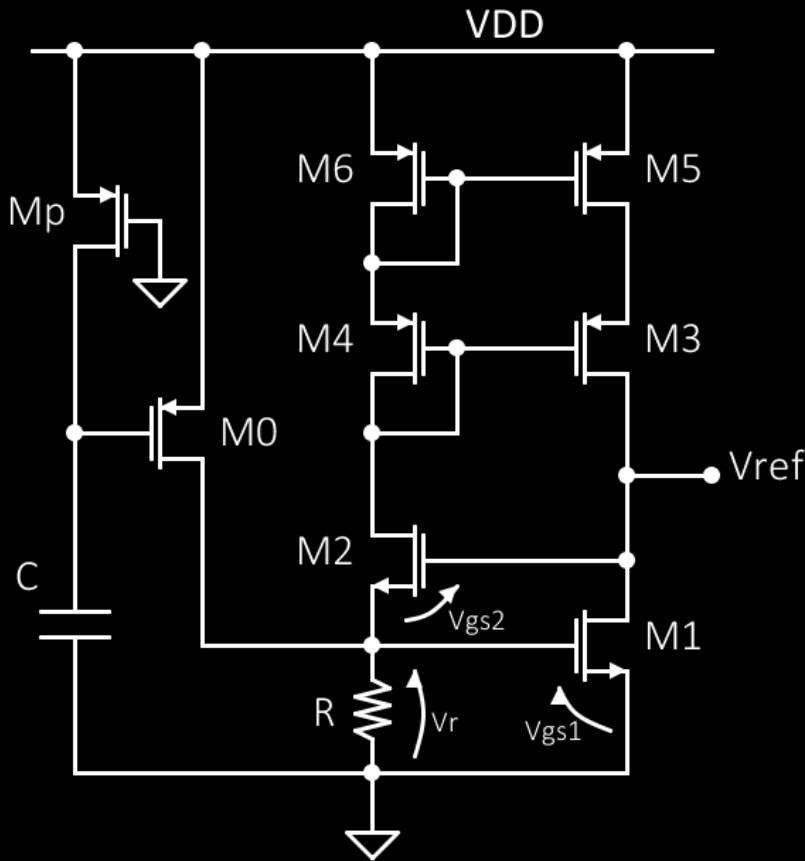
$$VDD - V_{ref} = 3.35 V @ VDD = 5 V$$

The second one has been taken as reference being more restrictive. Then, assuming to have half drop above each transistor:

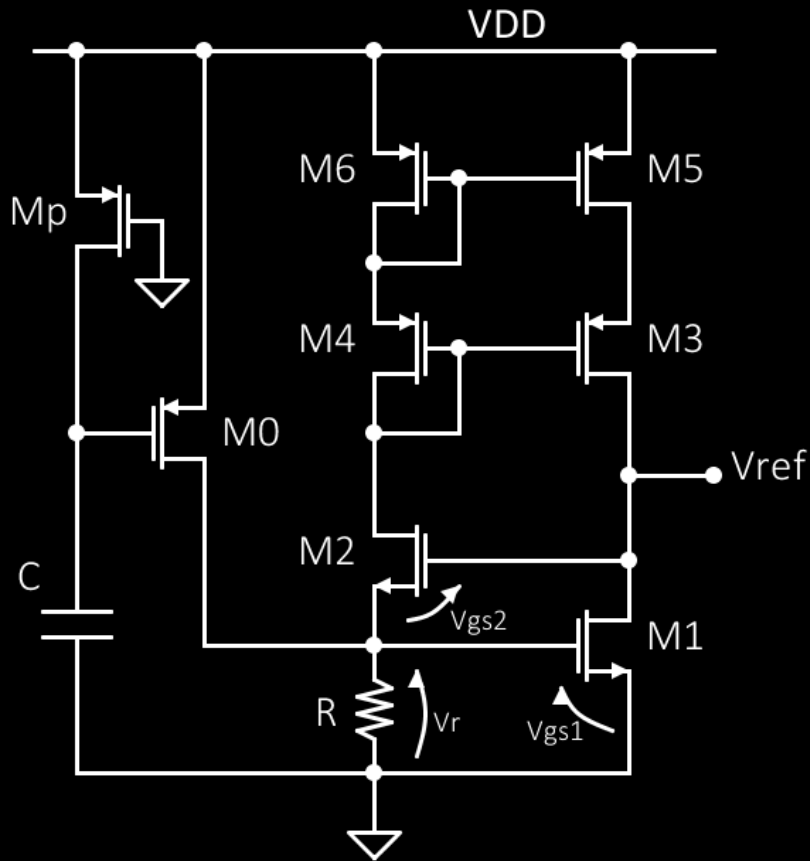
$$W/L|_6 = W/L|_5 = W/L|_4 = W/L|_3 > \frac{I_{branch}}{|K'_p| * \left(\frac{3.35}{2}\right)^2} \cong 0.08$$

Since this is a MINIMUM value, the actual dimension has been chosen in order to have almost-square transistor when their width is divided by 4 to implement a cascode 2D structure:

$$\begin{cases} W_6 = W_5 = W_4 = W_3 = 7.2 \mu m \\ L_6 = L_5 = L_4 = L_3 = 1.8 \mu m \end{cases} \rightarrow \begin{cases} W_6 = W_5 = W_4 = W_3 = 4 * 1.8 \mu m \\ L_6 = L_5 = L_4 = L_3 = 1.8 \mu m \end{cases}$$



- BIAS design I
- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- Reference design II
- **Reference design III**



The start-up circuit has been borrowed from the original article design being the chosen topology very simple. The capacitor C is  $100\text{ fF}$  and the transistor  $M_p$  is a long resistive device which charges the capacitor until the transistor  $M_0$  is shut down.

$$\begin{cases} L_p = 50\text{ }\mu\text{m} \\ W_p = 1.8\text{ }\mu\text{m} \end{cases}$$

The  $M_0$  dimension has been chosen in order to make it able to keep the VDD voltage on the resistor node until the capacitor reaches  $\frac{3}{4}$  of the power supply voltage. This condition requires the transistor to deliver  $25\text{ }\mu\text{A}$  to the resistor which sees a  $5\text{ V}$  drop at its terminals. It results:

$$W/L|_0 = \frac{25\text{ }\mu}{18.9\text{ }\mu * (1.25 - 0.9)^2} \cong 10$$

$$\begin{cases} L_0 = 1.8\text{ }\mu\text{m} \\ W_0 = 18\text{ }\mu\text{m} \end{cases} \rightarrow \begin{cases} L_0 = 1.8\text{ }\mu\text{m} \\ W_0 = 4 * 4.5\text{ }\mu\text{m} \end{cases}$$

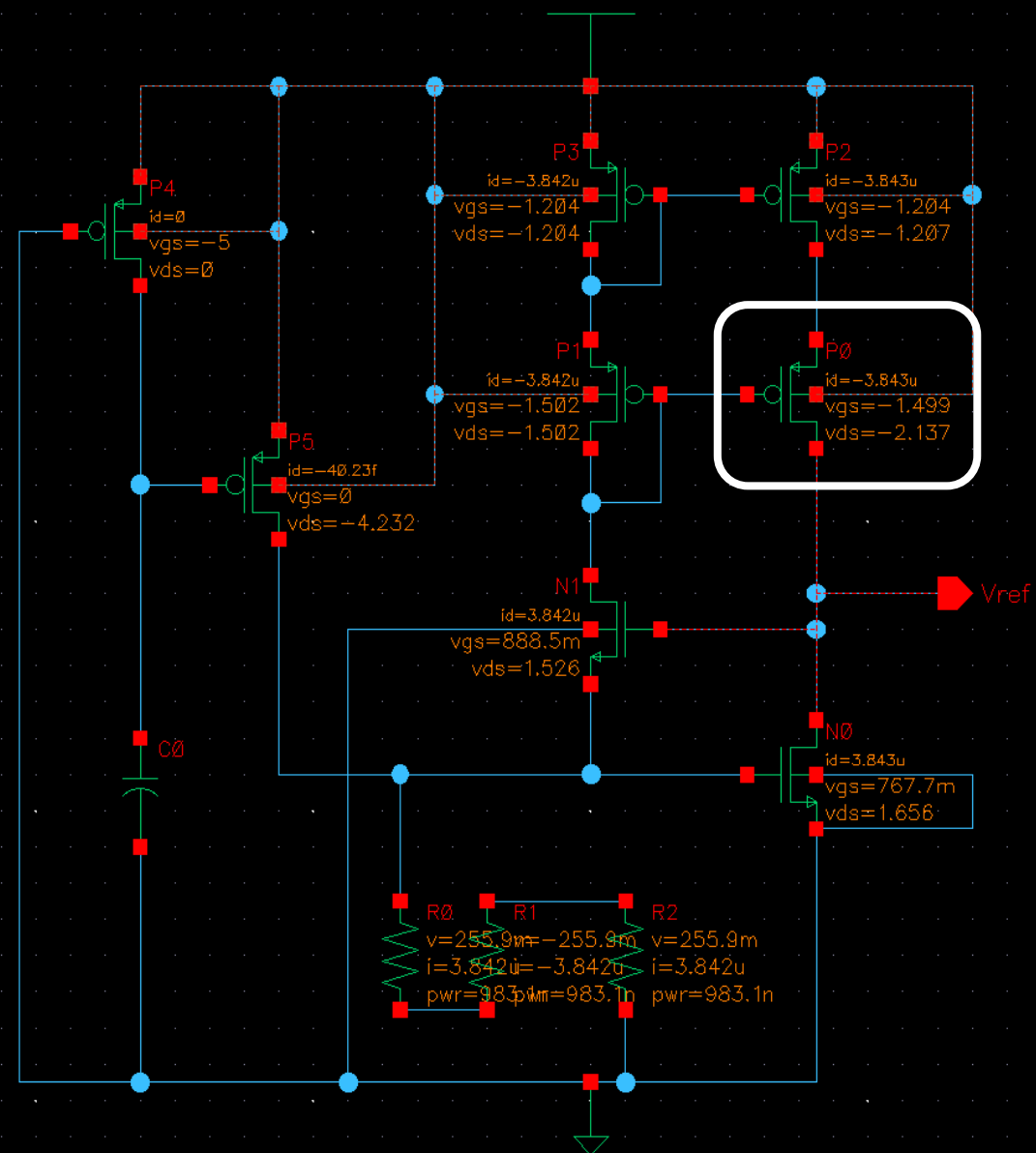
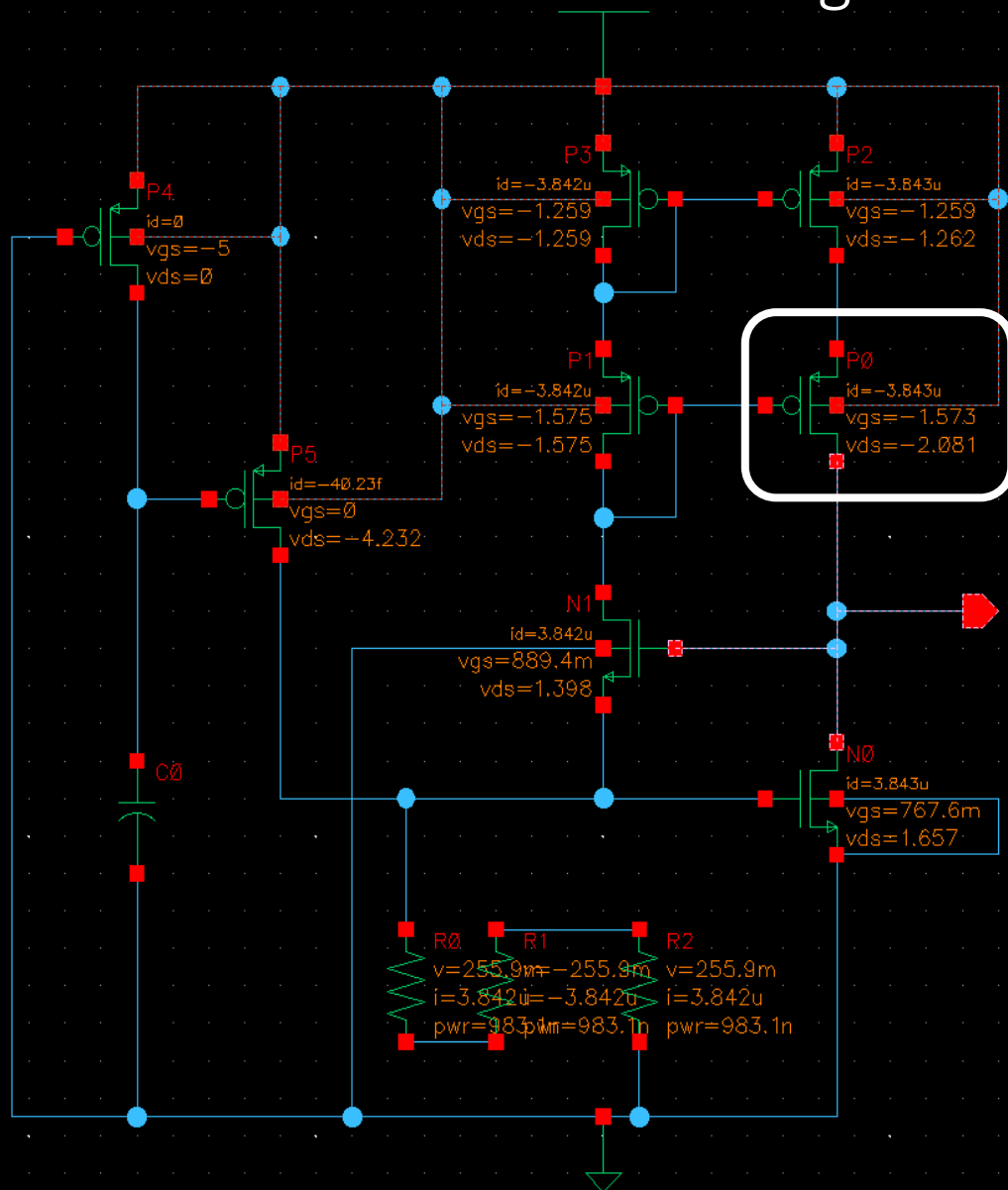
The above specification will be verified by means of a transient simulation.

- BIAS design II
- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- Reference design II
- Reference design III
- Reference DC Sim I**

# Voltage reference DC simulation

The only difference between these two circuits is the multiplier of the cascode current mirror.

All the transistors are saturated and the  $V_{gs}$  of the common centroid version is about 5% higher then the counterpart. This could be seen as a **smaller width equivalent transistor** driving the same current.

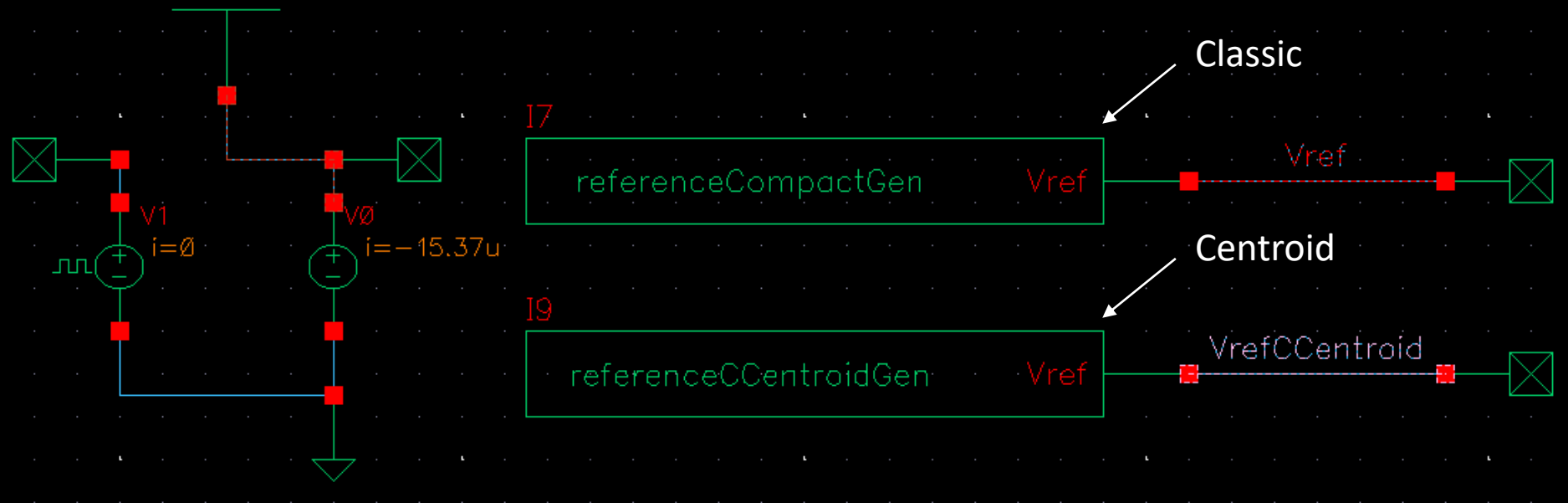


- BIAS design III
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- Reference design II
- Reference design III
- Reference DC Sim I
- **Reference DC Sim II**

# Voltage reference DC simulation environment

Both the implementations of the voltage reference generator have been placed into the same schematic in order to perform comparative simulations. The same will be done for the two layout produced in this work.

From now on these two version will be referred as Classic and Centroid as the image below shows.



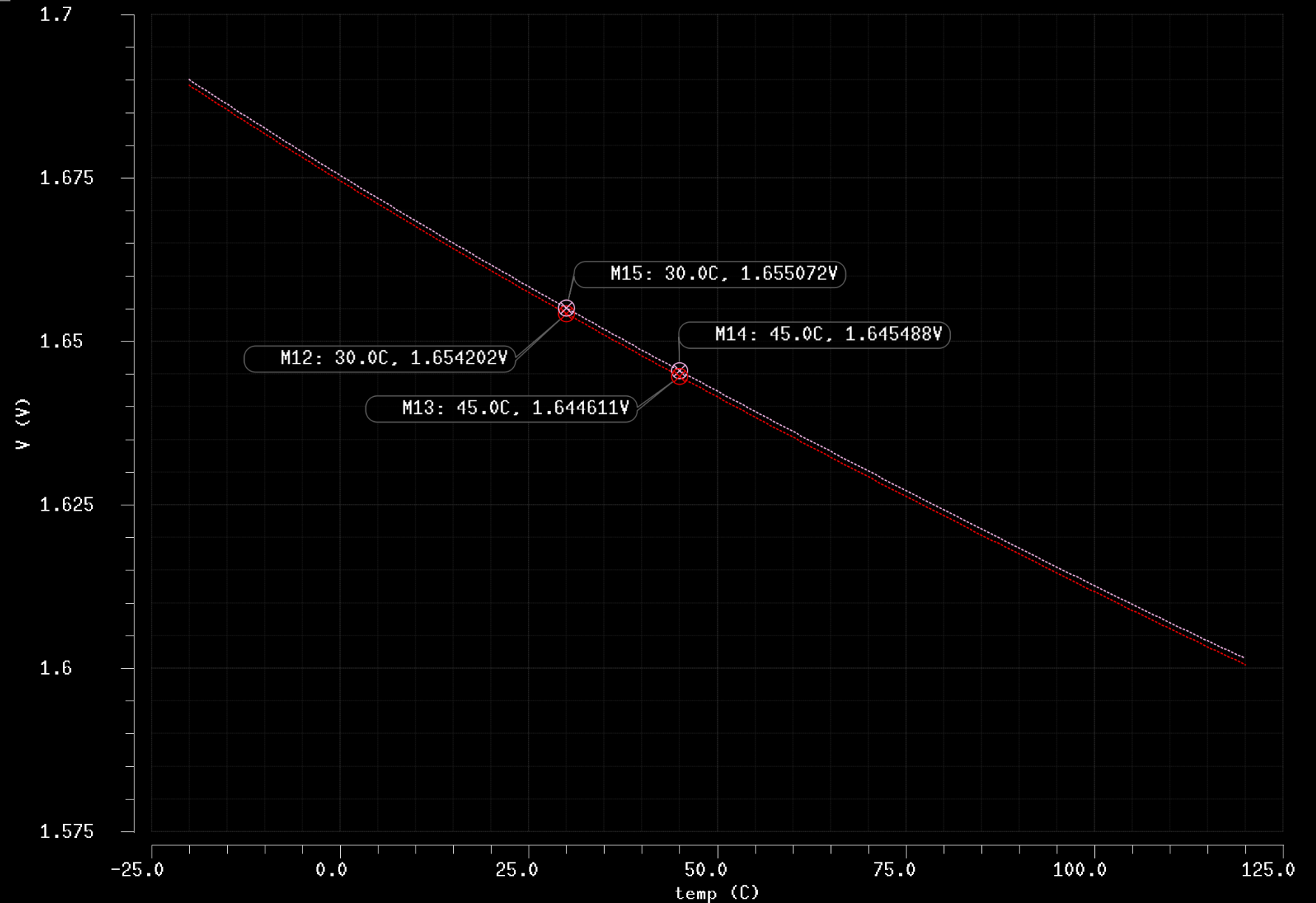
- BIAS DC sim
- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- Reference design II
- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- **Reference T Sweep**

| Name          |  |
|---------------|--|
| /Vref         |  |
| /VrefCentroid |  |

The classic version always presents a lower output due to the source-gate voltage difference previously seen by means of DC operation point. As for the BIAS circuit, the range between 30°C and 45°C has been highlighted. Even presenting a little offset the variation is similar and can be computed as:

$$\frac{1.655 - 1.645}{1.655} * 100 \cong 0.6\%$$

## Voltage reference temperature sweep

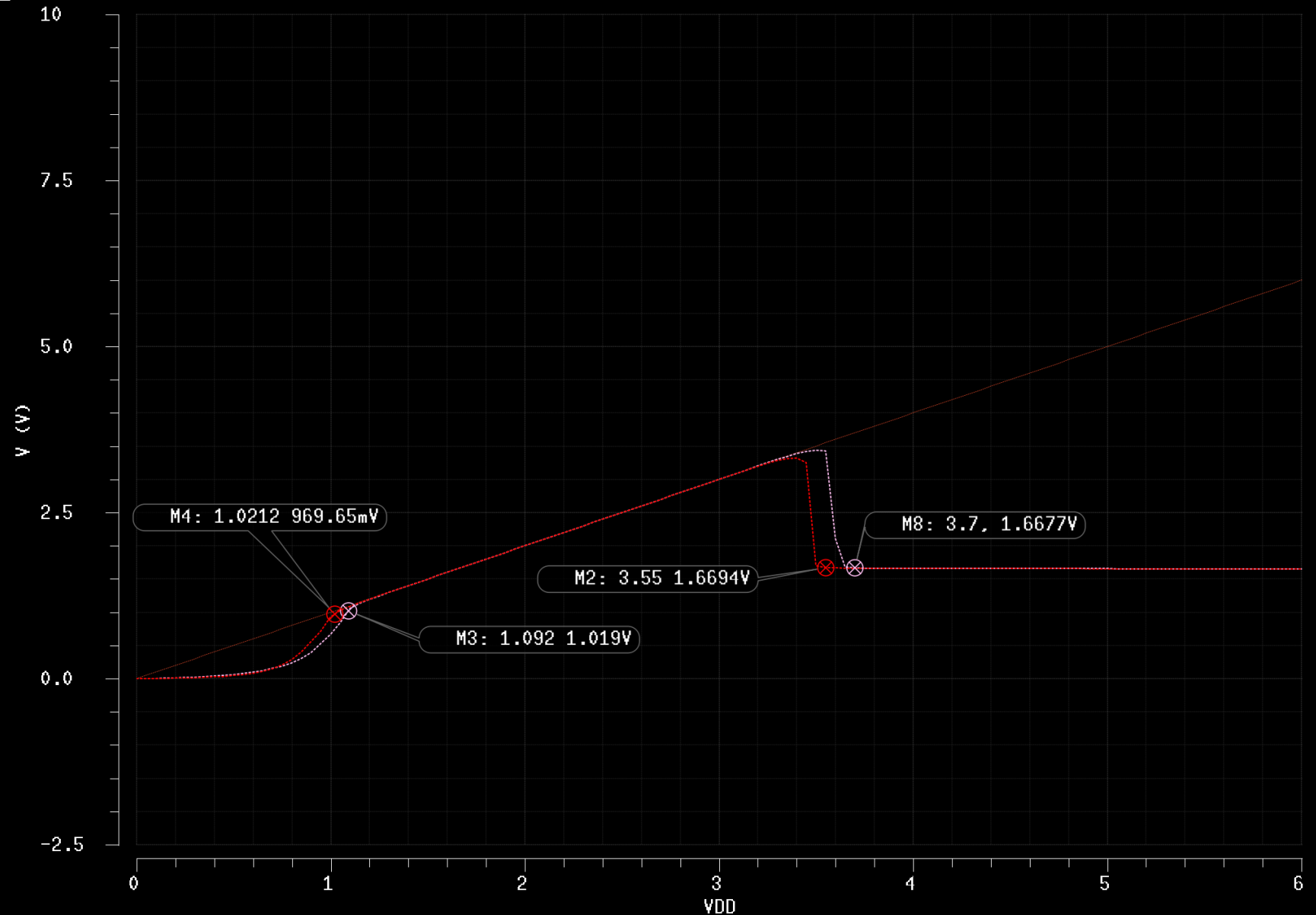


- BIAS DC adj I
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- Reference design II
- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- **Reference Line Reg**

| Name                                   |               |
|----------------------------------------|---------------|
| <span style="color: #8B0000;">—</span> | /vddI         |
| <span style="color: #FF0000;">—</span> | /Vref         |
| <span style="color: #FF00FF;">—</span> | /VrefCentroid |

Over the course of VDD growth the  $V_{ref}$  value follows the transistors power-on and the upper PMOS devices connect  $V_{ref}$  to VDD until the voltage is enough to bring the NMOS transistors into the saturation region. Then the output voltage is pulled down and it settles at the design value.

# Voltage reference Line regulation



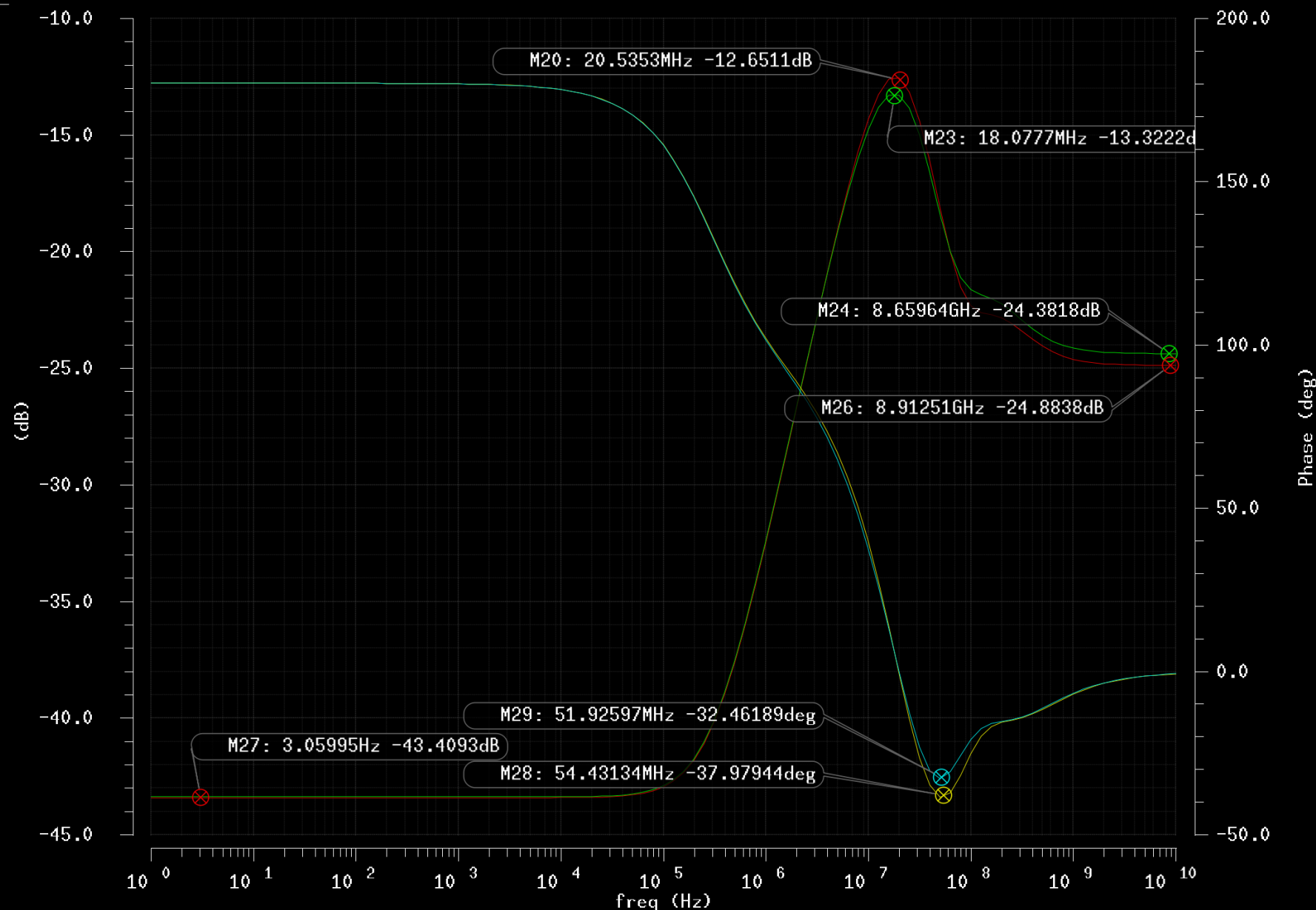
- BIAS DC adj II
- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- Reference design II
- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- **Reference PSRR**

| Name                     |  |
|--------------------------|--|
| PSRR Magnitude Classic   |  |
| PSRR Phase Classic       |  |
| PSRR Magnitude CCentroid |  |
| PSRR Phase CCentroid     |  |

All the points of interest are highlighted. The PSRR is about -13 dB when the frequency is around 20 MHz and falls down to -24 dB at higher frequency.

The DC PSRR is -43 dB and the Centroid circuit shows a 5% increment of peak PSRR performances.

# Voltage reference PSRR analysis



- BIAS Temp sweep
- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- Reference design II
- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- **Reference Tran I**

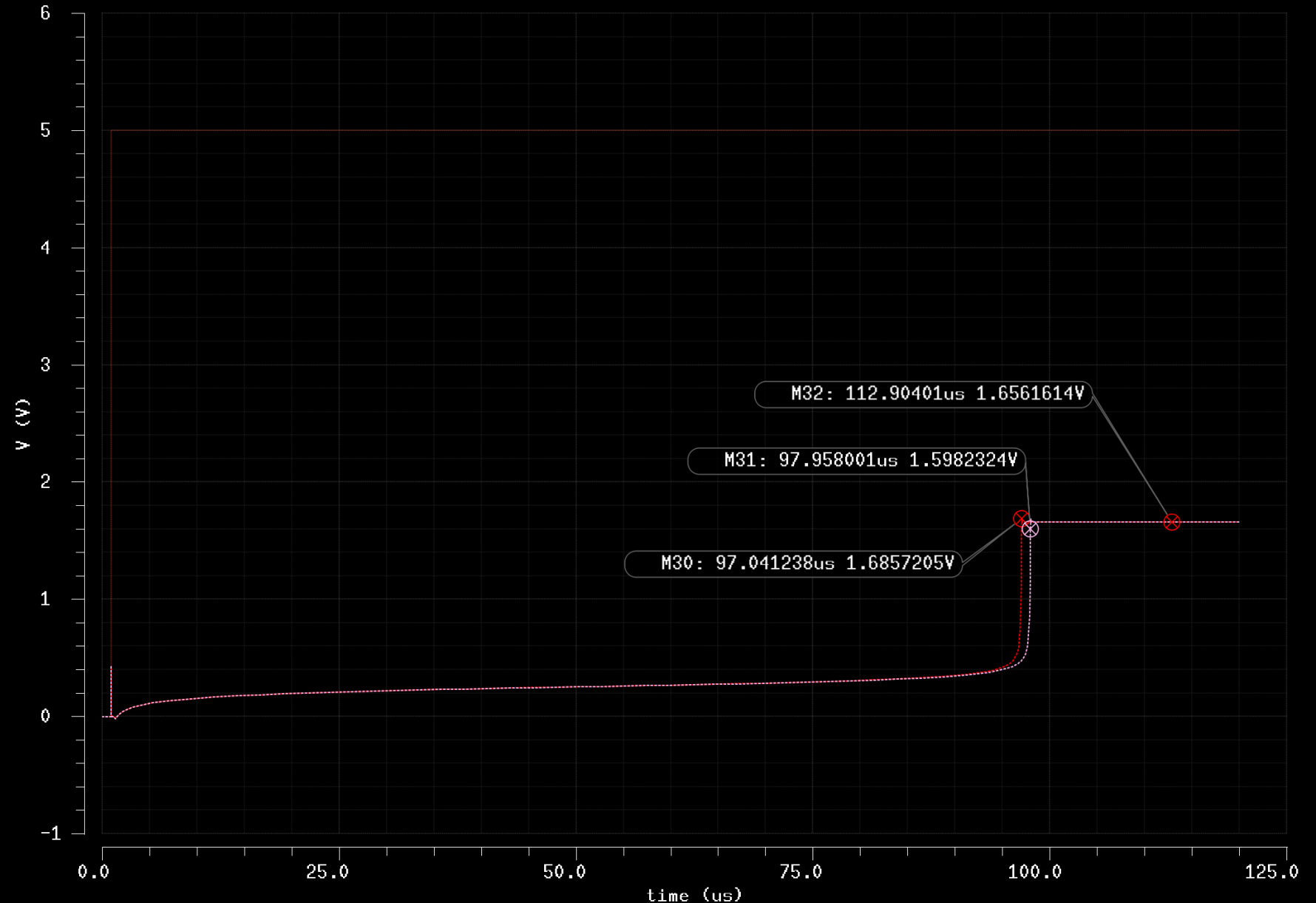
| Name                                 |                |
|--------------------------------------|----------------|
| <span style="color: blue;">█</span>  | /vddI          |
| <span style="color: red;">█</span>   | /Vref          |
| <span style="color: green;">█</span> | /VrefCCentroid |

The transient analysis is performed applying a step signal to VDD with a  $1\ \mu\text{s}$  delay. In the right plot are shown the start-up time of both the circuits:

- Classic:  $\approx 97\ \mu\text{s}$
- Centroid:  $\approx 98\ \mu\text{s}$

Both of them reaches the same reference voltage of 1.656 V.

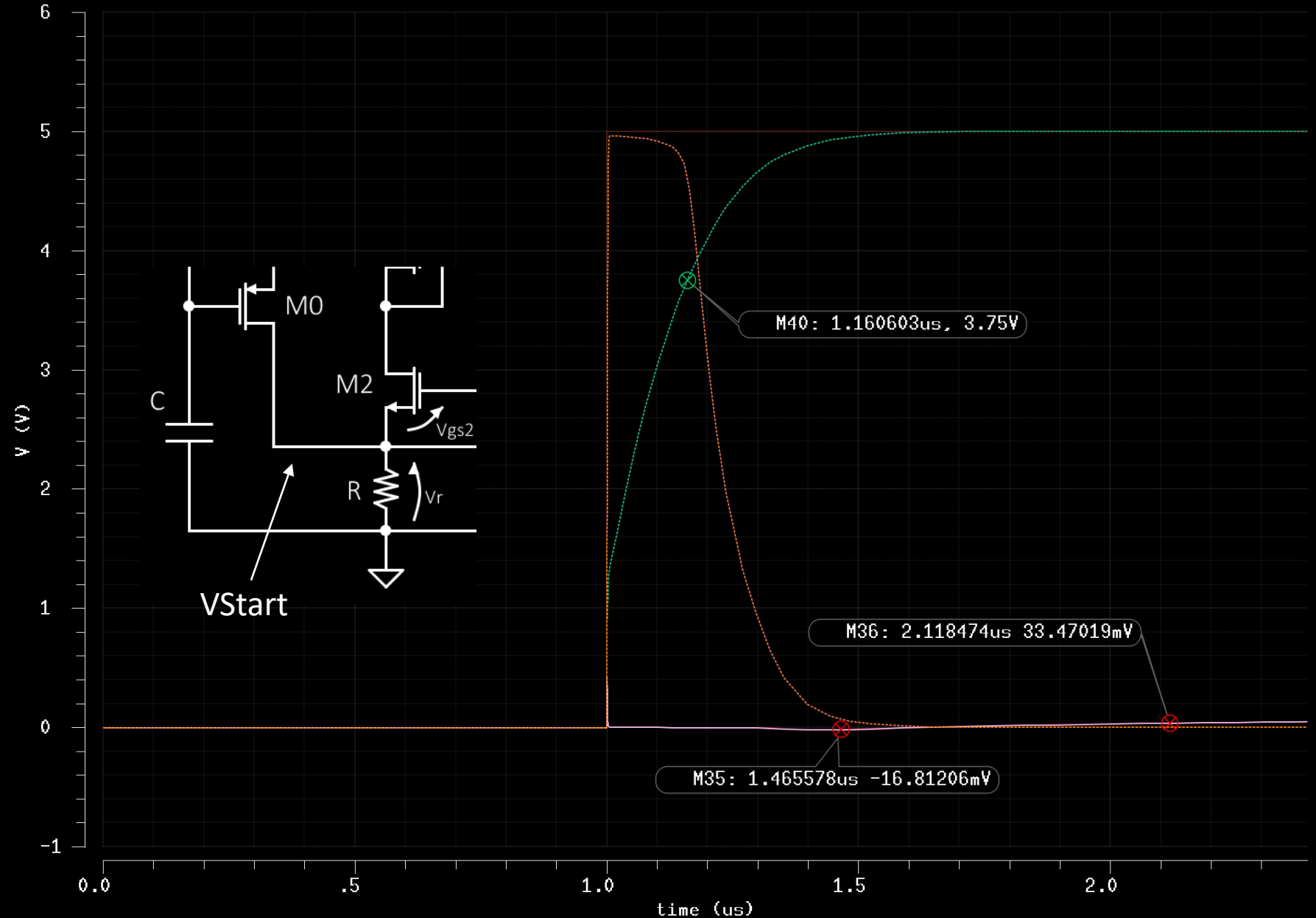
# Voltage reference transient analysis



- BIAS Line reg
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- Reference design II
- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- Reference Tran I
- **Reference Tran II**

| Name           |  |
|----------------|--|
| /vddI          |  |
| /Vref          |  |
| /VrefCCentroid |  |
| VStartCentroid |  |
| VCapCentroid   |  |
| VStartClassic  |  |
| VCapClassic    |  |

As designed, the start-up transistor ( $M_0$ ) keeps the high voltage on the R node until the capacitor voltage (green line) reaches  $\frac{3}{4}$  of VDD. Then the voltage across R drops down but in the meantime the current starts to flow into the circuit and the 0 V stability point is abandoned. Indeed, the  $V_{ref}$  voltage starts growing as the markers M35 and M36 show.

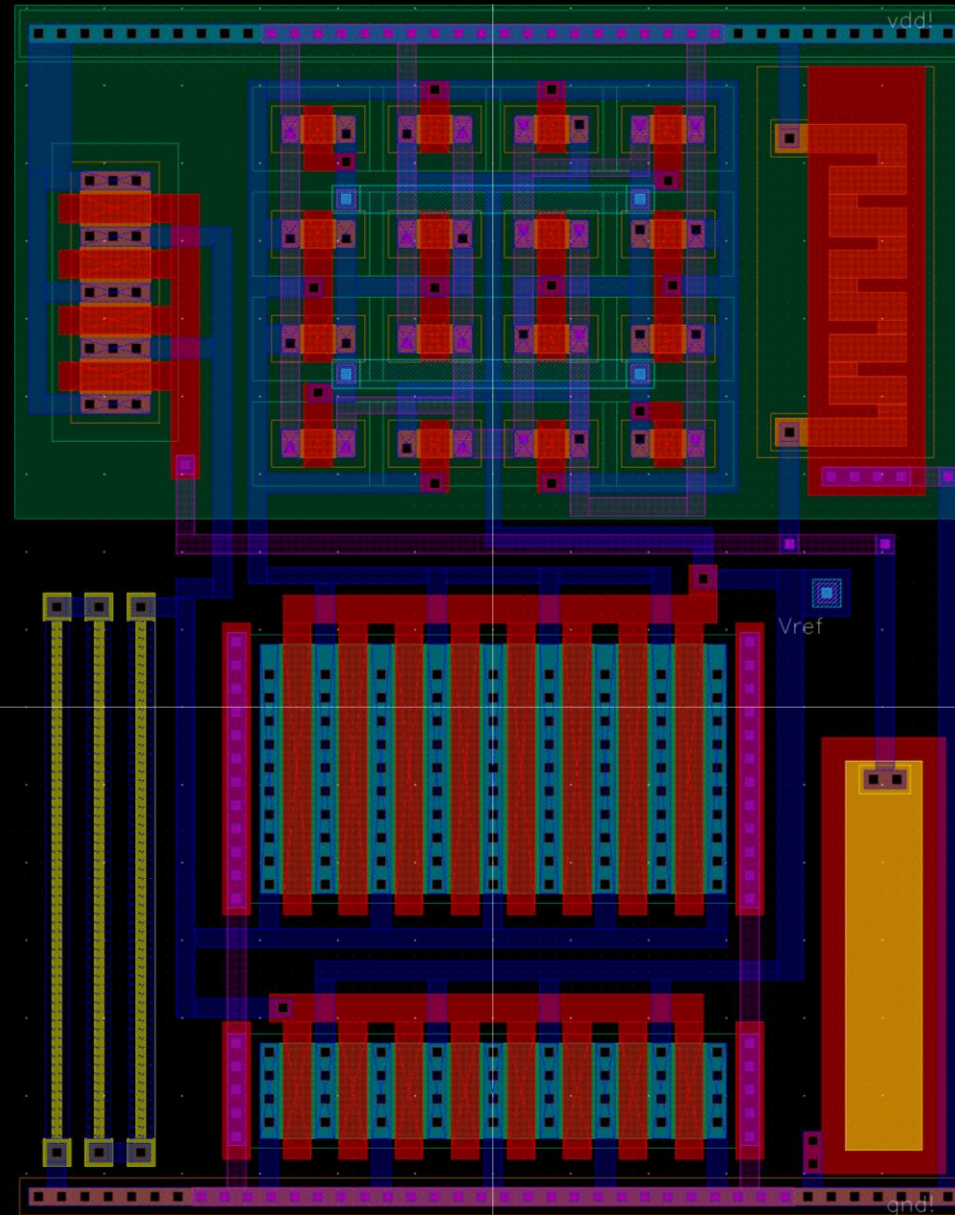


# Voltage reference LAYOUT

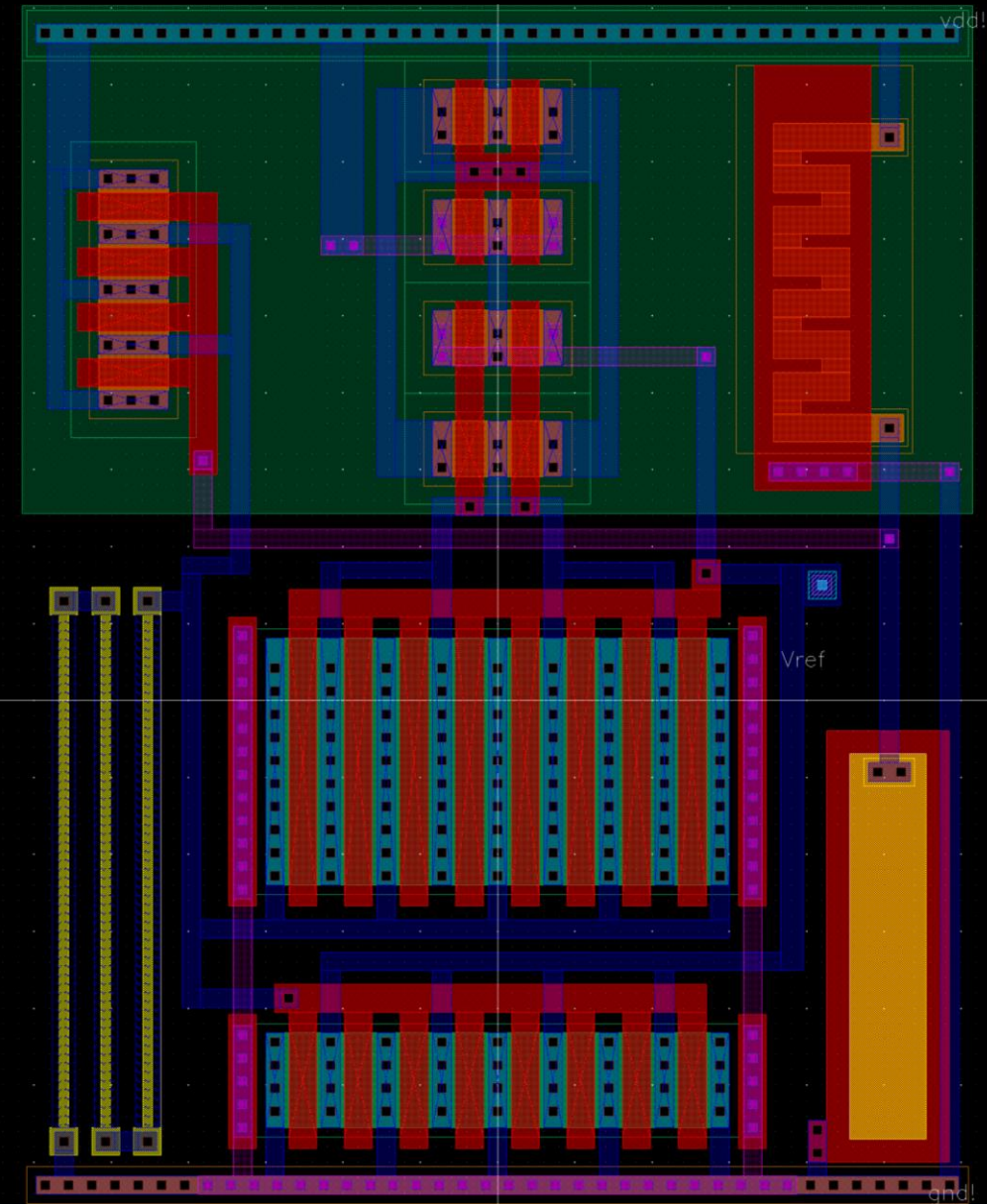
- BIAS PSRR
- BIAS transient I
- BIAS transient II
- Reference design I
- Reference design II
- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- Reference Tran I
- Reference Tran II
- **Reference Layout I**

As mentioned, two layout version have been implemented. The difference is only about the cascode current mirror. It will be discussed in a subsequent slide.

The total area is the same for both the cells:  
 $\approx 4787 \mu\text{m}^2$



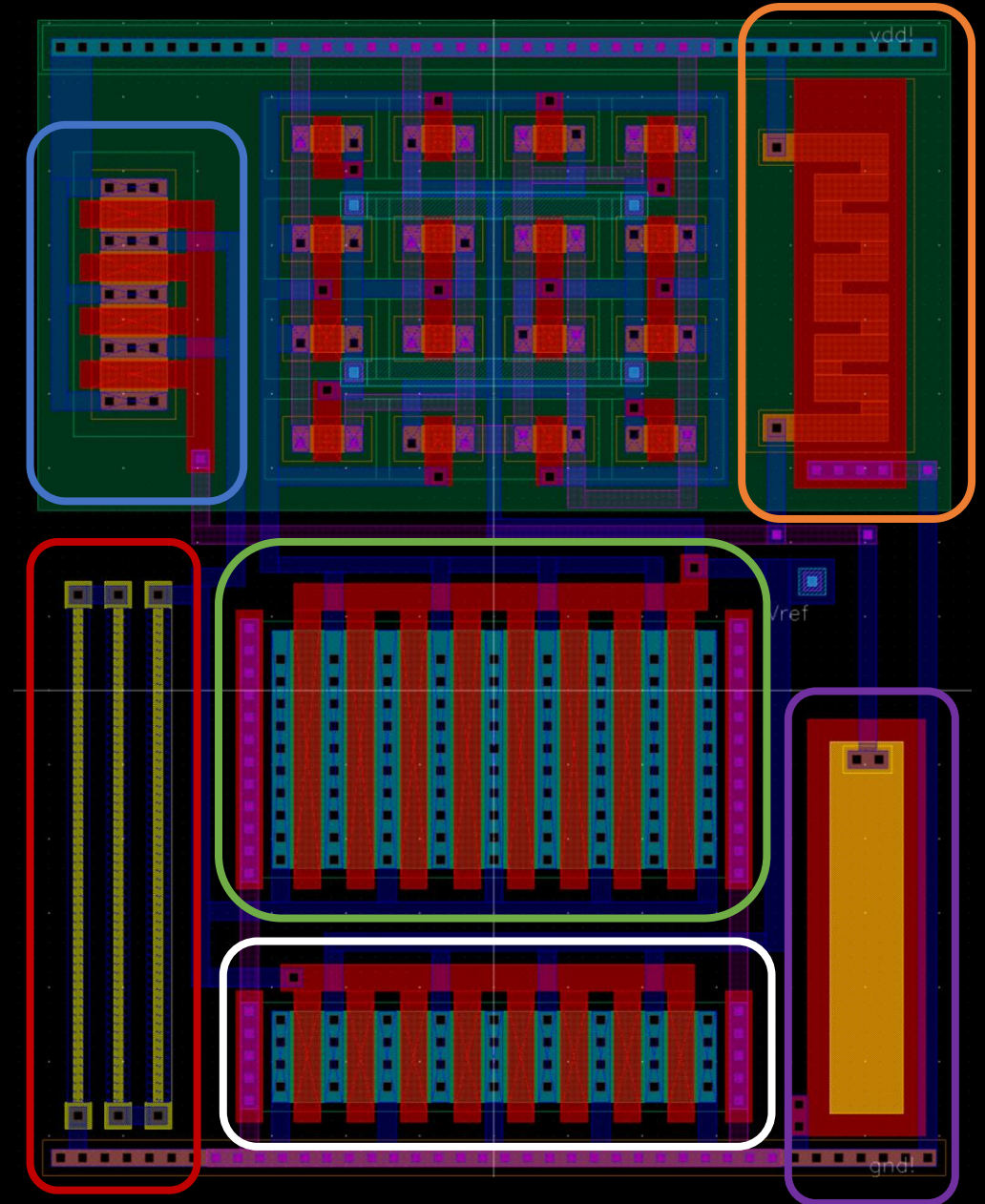
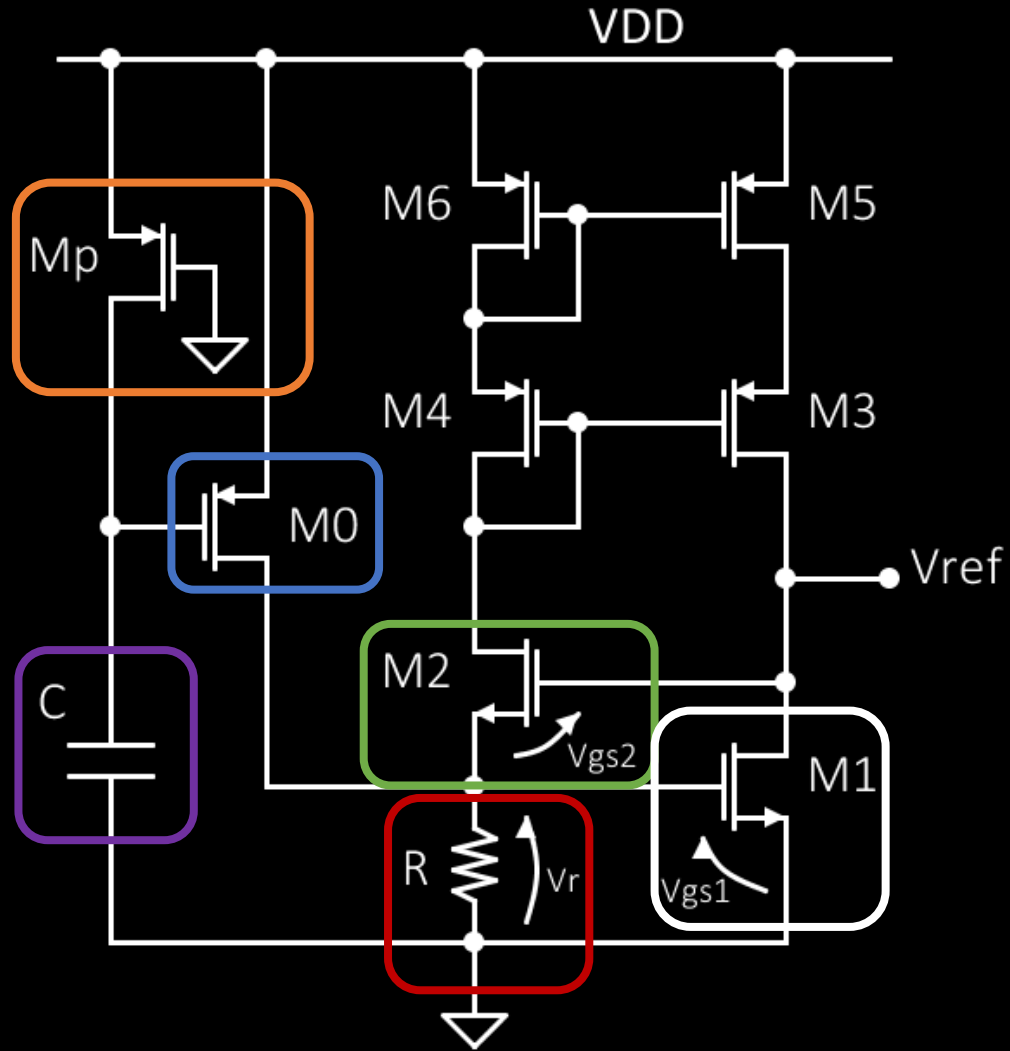
Centroid



Classic

- BIAS transient I
- BIAS transient II
- Reference design I
- Reference design II
- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- Reference Tran I
- Reference Tran II
- Reference Layout I
- **Reference Layout IIv**

# LAYOUT - Common parts



- BIAS transient II
- Reference design I
- Reference design II
- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- Reference Tran I
- Reference Tran II
- Reference Layout I
- Reference Layout II
- Reference Layout III**

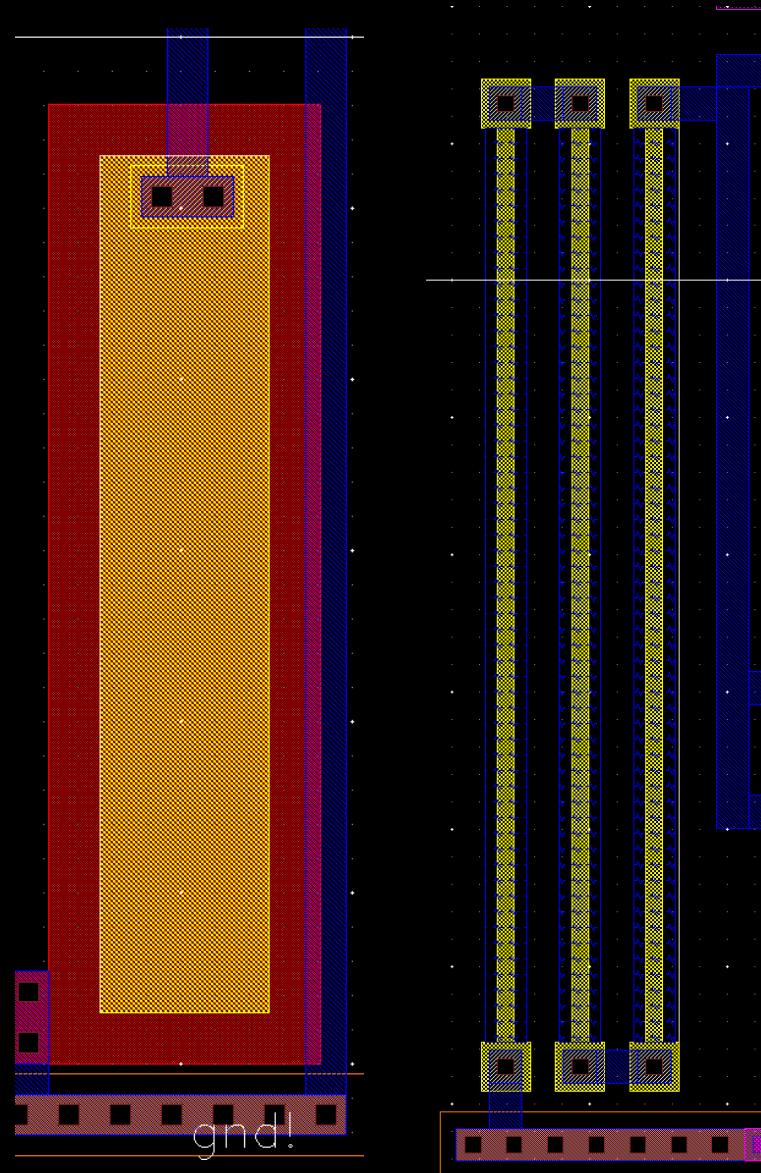
# LAYOUT - Long transistor design

In a classic BULK CMOS process the transistor width is usually defined by the active and the channel length is set by the poly which covers the active and select during the drain/source implantation process. Here the opposite process is exploited to fabricate a long channel transistor in a smaller area. The width is given by the active thickness and the poly covers everything but the drain and source regions.

- Reference design II
- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- Reference Tran I
- Reference Tran II
- Reference Layout I
- Reference Layout II
- Reference Layout III
- Reference Layout IV
- **Reference Layout V**

# LAYOUT - Resistors and capacitors

Precision capacitor with poly/poly2:  
as for the poly2 resistors one of the advantages of this capacitor implementation is the separation from bulk which can be easily source of noise. Moreover an etched layer fabrication should be more precise than an implantation based one. Here the real capacitor area is given by the poly2 layer and the extracted value is around 90 fF.

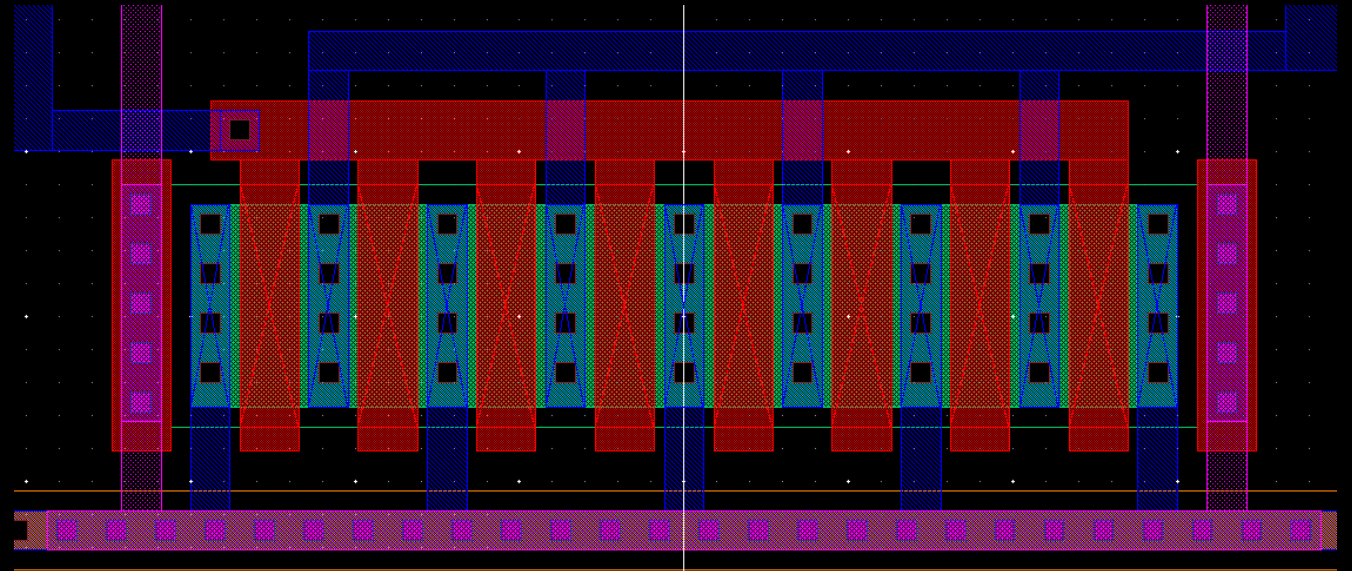


Precision resistor with poly2  
HiRes: this implementation comes with multiple advantages with respect to a N-WELL resistor, e.g. less parasitic capacitance due to larger and isolated distance from bulk, smaller area due to unnecessary insulation junction around it or an even larger sheet resistance ( $1076 \Omega/\square$ ).

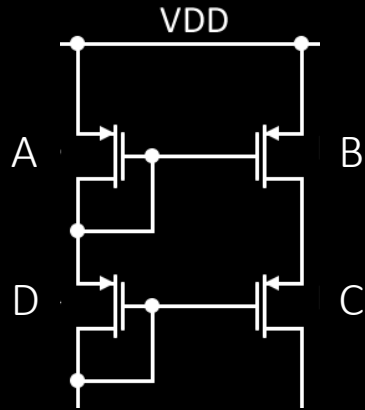
- Reference design I
- Reference design II
- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- Reference Tran I
- Reference Tran II
- Reference Layout I
- Reference Layout II
- Reference Layout III
- **Reference Layout IV**

# LAYOUT – Interdigitated structures

The large MOS is split into 8 smaller transistors with alternated drain and source implants. It follows that the multiple transistors show a sum of parasitic capacitance which should be lower than the equivalent single device. Here a couple of dummy gates have been attached to the extremities of the structure to let the last implants see the same neighbourhood of any other. These two additive gate are GND connected.



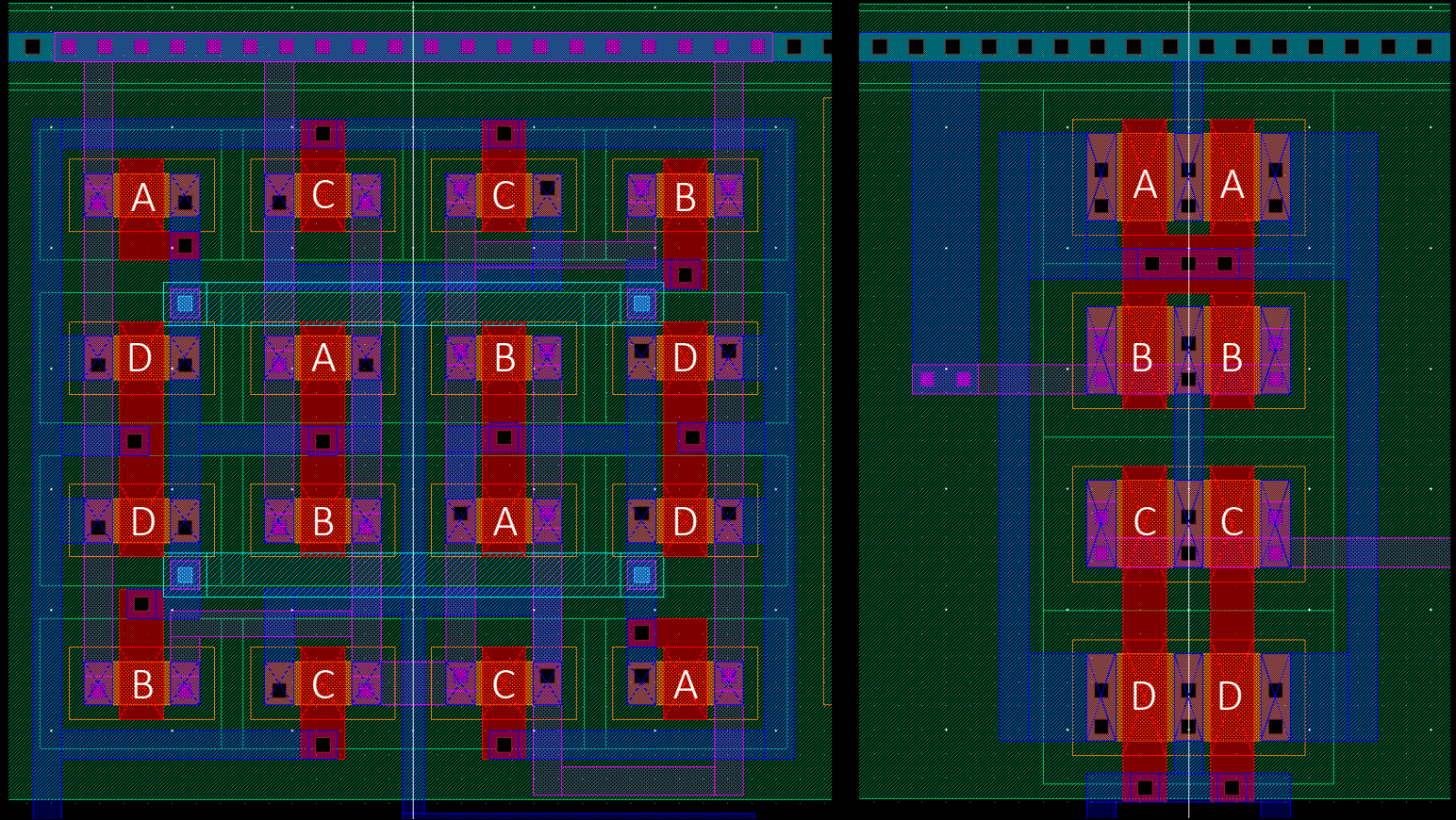
- Reference design II
- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- Reference Tran I
- Reference Tran II
- Reference Layout I
- Reference Layout II
- Reference Layout III
- Reference Layout IV
- **Reference Layout V**



## LAYOUT – Cascode current mirror

Here comes the main difference between the two proposed layouts.

A 2D common centroid structure with 16 transistors  $1.8 \mu m$  large makes use of 3 metal layers but allows an enhanced symmetry for temperature gradients cancelling in both directions. The second structure is simpler and does not require a third metal layer which in this case shall correspond to one less mask to be processed. Moreover, it is potentially less area consuming even if in this specific layout the cell width is imposed by the NMOS transistors.



- Reference design III
- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- Reference Tran I
- Reference Tran II
- Reference Layout I
- Reference Layout II
- Reference Layout III
- Reference Layout IV
- Reference Layout V
- **Ref extracted Tran**

| Name          |
|---------------|
| vddI          |
| Vref          |
| VrefCCentroid |

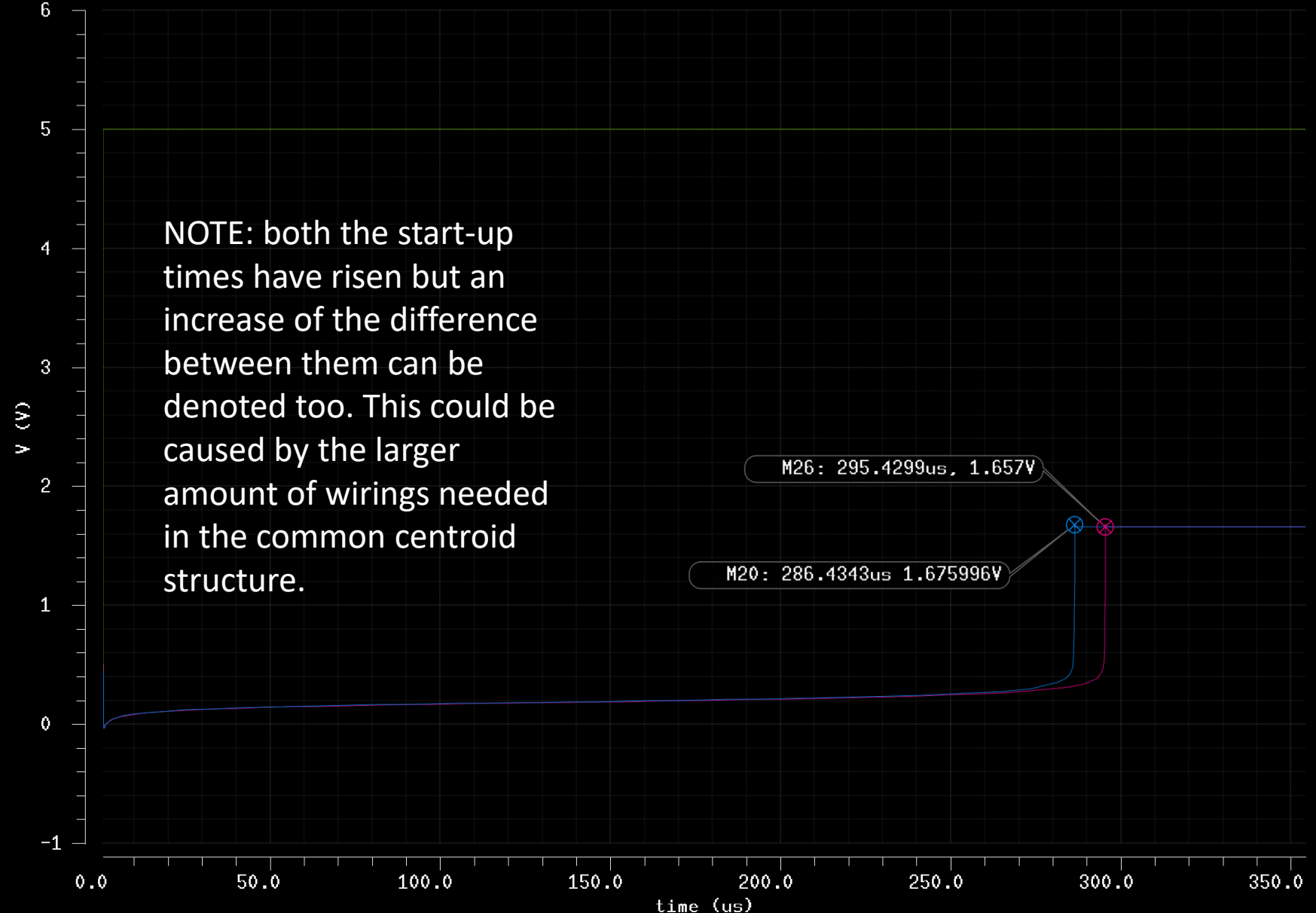
For comparison purposes could be useful to report here the schematic level start-up time previously obtained:

- Classic:  $\approx 97 \mu s$
- Centroid:  $\approx 98 \mu s$

As the markers show the time increased a lot but this was expected due to the real wiring and parasitic of a layout level simulation. The result is now:

- Classic:  $\approx 286 \mu s$
- Centroid:  $\approx 295 \mu s$

# EXTRACTED Transient Simulation



- Reference DC Sim I
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- Reference Tran I
- Reference Tran II
- Reference Layout I
- Reference Layout II
- Reference Layout III
- Reference Layout IV
- Reference Layout V
- Ref extracted Tran
- **Ref extracted PSRR**

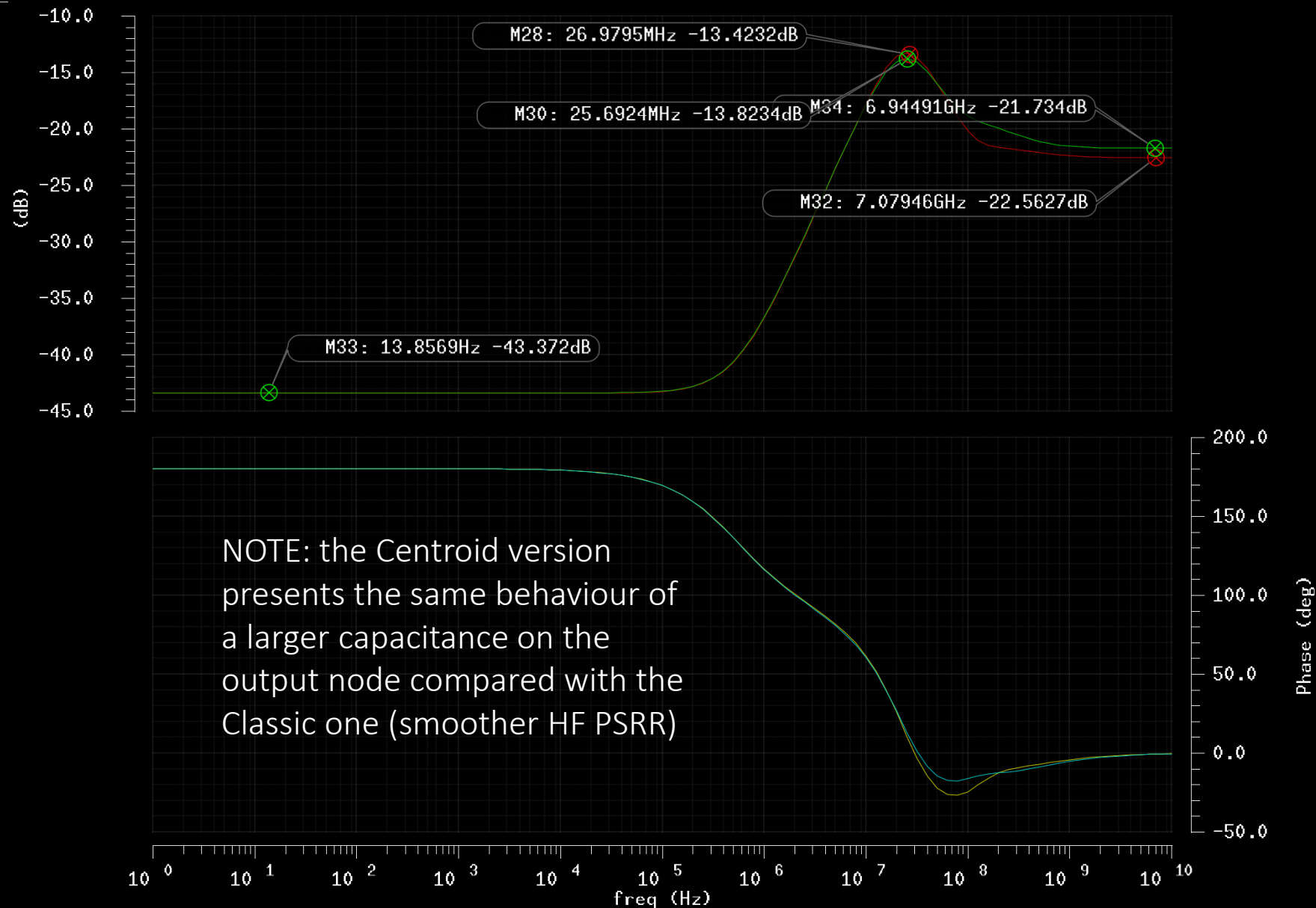
Comparing schematic vs. extracted simulations results, the latter behave better on the peak part and worst at higher frequencies.

PSRR Magnitude CCentroid  
PSRR Magnitude Classic

Schematic simulation results:

- Classic peak: -12.6 dB
- Centroid peak: -13.3 dB
- DC PSRR: -43.4 dB
- Classic HF PSRR: -24.8 dB
- Centroid HF PSRR: -24.3 dB

# EXTRACTED PSRR Simulation



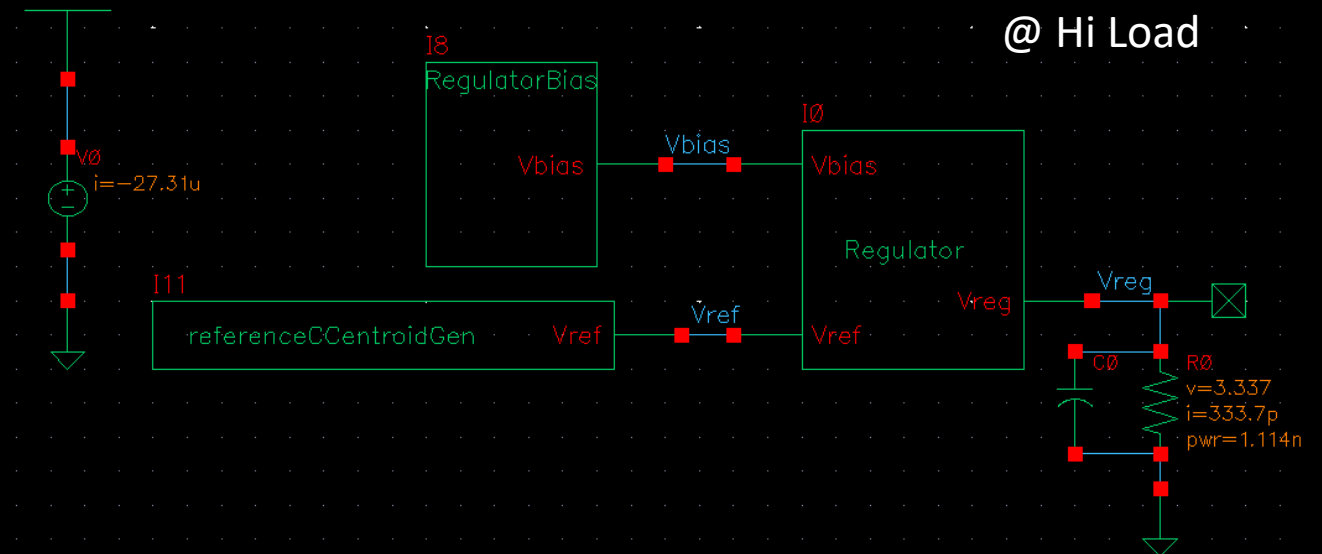
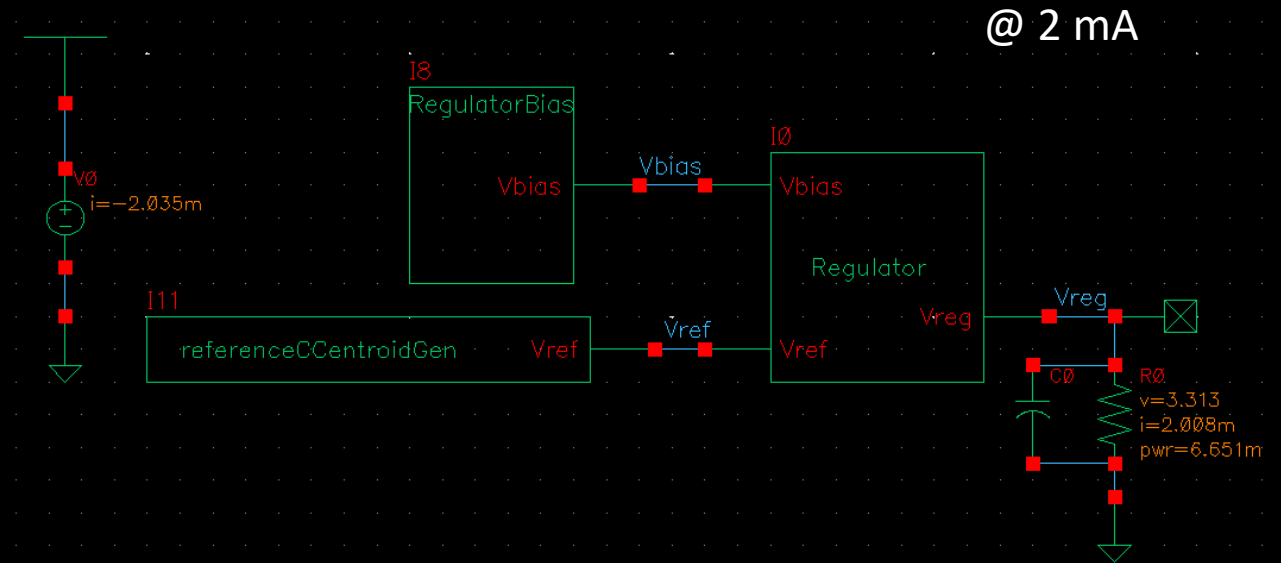
- Reference DC Sim II
- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- Reference Tran I
- Reference Tran II
- Reference Layout I
- Reference Layout II
- Reference Layout III
- Reference Layout IV
- Reference Layout V
- Ref extracted Tran
- Ref extracted PSRR
- Full DC Sim

# Complete system DC simulation

Here the complete designed circuit is tested when the load sinks the maximum current (2 mA) and when the load current is negligible. All the blocks are schematic level implementation except for the “referenceCCentroidGen” one.

The output voltage is correct when the current reaches its limit.

The high impedance DC simulation allows a quiescent current estimation which is about  $28 \mu A$ . Since the total current budget was set to  $30 \mu A$  the specification has been satisfied.

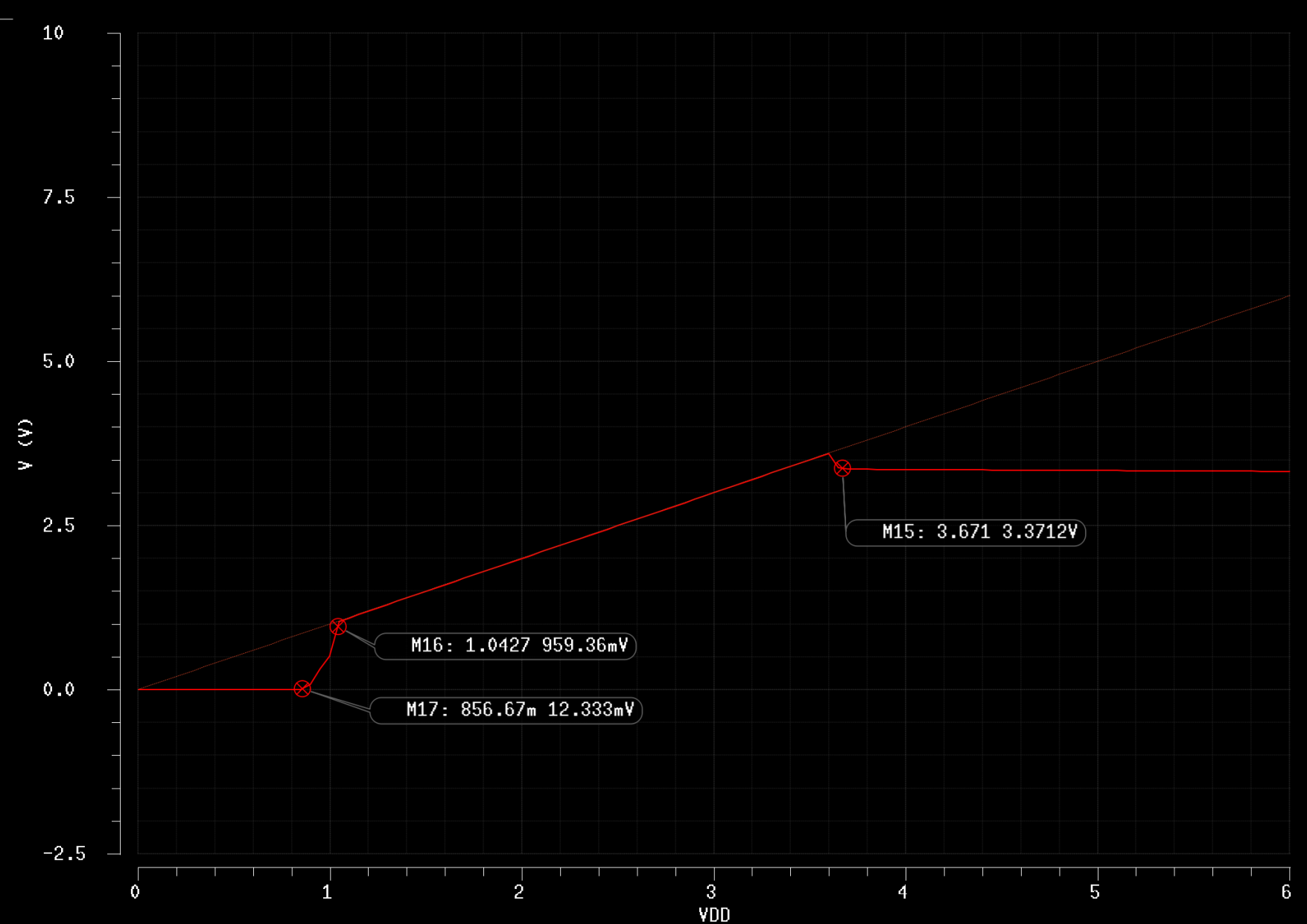


- Reference T Sweep
- Reference Line Reg
- Reference PSRR
- Reference Tran I
- Reference Tran II
- Reference Layout I
- Reference Layout II
- Reference Layout III
- Reference Layout IV
- Reference Layout V
- Ref extracted Tran
- Ref extracted PSRR
- Full DC Sim
- **Full Line Reg**

| Name                                         | V1s                                                                               |
|----------------------------------------------|-----------------------------------------------------------------------------------|
| <span style="color: #8B4513;">■</span> /vddI |  |
| <span style="color: #FF0000;">■</span> /Vreg |  |

# Complete system Line Regulation

The system starts to regulate the output voltage when VDD is above 3.7 V.



- Reference Line Reg
- Reference PSRR
- Reference Tran I
- Reference Tran II
- Reference Layout I
- Reference Layout II
- Reference Layout III
- Reference Layout IV
- Reference Layout V
- Ref extracted Tran
- Ref extracted PSRR
- Full DC Sim
- Full Line Reg
- **Full Load Reg**

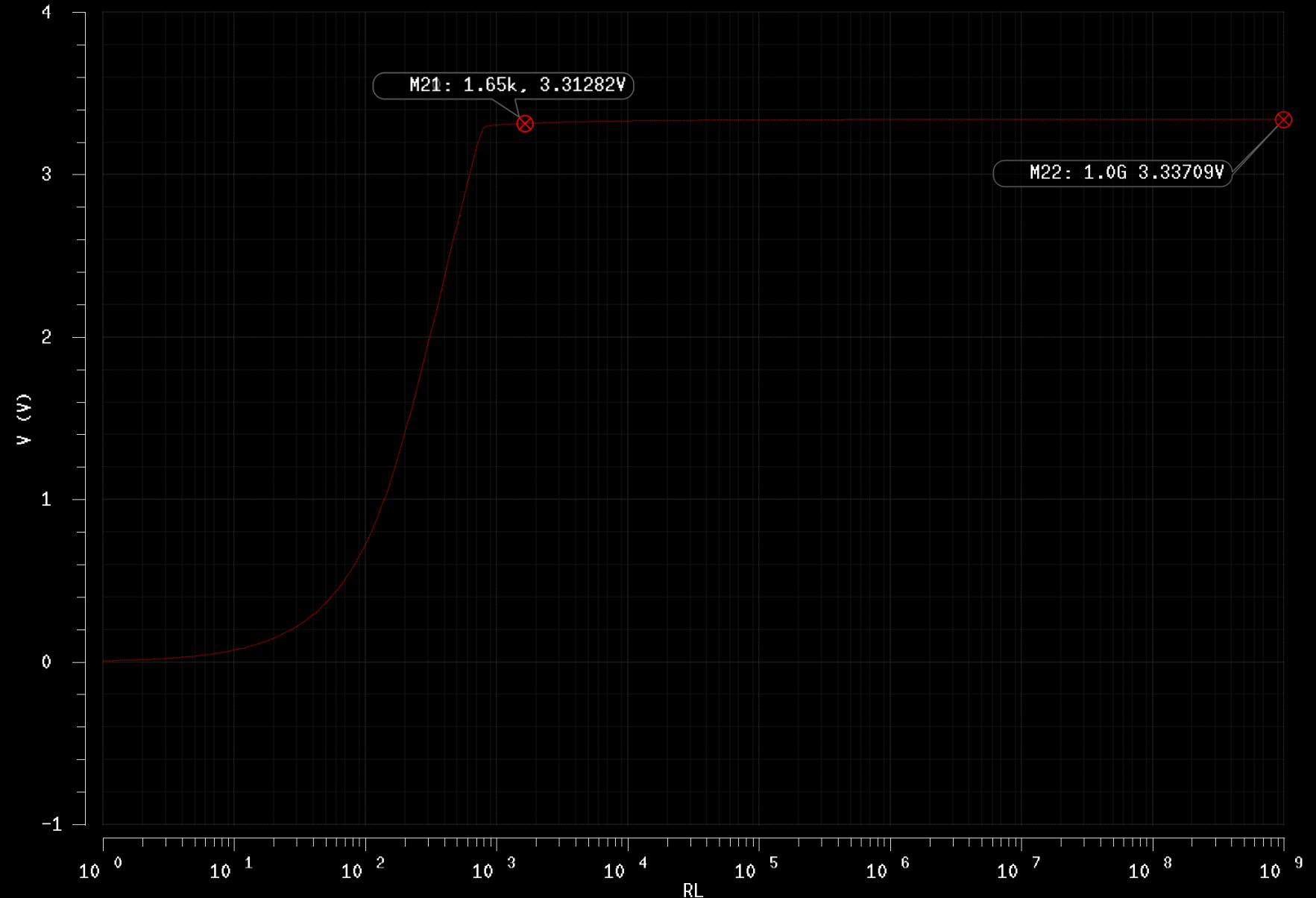
| Name                                     | Vis                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------|
| <span style="color: red;">■</span> /Vreg |  |

# Complete system Load Regulation

The variation due to load is:

$$\frac{3.337 - 3.313}{3.337} * 100 = 0.72\%$$

So the output voltage changes less the 1% when the current demand is increased up to its design limit.



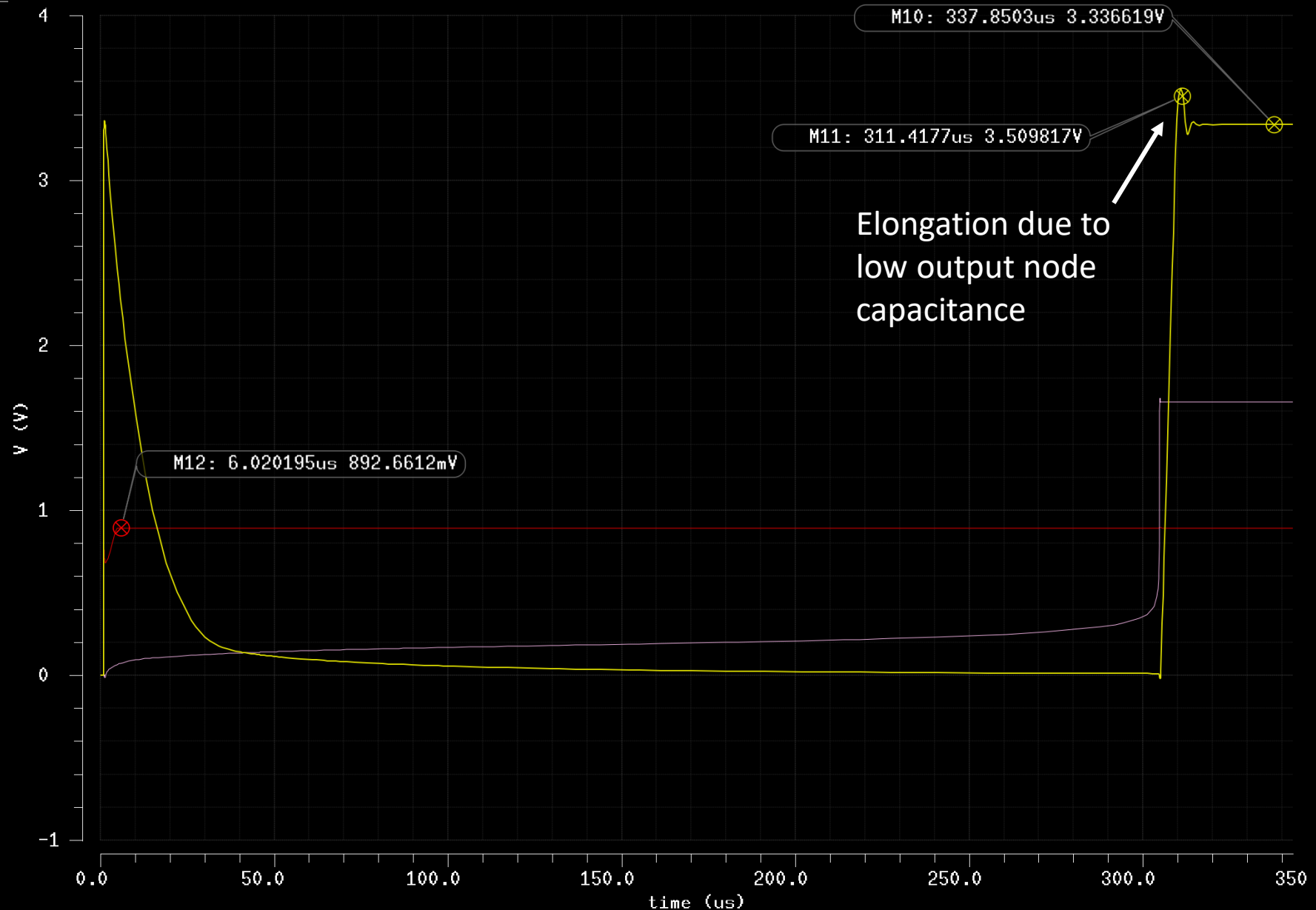
- Reference PSRR
  - Reference Tran I
  - Reference Tran II
  - Reference Layout I
  - Reference Layout II
  - Reference Layout III
  - Reference Layout IV
  - Reference Layout V
  - Ref extracted Tran
  - Ref extracted PSRR
  - Full DC Sim
  - Full Line Reg
  - Full Load Reg
  - **Full Transient I**
- | Name   | Color  |
|--------|--------|
| /Vreg  | Yellow |
| /Vbias | Red    |
| /Vref  | Purple |

The plot shows that the start-up time of the complete device is driven by the voltage reference circuit.

The BIAS circuit (red line) is ready after about  $6\ \mu\text{s}$  and before this moment the regulator output (yellow line) is pulled up by the PMOS pass-transistor. But when the BIAS is ready, the differential amplifier inside the LDO can start working and takes down  $V_{reg}$  since its inverting input is connected to the resistors divider at an higher voltage with respect to the non-inverting one.

NOTE: a low pass filter on the reference voltage output could smooth the system response and maybe improve PSRR too.

# Complete system Transient simulation



- Reference Tran I
- Reference Tran II
- Reference Layout I
- Reference Layout II
- Reference Layout III
- Reference Layout IV
- Reference Layout V
- Ref extracted Tran
- Ref extracted PSRR
- Full DC Sim
- Full Line Reg
- Full Load Reg
- Full Transient I
- **Full PSRR I**

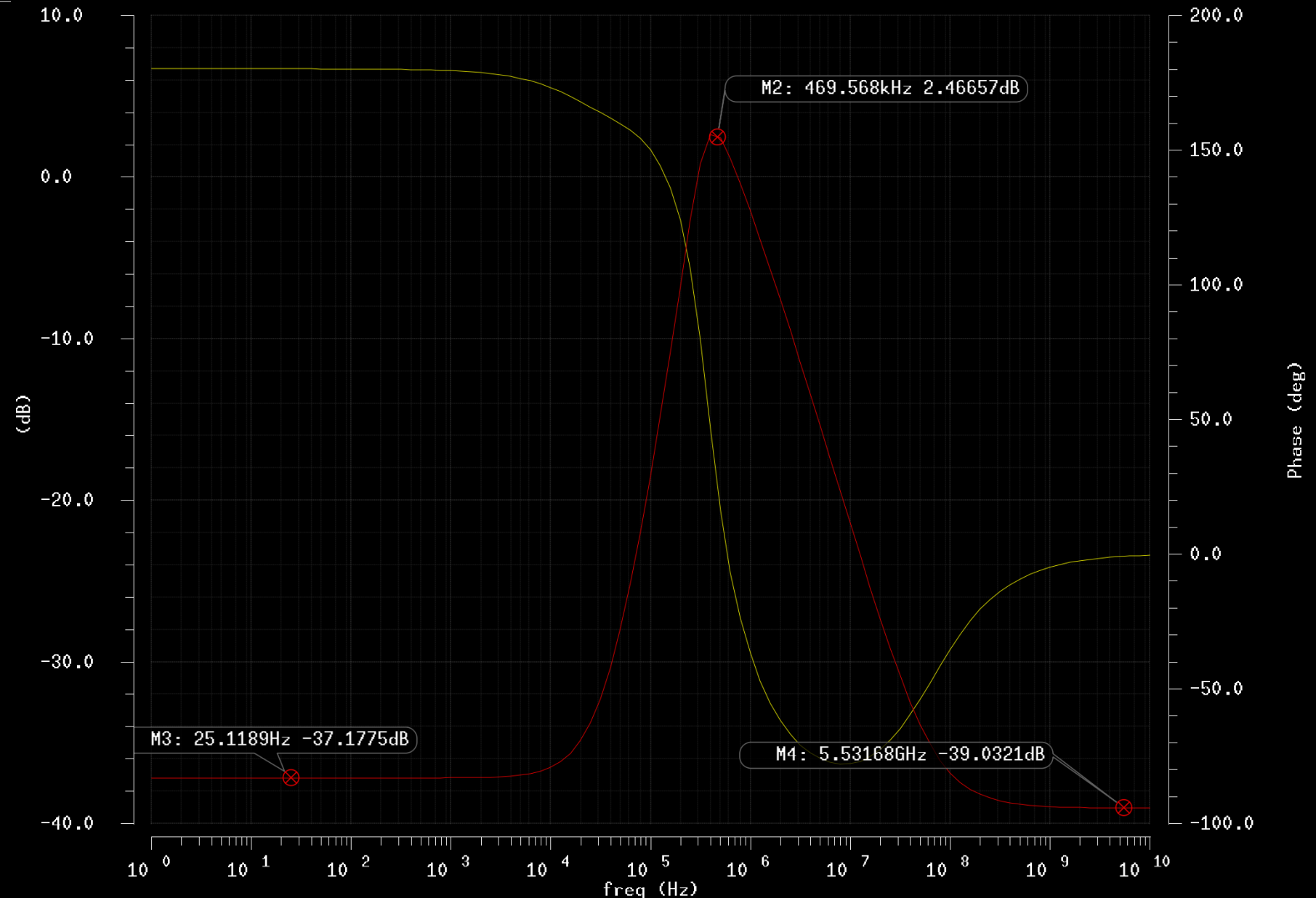
| Name           |  |
|----------------|--|
| PSRR Magnitude |  |
| PSRR Phase     |  |

The final PSRR performances are:

- -37 dB @ DC
- -39 dB @ High frequency
- 2.5 dB peak @470 MHz

The last parameter is an issue due to the inverted phase at the same very frequency. It was an expected result since it comes out from the LDO PSRR original behaviour.

# Complete system PSRR @ 23pF CL

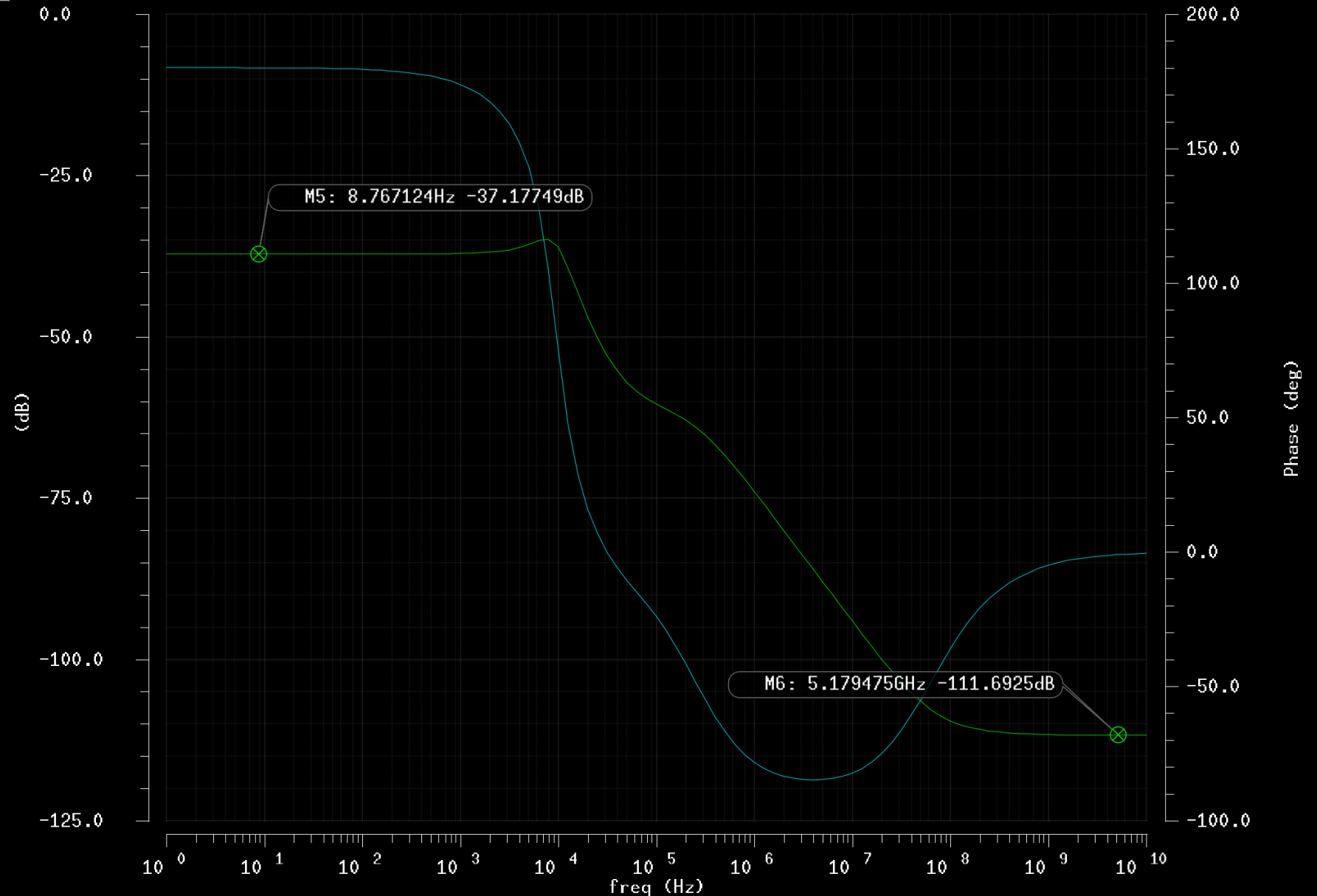


- Reference Tran II
- Reference Layout I
- Reference Layout II
- Reference Layout III
- Reference Layout IV
- Reference Layout V
- Ref extracted Tran
- Ref extracted PSRR
- Full DC Sim
- Full Line Reg
- Full Load Reg
- Full Transient I
- Full PSRR I
- **Full PSRR II**

| Name                                |                |
|-------------------------------------|----------------|
| <span style="color: red;">■</span>  | PSRR Magnitude |
| <span style="color: blue;">■</span> | PSRR Phase     |

One could think about placing an external capacitor which would make the system always stable with respect to the supply noise. The plot shows a PSRR starting from a -37 dB DC level and improving with frequency growth.

# Complete system PSRR @ 100nF CL



- Reference Layout I
- Reference Layout II
- Reference Layout III
- Reference Layout IV
- Reference Layout V
- Ref extracted Tran
- Ref extracted PSRR
- Full DC Sim
- Full Line Reg
- Full Load Reg
- Full Transient I
- Full PSRR I
- Full PSRR II
- **Final outcomes**

# Project outcomes

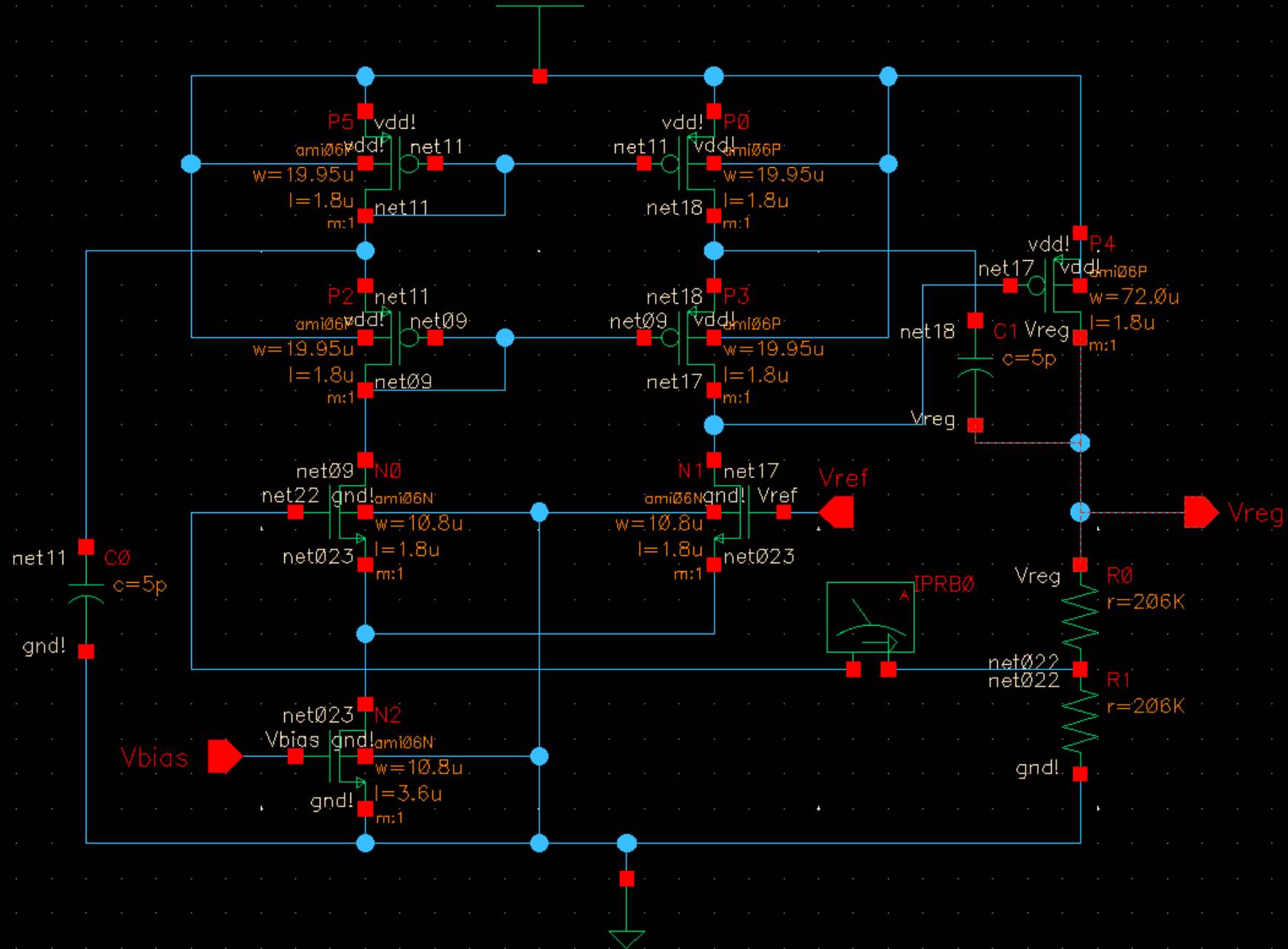
- The presented device seems to be compliant with all the design specifications.
- It shows very good Load and Line regulation (less than 1%) and drives a relatively large current.
- Its weakness is about the middle frequency (around  $500\text{ kHz}$ ) PSRR which shows a positive gain with inverted phase. Increasing the internal compensation capacitor seems to be not enough to stabilise it. Indeed, the peak is attenuated but remains positive with capacitance greater than  $30\text{ pF}$ .

An alternative to the integration of a huge capacitor is to consider an external output capacitor which would lead to a stable PSRR at any frequency.

A second possibility is to change a little bit the topology as the reference article authors did. Let's try to explore this second approach...

The modified topology is shown here and presents two main changes:

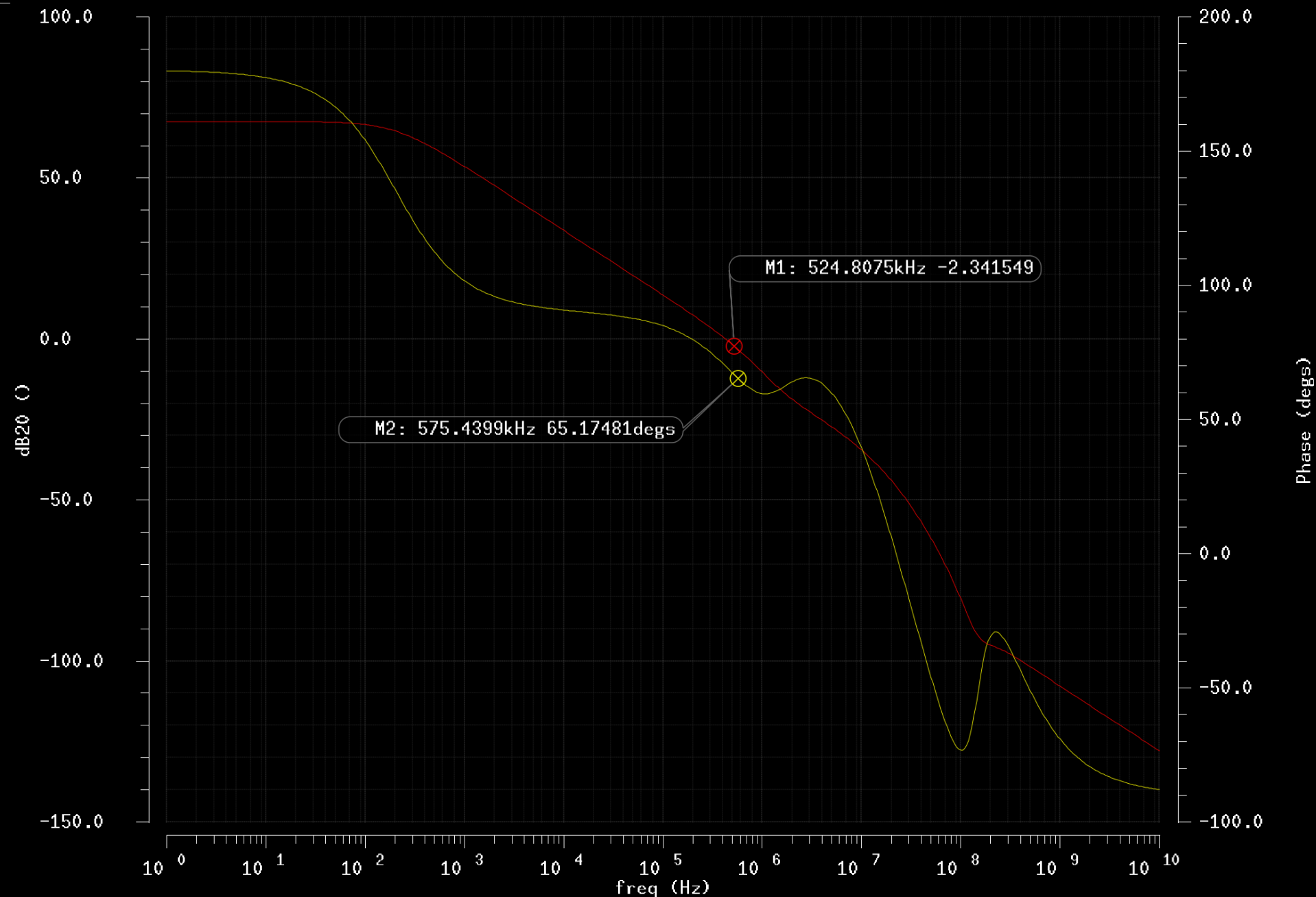
- the capacitors have been moved in the centre of the cascode current mirror;
- the cascode transistors width has been increased to get a larger gain which amplifies the compensation capacitance leading to better PSRR performances.



| Name     | Vis                                 |
|----------|-------------------------------------|
| loopGain | <input checked="" type="checkbox"/> |
| loopGain | <input checked="" type="checkbox"/> |

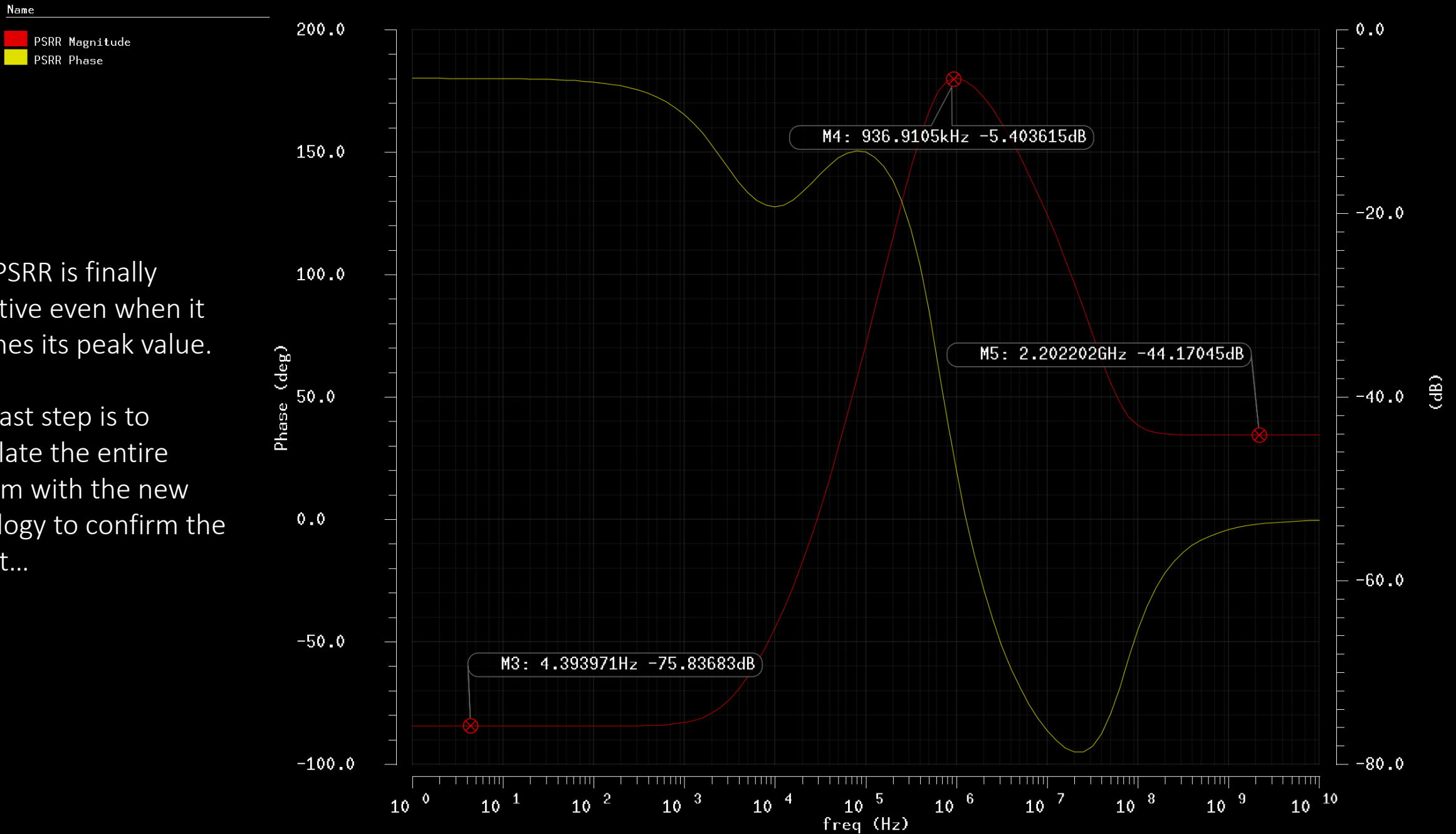
The phase margin results to be higher than before, moving from 61 dB to 65 dB.

The plot refers to an isolated analysis of the LDO schematic.



The PSRR is finally negative even when it reaches its peak value.

The last step is to simulate the entire system with the new topology to confirm the result...



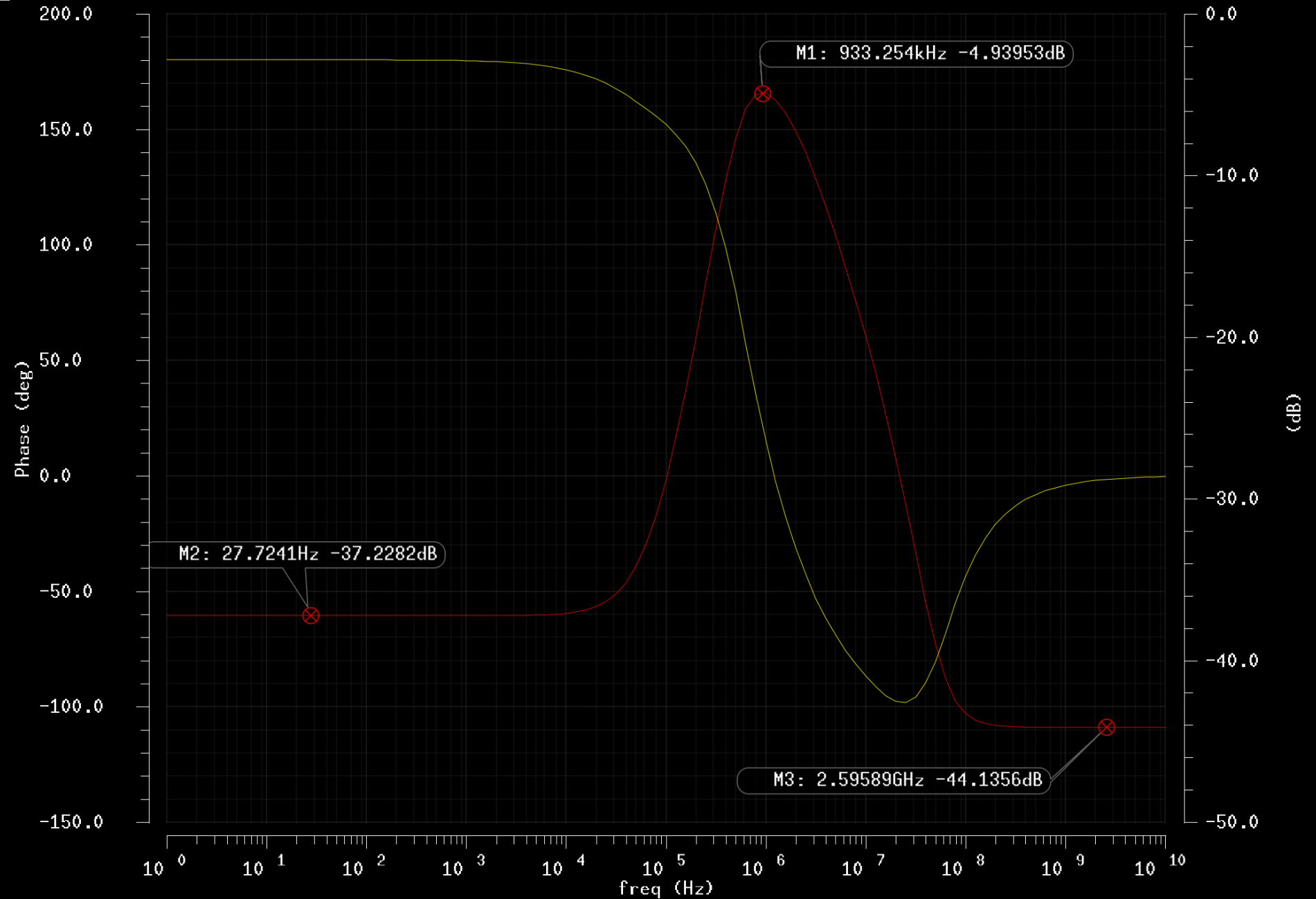
Name

PSRR Magnitude  
PSRR Phase

And with the present plot the behaviour can be confirmed for the final circuit too.

The resulting peak PSRR is:

- -5 dB @930 kHz



Thank you for your attention.